AI for Management Study Guide

Weeks 1 to 12 | Concepts, 50 MCQs, and 10 Online Assessment Questions per week

Weeks 1 to 12

Concepts explained in simple language

This PDF is arranged week by week. Each week contains a Concepts section with short, clear explanations and a full MCQs section written in exam-friendly language.

Week 1

Concepts

1. AI Types

ANI (Artificial Narrow Intelligence): Does one specific task (Siri, Netflix).
AGI: Human-level intelligence (not achieved yet).
ASI: Beyond human intelligence (theoretical).

2. AI Winter

Period where AI funding + interest dropped due to overpromises and failures.

3. AI in Strategy

AI creates advantage by:
Personalization at scale
Learning from data (feedback loops)
Automating complex tasks

4. Learning Types

Supervised: Has labels -> predicts output
Unsupervised: No labels -> finds patterns
Reinforcement: Learns via reward

5. AI Hallucination

AI gives confident but wrong answers.

6. Reasoning

Deductive: General -> specific (guaranteed correct if premise true)
Inductive: Specific -> general (not always certain)

7. Neuro-symbolic AI

Combines neural networks + logical reasoning.

8. AI Bias

AI can be biased due to biased data.

9. AI Deployment Insight

Success needs:
Training users
Process optimization
Not just building model

Example: In Week 1, a company can apply ai types together with ai winter to improve decision quality and business outcomes.

MCQs

1. AI systems like Siri belong to?

A) AGI
B) Sentient AI
C) ASI
D) ANI

Answer: D

Explanation: ANI is correct because it best fits the concept of ai systems like siri belong to.

2. What is Supervised learning uses?

A) No data
B) Random guesses
C) Labels
D) Only images

Answer: C

Explanation: Labels is correct because it best fits what is supervised learning uses.

3. What is Reinforcement learning uses?

A) Labels
B) Rewards
C) Rules
D) Patterns

Answer: B

Explanation: Rewards is correct because it best fits what is reinforcement learning uses.

4. What is Deductive reasoning?

A) Random
B) Specific -> general
C) None
D) General -> specific

Answer: D

Explanation: General -> specific is correct because it best fits what is deductive reasoning.

5. What is AI bias comes from?

A) Data
B) CPU
C) RAM
D) Hardware

Answer: A

Explanation: Data is correct because it best fits what is ai bias comes from.

6. What is ML limitation?

A) Easily fooled
B) No data
C) No errors
D) Perfect accuracy

Answer: A

Explanation: Easily fooled is correct because it best fits what is ml limitation.

7. What is Classification is?

A) Sorting
B) Clustering
C) Regression
D) Label prediction

Answer: D

Explanation: Label prediction is correct because it best fits what is classification is.

8. What is Backpropagation?

A) Forward pass
B) Stops training
C) Deletes nodes
D) Updates weights

Answer: D

Explanation: Updates weights is correct because it best fits what is backpropagation.

9. What is Hidden layers?

A) Remove data
B) Reduce power
C) Ignore input
D) Increase learning

Answer: D

Explanation: Increase learning is correct because it best fits what is hidden layers.

10. What is ANN learns?

A) No patterns
B) Rules
C) Complex patterns
D) Linear only

Answer: C

Explanation: Complex patterns is correct because it best fits what is ann learns.

11. What is Trend means?

A) Random
B) Error
C) Long-term movement
D) Noise

Answer: C

Explanation: Long-term movement is correct because it best fits what is trend means.

12. What is Cyclicity?

A) Short patterns
B) Trend
C) Noise
D) Long cycles

Answer: D

Explanation: Long cycles is correct because it best fits what is cyclicity.

13. What is White box model?

A) Interpretable
B) Complex
C) Unknown
D) Black

Answer: A

Explanation: Interpretable is correct because it best fits what is white box model.

14. What is Feedback loop helps?

A) Improve model
B) Stop model
C) Delete data
D) Reduce learning

Answer: A

Explanation: Improve model is correct because it best fits what is feedback loop helps.

15. What is Inductive reasoning?

A) Random
B) None
C) Specific -> general
D) General -> specific

Answer: C

Explanation: Specific -> general is correct because it best fits what is inductive reasoning.

16. What is Model generalization?

A) Stops
B) Memorization
C) Fails
D) Works on new data

Answer: D

Explanation: Works on new data is correct because it best fits what is model generalization.

17. What is Testing data?

A) Training
B) Input
C) Learning
D) Evaluation

Answer: D

Explanation: Evaluation is correct because it best fits what is testing data.

18. What is Learning goal?

A) Delete data
B) Random
C) Mapping input-output
D) Noise

Answer: C

Explanation: Mapping input-output is correct because it best fits what is learning goal.

19. What is ML needs?

A) No data
B) Only CPU
C) Data
D) Only GPU

Answer: C

Explanation: Data is correct because it best fits what is ml needs.

20. What is Feature?

A) Output
B) Input variable
C) Loss
D) Error

Answer: B

Explanation: Input variable is correct because it best fits what is feature.

21. What is Prediction?

A) Past
B) Future/output
C) Loss
D) Error

Answer: B

Explanation: Future/output is correct because it best fits what is prediction.

22. What is Optimization?

A) Stop
B) Random
C) Delete
D) Improve model

Answer: D

Explanation: Improve model is correct because it best fits what is optimization.

23. What is Learning rate?

A) Speed of learning
B) Output
C) Data size
D) Input

Answer: A

Explanation: Speed of learning is correct because it best fits what is learning rate.

24. What is Large learning rate?

A) Stable
B) Perfect
C) Slow
D) Unstable

Answer: D

Explanation: Unstable is correct because it best fits what is large learning rate.

25. What is Data quality affects?

A) GPU
B) CPU
C) Model performance
D) Nothing

Answer: C

Explanation: Model performance is correct because it best fits what is data quality affects.

26. What best describes AI Types in AI for management?

A) AI Types guarantees perfect outcomes in every scenario.
B) ANI (Artificial Narrow Intelligence): Does one specific task (Siri, Netflix).
C) AI Types removes the need for data quality checks and monitoring.
D) AI Types eliminates the need for managerial judgment and governance.

Answer: B

Explanation: ANI (Artificial Narrow Intelligence): Does one specific task (Siri, Netflix). is correct because it best fits what best describes ai types in ai for management.

27. Which statement about AI Winter is most accurate?

A) AI Winter eliminates the need for managerial judgment and governance.
B) Period where AI funding + interest dropped due to overpromises and failures.
C) AI Winter removes the need for data quality checks and monitoring.
D) AI Winter guarantees perfect outcomes in every scenario.

Answer: B

Explanation: Period where AI funding + interest dropped due to overpromises and failures. is correct because it best fits which statement about ai winter is most accurate.

28. How does AI in Strategy support managerial decision-making?

A) AI in Strategy removes the need for data quality checks and monitoring.
B) AI in Strategy guarantees perfect outcomes in every scenario.
C) Personalization at scale
D) AI in Strategy eliminates the need for managerial judgment and governance.

Answer: C

Explanation: Personalization at scale is correct because it best fits how does ai in strategy support managerial decision-making.

29. What is a key business implication of Learning Types?

A) Learning Types removes the need for data quality checks and monitoring.
B) Learning Types eliminates the need for managerial judgment and governance.
C) Supervised: Has labels -> predicts output
D) Learning Types guarantees perfect outcomes in every scenario.

Answer: C

Explanation: Supervised: Has labels -> predicts output is correct because it best fits what is a key business implication of learning types.

30. What best describes AI Hallucination in AI for management?

A) AI gives confident but wrong answers.
B) AI Hallucination guarantees perfect outcomes in every scenario.
C) AI Hallucination eliminates the need for managerial judgment and governance.
D) AI Hallucination removes the need for data quality checks and monitoring.

Answer: A

Explanation: AI gives confident but wrong answers. is correct because it best fits what best describes ai hallucination in ai for management.

31. Which statement about Reasoning is most accurate?

A) Reasoning guarantees perfect outcomes in every scenario.
B) Reasoning removes the need for data quality checks and monitoring.
C) Deductive: General -> specific (guaranteed correct if premise true)
D) Reasoning eliminates the need for managerial judgment and governance.

Answer: C

Explanation: Deductive: General -> specific (guaranteed correct if premise true) is correct because it best fits which statement about reasoning is most accurate.

32. How does Neuro-symbolic AI support managerial decision-making?

A) Neuro-symbolic AI removes the need for data quality checks and monitoring.
B) Combines neural networks + logical reasoning.
C) Neuro-symbolic AI eliminates the need for managerial judgment and governance.
D) Neuro-symbolic AI guarantees perfect outcomes in every scenario.

Answer: B

Explanation: Combines neural networks + logical reasoning. is correct because it best fits how does neuro-symbolic ai support managerial decision-making.

33. What is a key business implication of AI Bias?

A) AI can be biased due to biased data.
B) AI Bias eliminates the need for managerial judgment and governance.
C) AI Bias guarantees perfect outcomes in every scenario.
D) AI Bias removes the need for data quality checks and monitoring.

Answer: A

Explanation: AI can be biased due to biased data. is correct because it best fits what is a key business implication of ai bias.

34. What best describes AI Deployment Insight in AI for management?

A) AI Deployment Insight guarantees perfect outcomes in every scenario.
B) AI Deployment Insight eliminates the need for managerial judgment and governance.
C) AI Deployment Insight removes the need for data quality checks and monitoring.
D) Training users

Answer: D

Explanation: Training users is correct because it best fits what best describes ai deployment insight in ai for management.

35. Which statement about AI Types is most accurate?

A) ANI (Artificial Narrow Intelligence): Does one specific task (Siri, Netflix).
B) AI Types guarantees perfect outcomes in every scenario.
C) AI Types eliminates the need for managerial judgment and governance.
D) AI Types removes the need for data quality checks and monitoring.

Answer: A

Explanation: ANI (Artificial Narrow Intelligence): Does one specific task (Siri, Netflix). is correct because it best fits which statement about ai types is most accurate.

36. How does AI Winter support managerial decision-making?

A) AI Winter guarantees perfect outcomes in every scenario.
B) Period where AI funding + interest dropped due to overpromises and failures.
C) AI Winter eliminates the need for managerial judgment and governance.
D) AI Winter removes the need for data quality checks and monitoring.

Answer: B

Explanation: Period where AI funding + interest dropped due to overpromises and failures. is correct because it best fits how does ai winter support managerial decision-making.

37. What is a key business implication of AI in Strategy?

A) Personalization at scale
B) AI in Strategy guarantees perfect outcomes in every scenario.
C) AI in Strategy removes the need for data quality checks and monitoring.
D) AI in Strategy eliminates the need for managerial judgment and governance.

Answer: A

Explanation: Personalization at scale is correct because it best fits what is a key business implication of ai in strategy.

38. What best describes Learning Types in AI for management?

A) Supervised: Has labels -> predicts output
B) Learning Types eliminates the need for managerial judgment and governance.
C) Learning Types removes the need for data quality checks and monitoring.
D) Learning Types guarantees perfect outcomes in every scenario.

Answer: A

Explanation: Supervised: Has labels -> predicts output is correct because it best fits what best describes learning types in ai for management.

39. Which statement about AI Hallucination is most accurate?

A) AI Hallucination eliminates the need for managerial judgment and governance.
B) AI Hallucination removes the need for data quality checks and monitoring.
C) AI gives confident but wrong answers.
D) AI Hallucination guarantees perfect outcomes in every scenario.

Answer: C

Explanation: AI gives confident but wrong answers. is correct because it best fits which statement about ai hallucination is most accurate.

40. How does Reasoning support managerial decision-making?

A) Reasoning eliminates the need for managerial judgment and governance.
B) Reasoning guarantees perfect outcomes in every scenario.
C) Deductive: General -> specific (guaranteed correct if premise true)
D) Reasoning removes the need for data quality checks and monitoring.

Answer: C

Explanation: Deductive: General -> specific (guaranteed correct if premise true) is correct because it best fits how does reasoning support managerial decision-making.

41. What is a key business implication of Neuro-symbolic AI?

A) Neuro-symbolic AI removes the need for data quality checks and monitoring.
B) Neuro-symbolic AI eliminates the need for managerial judgment and governance.
C) Combines neural networks + logical reasoning.
D) Neuro-symbolic AI guarantees perfect outcomes in every scenario.

Answer: C

Explanation: Combines neural networks + logical reasoning. is correct because it best fits what is a key business implication of neuro-symbolic ai.

42. What best describes AI Bias in AI for management?

A) AI Bias guarantees perfect outcomes in every scenario.
B) AI Bias removes the need for data quality checks and monitoring.
C) AI Bias eliminates the need for managerial judgment and governance.
D) AI can be biased due to biased data.

Answer: D

Explanation: AI can be biased due to biased data. is correct because it best fits what best describes ai bias in ai for management.

43. Which statement about AI Deployment Insight is most accurate?

A) AI Deployment Insight removes the need for data quality checks and monitoring.
B) AI Deployment Insight eliminates the need for managerial judgment and governance.
C) AI Deployment Insight guarantees perfect outcomes in every scenario.
D) Training users

Answer: D

Explanation: Training users is correct because it best fits which statement about ai deployment insight is most accurate.

44. How does AI Types support managerial decision-making?

A) AI Types eliminates the need for managerial judgment and governance.
B) AI Types guarantees perfect outcomes in every scenario.
C) AI Types removes the need for data quality checks and monitoring.
D) ANI (Artificial Narrow Intelligence): Does one specific task (Siri, Netflix).

Answer: D

Explanation: ANI (Artificial Narrow Intelligence): Does one specific task (Siri, Netflix). is correct because it best fits how does ai types support managerial decision-making.

45. What is a key business implication of AI Winter?

A) AI Winter removes the need for data quality checks and monitoring.
B) AI Winter eliminates the need for managerial judgment and governance.
C) Period where AI funding + interest dropped due to overpromises and failures.
D) AI Winter guarantees perfect outcomes in every scenario.

Answer: C

Explanation: Period where AI funding + interest dropped due to overpromises and failures. is correct because it best fits what is a key business implication of ai winter.

46. What best describes AI in Strategy in AI for management?

A) AI in Strategy guarantees perfect outcomes in every scenario.
B) AI in Strategy removes the need for data quality checks and monitoring.
C) Personalization at scale
D) AI in Strategy eliminates the need for managerial judgment and governance.

Answer: C

Explanation: Personalization at scale is correct because it best fits what best describes ai in strategy in ai for management.

47. Which statement about Learning Types is most accurate?

A) Supervised: Has labels -> predicts output
B) Learning Types guarantees perfect outcomes in every scenario.
C) Learning Types eliminates the need for managerial judgment and governance.
D) Learning Types removes the need for data quality checks and monitoring.

Answer: A

Explanation: Supervised: Has labels -> predicts output is correct because it best fits which statement about learning types is most accurate.

48. How does AI Hallucination support managerial decision-making?

A) AI Hallucination removes the need for data quality checks and monitoring.
B) AI Hallucination guarantees perfect outcomes in every scenario.
C) AI Hallucination eliminates the need for managerial judgment and governance.
D) AI gives confident but wrong answers.

Answer: D

Explanation: AI gives confident but wrong answers. is correct because it best fits how does ai hallucination support managerial decision-making.

49. What is a key business implication of Reasoning?

A) Reasoning removes the need for data quality checks and monitoring.
B) Reasoning eliminates the need for managerial judgment and governance.
C) Deductive: General -> specific (guaranteed correct if premise true)
D) Reasoning guarantees perfect outcomes in every scenario.

Answer: C

Explanation: Deductive: General -> specific (guaranteed correct if premise true) is correct because it best fits what is a key business implication of reasoning.

50. What best describes Neuro-symbolic AI in AI for management?

A) Combines neural networks + logical reasoning.
B) Neuro-symbolic AI removes the need for data quality checks and monitoring.
C) Neuro-symbolic AI eliminates the need for managerial judgment and governance.
D) Neuro-symbolic AI guarantees perfect outcomes in every scenario.

Answer: A

Explanation: Combines neural networks + logical reasoning. is correct because it best fits what best describes neuro-symbolic ai in ai for management.

Online Assessment Questions

1. The type of AI that all current real-world systems belong to is:

A) Artificial Narrow Intelligence (ANI)
B) Sentient AI
C) Artificial General Intelligence (AGI)
D) Artificial Super Intelligence (ASI)

Answer: A

Explanation: Artificial Narrow Intelligence (ANI) is correct because it best fits the concept of the type of ai that all current real-world systems belong to is.

2. The period of reduced funding and interest in AI is called:

A) Singularity
B) AI Spring
C) AI Winter
D) Connectionist Boom

Answer: C

Explanation: AI Winter is correct because it best fits the concept of the period of reduced funding and interest in ai is called.

3. How does AI create competitive advantage? (Select all that apply)

A) Automating complex tasks
B) Hyper-personalization
C) Data-driven feedback loops
D) Removing AI systems

Answer: A, B, C

Explanation: Automating complex tasks, Hyper-personalization, Data-driven feedback loops are correct because each option captures a required part of how does ai create competitive advantage? (select all that apply).

4. The primary goal of an unsupervised learning algorithm is to:

A) Predict labels
B) Discover patterns in unlabeled data
C) Classify predefined data
D) Maximize reward

Answer: B

Explanation: Discover patterns in unlabeled data is correct because it best fits the concept of the primary goal of an unsupervised learning algorithm is to.

5. AI hallucination refers to:

A) Slow computation
B) Bias in AI
C) Easily tricked AI
D) Confident but incorrect output

Answer: D

Explanation: Confident but incorrect output is correct because it best fits the concept of ai hallucination refers to.

6. Which reasoning moves from general to specific?

A) Deductive
B) Abductive
C) Inductive
D) None

Answer: A

Explanation: Deductive is correct because it best fits which reasoning moves from general to specific.

7. The hybrid AI combining neural networks and logic is:

A) Probabilistic
B) Neuro-symbolic
C) Symbolist
D) Evolutionary

Answer: B

Explanation: Neuro-symbolic is correct because it best fits the concept of the hybrid ai combining neural networks and logic is.

8. Leadership volatility in OpenAI (late 2023) was due to:

A) Cloud conflicts
B) Safety vs speed disagreements
C) Logo issues
D) Regulations

Answer: B

Explanation: Safety vs speed disagreements is correct because it best fits the concept of leadership volatility in openai (late 2023) was due to.

9. AI systems are completely unbiased.

A) False
B) True

Answer: A

Explanation: False is correct because it best fits the concept of ai systems are completely unbiased.

10. Successful AI deployment requires:

A) Only model building
B) Marketing
C) Better hardware
D) Training users and process optimization

Answer: D

Explanation: Training users and process optimization is correct because it best fits the concept of successful ai deployment requires.

Week 2

Concepts

1. ML Limitations

Easily fooled by small input changes
Needs supervision
Lacks common sense

2. Overfitting

Model memorizes noise -> poor generalization

3. Supervised Learning Characteristics

Uses labeled data
Used in classification/regression
Learns input -> output mapping

4. Perceptron Limitation

Works only if data is linearly separable

5. Backpropagation

Solves multilayer learning
Updates weights using error

6. Error Function

Half squared error between predicted and actual

7. Hidden Layers

More layers -> more power
But increases overfitting risk

8. ANN Advantage

Hidden layers -> capture complex patterns

9. Time Series Components

Trend
Seasonality
Cyclicity
Randomness

10. White Box vs Neural Network

White box -> interpretable
Neural network -> black box

Example: In Week 2, a company can apply ml limitations together with overfitting to improve decision quality and business outcomes.

MCQs

1. What is AI Winter refers to?

A) Funding decline
B) Hardware upgrade
C) Model training
D) Growth phase

Answer: A

Explanation: Funding decline is correct because it best fits what is ai winter refers to.

2. What is Unsupervised learning?

A) Uses labels
B) Finds patterns
C) Uses reward
D) Predicts output

Answer: B

Explanation: Finds patterns is correct because it best fits what is unsupervised learning.

3. What is AI hallucination is?

A) Overfitting
B) Learning
C) Wrong but confident answer
D) Bias

Answer: C

Explanation: Wrong but confident answer is correct because it best fits what is ai hallucination is.

4. What is Neuro-symbolic AI?

A) Only rules
B) Only neural
C) Hybrid
D) None

Answer: C

Explanation: Hybrid is correct because it best fits what is neuro-symbolic ai.

5. What is Deployment success depends on?

A) Cost
B) Training + process
C) Data only
D) Only model

Answer: B

Explanation: Training + process is correct because it best fits what is deployment success depends on.

6. What is Overfitting means?

A) Under learning
B) No learning
C) Memorizing noise
D) Perfect learning

Answer: C

Explanation: Memorizing noise is correct because it best fits what is overfitting means.

7. What is Perceptron fails when?

A) Non-linear
B) Clean data
C) Large data
D) Data small

Answer: A

Explanation: Non-linear is correct because it best fits what is perceptron fails when.

8. What is Error function uses?

A) Squared error
B) Random
C) Absolute error
D) Log

Answer: A

Explanation: Squared error is correct because it best fits what is error function uses.

9. What is More layers cause?

A) Accuracy drop always
B) No change
C) Underfitting
D) Overfitting risk

Answer: D

Explanation: Overfitting risk is correct because it best fits what is more layers cause.

10. What is Time series includes?

A) None
B) Only trend
C) Only seasonality
D) Multiple components

Answer: D

Explanation: Multiple components is correct because it best fits what is time series includes.

11. What is Seasonality?

A) Trend
B) Periodic pattern
C) Random
D) Noise

Answer: B

Explanation: Periodic pattern is correct because it best fits what is seasonality.

12. What is Randomness?

A) Noise
B) Trend
C) Cycle
D) Predictable

Answer: A

Explanation: Noise is correct because it best fits what is randomness.

13. What is Neural network?

A) Black box
B) Linear
C) Simple
D) Transparent

Answer: A

Explanation: Black box is correct because it best fits what is neural network.

14. What is AI replaces?

A) Simple tasks only
B) Nothing
C) Complex tasks
D) Humans always

Answer: C

Explanation: Complex tasks is correct because it best fits what is ai replaces.

15. What is Bias means?

A) Accuracy
B) Unfair pattern
C) Loss
D) Fair

Answer: B

Explanation: Unfair pattern is correct because it best fits what is bias means.

16. What is Training data?

A) Learning
B) Error
C) Output
D) Testing

Answer: A

Explanation: Learning is correct because it best fits what is training data.

17. What is ANN stands for?

A) None
B) Artificial Neural Network
C) AI Neural Node
D) Automated Node Network

Answer: B

Explanation: Artificial Neural Network is correct because it best fits what is ann stands for.

18. What is Overfitting reduces?

A) Training accuracy
B) Both
C) Test accuracy
D) None

Answer: C

Explanation: Test accuracy is correct because it best fits what is overfitting reduces.

19. What is Labels are?

A) Random
B) Inputs
C) Outputs
D) Hidden

Answer: C

Explanation: Outputs is correct because it best fits what is labels are.

20. What is Model?

A) Function
B) Data
C) Noise
D) Error

Answer: A

Explanation: Function is correct because it best fits what is model.

21. What is Loss?

A) Gain
B) Error measure
C) Input
D) Data

Answer: B

Explanation: Error measure is correct because it best fits what is loss.

22. What is Gradient?

A) Random
B) Direction
C) Error
D) Loss

Answer: B

Explanation: Direction is correct because it best fits what is gradient.

23. What is Small learning rate?

A) Random
B) Fast
C) Slow
D) No learning

Answer: C

Explanation: Slow is correct because it best fits what is small learning rate.

24. What is Model complexity?

A) No layers
B) Random
C) Low layers
D) High layers

Answer: D

Explanation: High layers is correct because it best fits what is model complexity.

25. What is AI goal?

A) Replace all
B) None
C) Stop humans
D) Assist + automate

Answer: D

Explanation: Assist + automate is correct because it best fits what is ai goal.

26. What best describes ML Limitations in AI for management?

A) ML Limitations guarantees perfect outcomes in every scenario.
B) ML Limitations removes the need for data quality checks and monitoring.
C) ML Limitations eliminates the need for managerial judgment and governance.
D) Easily fooled by small input changes

Answer: D

Explanation: Easily fooled by small input changes is correct because it best fits what best describes ml limitations in ai for management.

27. Which statement about Overfitting is most accurate?

A) Overfitting eliminates the need for managerial judgment and governance.
B) Overfitting guarantees perfect outcomes in every scenario.
C) Overfitting removes the need for data quality checks and monitoring.
D) Model memorizes noise -> poor generalization

Answer: D

Explanation: Model memorizes noise -> poor generalization is correct because it best fits which statement about overfitting is most accurate.

28. How does Supervised Learning Characteristics support managerial decision-making?

A) Uses labeled data
B) Supervised Learning Characteristics eliminates the need for managerial judgment and governance.
C) Supervised Learning Characteristics removes the need for data quality checks and monitoring.
D) Supervised Learning Characteristics guarantees perfect outcomes in every scenario.

Answer: A

Explanation: Uses labeled data is correct because it best fits how does supervised learning characteristics support managerial decision-making.

29. What is a key business implication of Perceptron Limitation?

A) Perceptron Limitation eliminates the need for managerial judgment and governance.
B) Perceptron Limitation removes the need for data quality checks and monitoring.
C) Perceptron Limitation guarantees perfect outcomes in every scenario.
D) Works only if data is linearly separable

Answer: D

Explanation: Works only if data is linearly separable is correct because it best fits what is a key business implication of perceptron limitation.

30. What best describes Backpropagation in AI for management?

A) Backpropagation eliminates the need for managerial judgment and governance.
B) Backpropagation guarantees perfect outcomes in every scenario.
C) Backpropagation removes the need for data quality checks and monitoring.
D) Solves multilayer learning

Answer: D

Explanation: Solves multilayer learning is correct because it best fits what best describes backpropagation in ai for management.

31. Which statement about Error Function is most accurate?

A) Error Function eliminates the need for managerial judgment and governance.
B) Half squared error between predicted and actual
C) Error Function guarantees perfect outcomes in every scenario.
D) Error Function removes the need for data quality checks and monitoring.

Answer: B

Explanation: Half squared error between predicted and actual is correct because it best fits which statement about error function is most accurate.

32. How does Hidden Layers support managerial decision-making?

A) More layers -> more power
B) Hidden Layers eliminates the need for managerial judgment and governance.
C) Hidden Layers guarantees perfect outcomes in every scenario.
D) Hidden Layers removes the need for data quality checks and monitoring.

Answer: A

Explanation: More layers -> more power is correct because it best fits how does hidden layers support managerial decision-making.

33. What is a key business implication of ANN Advantage?

A) ANN Advantage eliminates the need for managerial judgment and governance.
B) Hidden layers -> capture complex patterns
C) ANN Advantage guarantees perfect outcomes in every scenario.
D) ANN Advantage removes the need for data quality checks and monitoring.

Answer: B

Explanation: Hidden layers -> capture complex patterns is correct because it best fits what is a key business implication of ann advantage.

34. What best describes Time Series Components in AI for management?

A) Time Series Components guarantees perfect outcomes in every scenario.
B) Time Series Components removes the need for data quality checks and monitoring.
C) Time Series Components eliminates the need for managerial judgment and governance.
D) Time Series Components

Answer: D

Explanation: Time Series Components is correct because it best fits what best describes time series components in ai for management.

35. Which statement about White Box vs Neural Network is most accurate?

A) White Box vs Neural Network guarantees perfect outcomes in every scenario.
B) White box -> interpretable
C) White Box vs Neural Network eliminates the need for managerial judgment and governance.
D) White Box vs Neural Network removes the need for data quality checks and monitoring.

Answer: B

Explanation: White box -> interpretable is correct because it best fits which statement about white box vs neural network is most accurate.

36. How does ML Limitations support managerial decision-making?

A) ML Limitations eliminates the need for managerial judgment and governance.
B) Easily fooled by small input changes
C) ML Limitations guarantees perfect outcomes in every scenario.
D) ML Limitations removes the need for data quality checks and monitoring.

Answer: B

Explanation: Easily fooled by small input changes is correct because it best fits how does ml limitations support managerial decision-making.

37. What is a key business implication of Overfitting?

A) Model memorizes noise -> poor generalization
B) Overfitting guarantees perfect outcomes in every scenario.
C) Overfitting eliminates the need for managerial judgment and governance.
D) Overfitting removes the need for data quality checks and monitoring.

Answer: A

Explanation: Model memorizes noise -> poor generalization is correct because it best fits what is a key business implication of overfitting.

38. What best describes Supervised Learning Characteristics in AI for management?

A) Supervised Learning Characteristics removes the need for data quality checks and monitoring.
B) Supervised Learning Characteristics eliminates the need for managerial judgment and governance.
C) Supervised Learning Characteristics guarantees perfect outcomes in every scenario.
D) Uses labeled data

Answer: D

Explanation: Uses labeled data is correct because it best fits what best describes supervised learning characteristics in ai for management.

39. Which statement about Perceptron Limitation is most accurate?

A) Works only if data is linearly separable
B) Perceptron Limitation eliminates the need for managerial judgment and governance.
C) Perceptron Limitation guarantees perfect outcomes in every scenario.
D) Perceptron Limitation removes the need for data quality checks and monitoring.

Answer: A

Explanation: Works only if data is linearly separable is correct because it best fits which statement about perceptron limitation is most accurate.

40. How does Backpropagation support managerial decision-making?

A) Backpropagation guarantees perfect outcomes in every scenario.
B) Backpropagation eliminates the need for managerial judgment and governance.
C) Solves multilayer learning
D) Backpropagation removes the need for data quality checks and monitoring.

Answer: C

Explanation: Solves multilayer learning is correct because it best fits how does backpropagation support managerial decision-making.

41. What is a key business implication of Error Function?

A) Half squared error between predicted and actual
B) Error Function removes the need for data quality checks and monitoring.
C) Error Function guarantees perfect outcomes in every scenario.
D) Error Function eliminates the need for managerial judgment and governance.

Answer: A

Explanation: Half squared error between predicted and actual is correct because it best fits what is a key business implication of error function.

42. What best describes Hidden Layers in AI for management?

A) More layers -> more power
B) Hidden Layers guarantees perfect outcomes in every scenario.
C) Hidden Layers eliminates the need for managerial judgment and governance.
D) Hidden Layers removes the need for data quality checks and monitoring.

Answer: A

Explanation: More layers -> more power is correct because it best fits what best describes hidden layers in ai for management.

43. Which statement about ANN Advantage is most accurate?

A) ANN Advantage removes the need for data quality checks and monitoring.
B) ANN Advantage eliminates the need for managerial judgment and governance.
C) Hidden layers -> capture complex patterns
D) ANN Advantage guarantees perfect outcomes in every scenario.

Answer: C

Explanation: Hidden layers -> capture complex patterns is correct because it best fits which statement about ann advantage is most accurate.

44. How does Time Series Components support managerial decision-making?

A) Time Series Components guarantees perfect outcomes in every scenario.
B) Time Series Components eliminates the need for managerial judgment and governance.
C) Time Series Components
D) Time Series Components removes the need for data quality checks and monitoring.

Answer: C

Explanation: Time Series Components is correct because it best fits how does time series components support managerial decision-making.

45. What is a key business implication of White Box vs Neural Network?

A) White Box vs Neural Network removes the need for data quality checks and monitoring.
B) White box -> interpretable
C) White Box vs Neural Network eliminates the need for managerial judgment and governance.
D) White Box vs Neural Network guarantees perfect outcomes in every scenario.

Answer: B

Explanation: White box -> interpretable is correct because it best fits what is a key business implication of white box vs neural network.

46. What is one important AI-for-management consideration in Week 2 scenario 46?

A) It should rely only on hardware changes without process updates.
B) It should be aligned with business outcomes and decision quality.
C) It should ignore governance, validation, and monitoring.
D) It should replace managerial judgment in all cases.

Answer: B

Explanation: It should be aligned with business outcomes and decision quality. is correct because it best fits what is one important ai-for-management consideration in week 2 scenario 46.

47. What is one important AI-for-management consideration in Week 2 scenario 47?

A) It should ignore governance, validation, and monitoring.
B) It should replace managerial judgment in all cases.
C) It should rely only on hardware changes without process updates.
D) It should be aligned with business outcomes and decision quality.

Answer: D

Explanation: It should be aligned with business outcomes and decision quality. is correct because it best fits what is one important ai-for-management consideration in week 2 scenario 47.

48. What is one important AI-for-management consideration in Week 2 scenario 48?

A) It should rely only on hardware changes without process updates.
B) It should be aligned with business outcomes and decision quality.
C) It should replace managerial judgment in all cases.
D) It should ignore governance, validation, and monitoring.

Answer: B

Explanation: It should be aligned with business outcomes and decision quality. is correct because it best fits what is one important ai-for-management consideration in week 2 scenario 48.

49. What is one important AI-for-management consideration in Week 2 scenario 49?

A) It should rely only on hardware changes without process updates.
B) It should be aligned with business outcomes and decision quality.
C) It should replace managerial judgment in all cases.
D) It should ignore governance, validation, and monitoring.

Answer: B

Explanation: It should be aligned with business outcomes and decision quality. is correct because it best fits what is one important ai-for-management consideration in week 2 scenario 49.

50. What is one important AI-for-management consideration in Week 2 scenario 50?

A) It should be aligned with business outcomes and decision quality.
B) It should replace managerial judgment in all cases.
C) It should rely only on hardware changes without process updates.
D) It should ignore governance, validation, and monitoring.

Answer: A

Explanation: It should be aligned with business outcomes and decision quality. is correct because it best fits what is one important ai-for-management consideration in week 2 scenario 50.

Online Assessment Questions

1. A key limitation of ML systems is:

A) Easily misled by small changes
B) Deep reasoning
C) Generalization with small data
D) No supervision needed

Answer: A

Explanation: Easily misled by small changes is correct because it best fits the concept of a key limitation of ml systems is.

2. When a model fits noise instead of pattern, it is called:

A) Overfitting
B) Underfitting
C) Data leakage
D) Regularization

Answer: A

Explanation: Overfitting is correct because it best fits when a model fits noise instead of pattern, it is called.

3. Characteristics of supervised learning: (Select all)

A) Learns input-output mapping
B) No guidance
C) Uses labeled data
D) Regression/classification

Answer: A, C, D

Explanation: Learns input-output mapping, Uses labeled data, Regression/classification are correct because each option captures a required part of the concept of characteristics of supervised learning: (select all).

4. Perceptron fails when:

A) Error wrong
B) Data not linearly separable
C) Learning rate high
D) No gradients

Answer: B

Explanation: Data not linearly separable is correct because it best fits the concept of perceptron fails when.

5. Backpropagation was introduced because:

A) Bias issue
B) No activation
C) Instability
D) Nonlinear problems

Answer: D

Explanation: Nonlinear problems is correct because it best fits the concept of backpropagation was introduced because.

6. Error function E(w) is:

A) Half squared error
B) Cross entropy
C) Absolute error
D) Product

Answer: A

Explanation: Half squared error is correct because it best fits the concept of error function e(w) is.

7. Increasing hidden layer size leads to:

A) Single output
B) Less power
C) Linear activation
D) More overfitting risk

Answer: D

Explanation: More overfitting risk is correct because it best fits the concept of increasing hidden layer size leads to.

8. Adding hidden layers allows:

A) Complex pattern recognition
B) Linear separation
C) No bias
D) Zero error

Answer: A

Explanation: Complex pattern recognition is correct because it best fits the concept of adding hidden layers allows.

9. Components of time series:

A) Seasonality
B) Trend
C) Randomness
D) Cyclicity

Answer: A, B, C, D

Explanation: Seasonality, Trend, Randomness, Cyclicity are correct because each option captures a required part of the concept of components of time series.

10. Difference between white box and neural networks:

A) Different languages
B) Neural networks not interpretable
C) White box better always
D) White box limited

Answer: B

Explanation: Neural networks not interpretable is correct because it best fits the concept of difference between white box and neural networks.

Week 3

Concepts

1. Porter’s Strategies & Value Chain

Cost Leadership: Sell products cheaper than competitors while maintaining quality.

Differentiation: Offer unique features customers value.

Value Chain: Business activities split into Primary (inbound, operations, outbound, marketing, service) and Support (HR, tech, procurement).

IT was originally considered a support activity.

2. IT Strategic Grid

Categorizes IT impact:

Strategic: High future + current impact

Factory: Low future, high current (efficiency focus)

Support: Low both

Turnaround: High future, low current

3. Business Model Evolution

After dot-com crash -> investors wanted clear monetization.

Focus shifted from "growth" to profitability proof.

4. Platform & Network Effects

Platform: Connects producers & consumers.

Network Effect: More users -> more value.

Governance: Controlled by platform owner.

Frictionless entry: Easy onboarding -> rapid growth.

5. Platform Monetization

Virtual goods, ads, subscriptions, e-commerce.

Freemium: Free basic + paid premium.

Cost advantage: users create content -> lower cost.

6. Ecosystem Model

Multiple stakeholders interact.

Value created via orchestrated interactions, not linear chain.

7. AI Business Models

AI as Service

AI Platforms

AI Data Monetization: Data -> value -> revenue

8. Recommendation Systems

Content-based: Based on item features.

Collaborative: Based on similar users.

Content-based tracks user behavior (watch time, clicks).

9. GPU & AI Growth

GPUs enable fast parallel computation.

Key for deep learning training.

Example: In Week 3, a company can apply porter's strategies & value chain together with it strategic grid to improve decision quality and business outcomes.

MCQs

1. Which strategy focuses on lowest cost?

A) Cost Leadership
B) Focus
C) Turnaround
D) Differentiation

Answer: A

Explanation: Cost Leadership is correct because it best fits which strategy focuses on lowest cost.

2. High operational impact, low strategic?

A) Strategic
B) Support
C) Turnaround
D) Factory

Answer: D

Explanation: Factory is correct because it best fits the concept of high operational impact, low strategic.

3. What is Governance is controlled by?

A) Platform owner
B) Users
C) Government
D) Sellers

Answer: A

Explanation: Platform owner is correct because it best fits what is governance is controlled by.

4. What best describes Porter's Strategies & Value Chain in AI for management?

A) Porter's Strategies & Value Chain eliminates the need for managerial judgment and governance.
B) Porter's Strategies & Value Chain guarantees perfect outcomes in every scenario.
C) Cost Leadership: Sell products cheaper than competitors while maintaining quality.
D) Porter's Strategies & Value Chain removes the need for data quality checks and monitoring.

Answer: C

Explanation: Cost Leadership: Sell products cheaper than competitors while maintaining quality. is correct because it best fits what best describes porter's strategies & value chain in ai for management.

5. Which statement about IT Strategic Grid is most accurate?

A) IT Strategic Grid guarantees perfect outcomes in every scenario.
B) IT Strategic Grid eliminates the need for managerial judgment and governance.
C) Strategic: High future + current impact
D) IT Strategic Grid removes the need for data quality checks and monitoring.

Answer: C

Explanation: Strategic: High future + current impact is correct because it best fits which statement about it strategic grid is most accurate.

6. How does Business Model Evolution support managerial decision-making?

A) Business Model Evolution guarantees perfect outcomes in every scenario.
B) Business Model Evolution eliminates the need for managerial judgment and governance.
C) After dot-com crash -> investors wanted clear monetization.
D) Business Model Evolution removes the need for data quality checks and monitoring.

Answer: C

Explanation: After dot-com crash -> investors wanted clear monetization. is correct because it best fits how does business model evolution support managerial decision-making.

7. What is a key business implication of Platform & Network Effects?

A) Platform: Connects producers & consumers.
B) Platform & Network Effects eliminates the need for managerial judgment and governance.
C) Platform & Network Effects removes the need for data quality checks and monitoring.
D) Platform & Network Effects guarantees perfect outcomes in every scenario.

Answer: A

Explanation: Platform: Connects producers & consumers. is correct because it best fits what is a key business implication of platform & network effects.

8. What best describes Platform Monetization in AI for management?

A) Platform Monetization removes the need for data quality checks and monitoring.
B) Platform Monetization eliminates the need for managerial judgment and governance.
C) Platform Monetization guarantees perfect outcomes in every scenario.
D) Virtual goods, ads, subscriptions, e-commerce.

Answer: D

Explanation: Virtual goods, ads, subscriptions, e-commerce. is correct because it best fits what best describes platform monetization in ai for management.

9. Which statement about Ecosystem Model is most accurate?

A) Ecosystem Model removes the need for data quality checks and monitoring.
B) Ecosystem Model eliminates the need for managerial judgment and governance.
C) Ecosystem Model guarantees perfect outcomes in every scenario.
D) Multiple stakeholders interact.

Answer: D

Explanation: Multiple stakeholders interact. is correct because it best fits which statement about ecosystem model is most accurate.

10. How does AI Business Models support managerial decision-making?

A) AI Business Models removes the need for data quality checks and monitoring.
B) AI Business Models guarantees perfect outcomes in every scenario.
C) AI as Service
D) AI Business Models eliminates the need for managerial judgment and governance.

Answer: C

Explanation: AI as Service is correct because it best fits how does ai business models support managerial decision-making.

11. What is a key business implication of Recommendation Systems?

A) Recommendation Systems guarantees perfect outcomes in every scenario.
B) Content-based: Based on item features.
C) Recommendation Systems removes the need for data quality checks and monitoring.
D) Recommendation Systems eliminates the need for managerial judgment and governance.

Answer: B

Explanation: Content-based: Based on item features. is correct because it best fits what is a key business implication of recommendation systems.

12. What best describes GPU & AI Growth in AI for management?

A) GPU & AI Growth removes the need for data quality checks and monitoring.
B) GPU & AI Growth eliminates the need for managerial judgment and governance.
C) GPU & AI Growth guarantees perfect outcomes in every scenario.
D) GPUs enable fast parallel computation.

Answer: D

Explanation: GPUs enable fast parallel computation. is correct because it best fits what best describes gpu & ai growth in ai for management.

13. Which statement about Porter's Strategies & Value Chain is most accurate?

A) Porter's Strategies & Value Chain eliminates the need for managerial judgment and governance.
B) Cost Leadership: Sell products cheaper than competitors while maintaining quality.
C) Porter's Strategies & Value Chain removes the need for data quality checks and monitoring.
D) Porter's Strategies & Value Chain guarantees perfect outcomes in every scenario.

Answer: B

Explanation: Cost Leadership: Sell products cheaper than competitors while maintaining quality. is correct because it best fits which statement about porter's strategies & value chain is most accurate.

14. How does IT Strategic Grid support managerial decision-making?

A) IT Strategic Grid eliminates the need for managerial judgment and governance.
B) IT Strategic Grid guarantees perfect outcomes in every scenario.
C) Strategic: High future + current impact
D) IT Strategic Grid removes the need for data quality checks and monitoring.

Answer: C

Explanation: Strategic: High future + current impact is correct because it best fits how does it strategic grid support managerial decision-making.

15. What is a key business implication of Business Model Evolution?

A) Business Model Evolution removes the need for data quality checks and monitoring.
B) Business Model Evolution eliminates the need for managerial judgment and governance.
C) After dot-com crash -> investors wanted clear monetization.
D) Business Model Evolution guarantees perfect outcomes in every scenario.

Answer: C

Explanation: After dot-com crash -> investors wanted clear monetization. is correct because it best fits what is a key business implication of business model evolution.

16. What best describes Platform & Network Effects in AI for management?

A) Platform & Network Effects guarantees perfect outcomes in every scenario.
B) Platform & Network Effects removes the need for data quality checks and monitoring.
C) Platform: Connects producers & consumers.
D) Platform & Network Effects eliminates the need for managerial judgment and governance.

Answer: C

Explanation: Platform: Connects producers & consumers. is correct because it best fits what best describes platform & network effects in ai for management.

17. Which statement about Platform Monetization is most accurate?

A) Platform Monetization removes the need for data quality checks and monitoring.
B) Virtual goods, ads, subscriptions, e-commerce.
C) Platform Monetization eliminates the need for managerial judgment and governance.
D) Platform Monetization guarantees perfect outcomes in every scenario.

Answer: B

Explanation: Virtual goods, ads, subscriptions, e-commerce. is correct because it best fits which statement about platform monetization is most accurate.

18. How does Ecosystem Model support managerial decision-making?

A) Multiple stakeholders interact.
B) Ecosystem Model removes the need for data quality checks and monitoring.
C) Ecosystem Model guarantees perfect outcomes in every scenario.
D) Ecosystem Model eliminates the need for managerial judgment and governance.

Answer: A

Explanation: Multiple stakeholders interact. is correct because it best fits how does ecosystem model support managerial decision-making.

19. What is a key business implication of AI Business Models?

A) AI Business Models eliminates the need for managerial judgment and governance.
B) AI Business Models guarantees perfect outcomes in every scenario.
C) AI Business Models removes the need for data quality checks and monitoring.
D) AI as Service

Answer: D

Explanation: AI as Service is correct because it best fits what is a key business implication of ai business models.

20. What best describes Recommendation Systems in AI for management?

A) Recommendation Systems eliminates the need for managerial judgment and governance.
B) Content-based: Based on item features.
C) Recommendation Systems removes the need for data quality checks and monitoring.
D) Recommendation Systems guarantees perfect outcomes in every scenario.

Answer: B

Explanation: Content-based: Based on item features. is correct because it best fits what best describes recommendation systems in ai for management.

21. Which statement about GPU & AI Growth is most accurate?

A) GPU & AI Growth guarantees perfect outcomes in every scenario.
B) GPUs enable fast parallel computation.
C) GPU & AI Growth eliminates the need for managerial judgment and governance.
D) GPU & AI Growth removes the need for data quality checks and monitoring.

Answer: B

Explanation: GPUs enable fast parallel computation. is correct because it best fits which statement about gpu & ai growth is most accurate.

22. How does Porter's Strategies & Value Chain support managerial decision-making?

A) Porter's Strategies & Value Chain guarantees perfect outcomes in every scenario.
B) Cost Leadership: Sell products cheaper than competitors while maintaining quality.
C) Porter's Strategies & Value Chain eliminates the need for managerial judgment and governance.
D) Porter's Strategies & Value Chain removes the need for data quality checks and monitoring.

Answer: B

Explanation: Cost Leadership: Sell products cheaper than competitors while maintaining quality. is correct because it best fits how does porter's strategies & value chain support managerial decision-making.

23. What is a key business implication of IT Strategic Grid?

A) IT Strategic Grid removes the need for data quality checks and monitoring.
B) IT Strategic Grid guarantees perfect outcomes in every scenario.
C) Strategic: High future + current impact
D) IT Strategic Grid eliminates the need for managerial judgment and governance.

Answer: C

Explanation: Strategic: High future + current impact is correct because it best fits what is a key business implication of it strategic grid.

24. What best describes Business Model Evolution in AI for management?

A) Business Model Evolution eliminates the need for managerial judgment and governance.
B) Business Model Evolution removes the need for data quality checks and monitoring.
C) Business Model Evolution guarantees perfect outcomes in every scenario.
D) After dot-com crash -> investors wanted clear monetization.

Answer: D

Explanation: After dot-com crash -> investors wanted clear monetization. is correct because it best fits what best describes business model evolution in ai for management.

25. Which statement about Platform & Network Effects is most accurate?

A) Platform & Network Effects removes the need for data quality checks and monitoring.
B) Platform & Network Effects eliminates the need for managerial judgment and governance.
C) Platform: Connects producers & consumers.
D) Platform & Network Effects guarantees perfect outcomes in every scenario.

Answer: C

Explanation: Platform: Connects producers & consumers. is correct because it best fits which statement about platform & network effects is most accurate.

26. How does Platform Monetization support managerial decision-making?

A) Platform Monetization guarantees perfect outcomes in every scenario.
B) Platform Monetization removes the need for data quality checks and monitoring.
C) Platform Monetization eliminates the need for managerial judgment and governance.
D) Virtual goods, ads, subscriptions, e-commerce.

Answer: D

Explanation: Virtual goods, ads, subscriptions, e-commerce. is correct because it best fits how does platform monetization support managerial decision-making.

27. What is a key business implication of Ecosystem Model?

A) Ecosystem Model guarantees perfect outcomes in every scenario.
B) Ecosystem Model removes the need for data quality checks and monitoring.
C) Multiple stakeholders interact.
D) Ecosystem Model eliminates the need for managerial judgment and governance.

Answer: C

Explanation: Multiple stakeholders interact. is correct because it best fits what is a key business implication of ecosystem model.

28. What best describes AI Business Models in AI for management?

A) AI Business Models eliminates the need for managerial judgment and governance.
B) AI Business Models guarantees perfect outcomes in every scenario.
C) AI Business Models removes the need for data quality checks and monitoring.
D) AI as Service

Answer: D

Explanation: AI as Service is correct because it best fits what best describes ai business models in ai for management.

29. Which statement about Recommendation Systems is most accurate?

A) Recommendation Systems guarantees perfect outcomes in every scenario.
B) Recommendation Systems eliminates the need for managerial judgment and governance.
C) Content-based: Based on item features.
D) Recommendation Systems removes the need for data quality checks and monitoring.

Answer: C

Explanation: Content-based: Based on item features. is correct because it best fits which statement about recommendation systems is most accurate.

30. How does GPU & AI Growth support managerial decision-making?

A) GPU & AI Growth eliminates the need for managerial judgment and governance.
B) GPU & AI Growth guarantees perfect outcomes in every scenario.
C) GPUs enable fast parallel computation.
D) GPU & AI Growth removes the need for data quality checks and monitoring.

Answer: C

Explanation: GPUs enable fast parallel computation. is correct because it best fits how does gpu & ai growth support managerial decision-making.

31. What is a key business implication of Porter's Strategies & Value Chain?

A) Porter's Strategies & Value Chain removes the need for data quality checks and monitoring.
B) Porter's Strategies & Value Chain eliminates the need for managerial judgment and governance.
C) Cost Leadership: Sell products cheaper than competitors while maintaining quality.
D) Porter's Strategies & Value Chain guarantees perfect outcomes in every scenario.

Answer: C

Explanation: Cost Leadership: Sell products cheaper than competitors while maintaining quality. is correct because it best fits what is a key business implication of porter's strategies & value chain.

32. What best describes IT Strategic Grid in AI for management?

A) IT Strategic Grid guarantees perfect outcomes in every scenario.
B) IT Strategic Grid removes the need for data quality checks and monitoring.
C) Strategic: High future + current impact
D) IT Strategic Grid eliminates the need for managerial judgment and governance.

Answer: C

Explanation: Strategic: High future + current impact is correct because it best fits what best describes it strategic grid in ai for management.

33. Which statement about Business Model Evolution is most accurate?

A) Business Model Evolution eliminates the need for managerial judgment and governance.
B) Business Model Evolution removes the need for data quality checks and monitoring.
C) After dot-com crash -> investors wanted clear monetization.
D) Business Model Evolution guarantees perfect outcomes in every scenario.

Answer: C

Explanation: After dot-com crash -> investors wanted clear monetization. is correct because it best fits which statement about business model evolution is most accurate.

34. How does Platform & Network Effects support managerial decision-making?

A) Platform & Network Effects guarantees perfect outcomes in every scenario.
B) Platform: Connects producers & consumers.
C) Platform & Network Effects removes the need for data quality checks and monitoring.
D) Platform & Network Effects eliminates the need for managerial judgment and governance.

Answer: B

Explanation: Platform: Connects producers & consumers. is correct because it best fits how does platform & network effects support managerial decision-making.

35. What is a key business implication of Platform Monetization?

A) Platform Monetization eliminates the need for managerial judgment and governance.
B) Platform Monetization removes the need for data quality checks and monitoring.
C) Virtual goods, ads, subscriptions, e-commerce.
D) Platform Monetization guarantees perfect outcomes in every scenario.

Answer: C

Explanation: Virtual goods, ads, subscriptions, e-commerce. is correct because it best fits what is a key business implication of platform monetization.

36. What best describes Ecosystem Model in AI for management?

A) Ecosystem Model eliminates the need for managerial judgment and governance.
B) Ecosystem Model removes the need for data quality checks and monitoring.
C) Multiple stakeholders interact.
D) Ecosystem Model guarantees perfect outcomes in every scenario.

Answer: C

Explanation: Multiple stakeholders interact. is correct because it best fits what best describes ecosystem model in ai for management.

37. Which statement about AI Business Models is most accurate?

A) AI Business Models guarantees perfect outcomes in every scenario.
B) AI as Service
C) AI Business Models eliminates the need for managerial judgment and governance.
D) AI Business Models removes the need for data quality checks and monitoring.

Answer: B

Explanation: AI as Service is correct because it best fits which statement about ai business models is most accurate.

38. How does Recommendation Systems support managerial decision-making?

A) Recommendation Systems guarantees perfect outcomes in every scenario.
B) Content-based: Based on item features.
C) Recommendation Systems eliminates the need for managerial judgment and governance.
D) Recommendation Systems removes the need for data quality checks and monitoring.

Answer: B

Explanation: Content-based: Based on item features. is correct because it best fits how does recommendation systems support managerial decision-making.

39. What is a key business implication of GPU & AI Growth?

A) GPU & AI Growth removes the need for data quality checks and monitoring.
B) GPU & AI Growth eliminates the need for managerial judgment and governance.
C) GPUs enable fast parallel computation.
D) GPU & AI Growth guarantees perfect outcomes in every scenario.

Answer: C

Explanation: GPUs enable fast parallel computation. is correct because it best fits what is a key business implication of gpu & ai growth.

40. What is one important AI-for-management consideration in Week 3 scenario 40?

A) It should be aligned with business outcomes and decision quality.
B) It should rely only on hardware changes without process updates.
C) It should replace managerial judgment in all cases.
D) It should ignore governance, validation, and monitoring.

Answer: A

Explanation: It should be aligned with business outcomes and decision quality. is correct because it best fits what is one important ai-for-management consideration in week 3 scenario 40.

41. What is one important AI-for-management consideration in Week 3 scenario 41?

A) It should ignore governance, validation, and monitoring.
B) It should be aligned with business outcomes and decision quality.
C) It should replace managerial judgment in all cases.
D) It should rely only on hardware changes without process updates.

Answer: B

Explanation: It should be aligned with business outcomes and decision quality. is correct because it best fits what is one important ai-for-management consideration in week 3 scenario 41.

42. What is one important AI-for-management consideration in Week 3 scenario 42?

A) It should rely only on hardware changes without process updates.
B) It should ignore governance, validation, and monitoring.
C) It should replace managerial judgment in all cases.
D) It should be aligned with business outcomes and decision quality.

Answer: D

Explanation: It should be aligned with business outcomes and decision quality. is correct because it best fits what is one important ai-for-management consideration in week 3 scenario 42.

43. What is one important AI-for-management consideration in Week 3 scenario 43?

A) It should replace managerial judgment in all cases.
B) It should be aligned with business outcomes and decision quality.
C) It should ignore governance, validation, and monitoring.
D) It should rely only on hardware changes without process updates.

Answer: B

Explanation: It should be aligned with business outcomes and decision quality. is correct because it best fits what is one important ai-for-management consideration in week 3 scenario 43.

44. What is one important AI-for-management consideration in Week 3 scenario 44?

A) It should rely only on hardware changes without process updates.
B) It should be aligned with business outcomes and decision quality.
C) It should ignore governance, validation, and monitoring.
D) It should replace managerial judgment in all cases.

Answer: B

Explanation: It should be aligned with business outcomes and decision quality. is correct because it best fits what is one important ai-for-management consideration in week 3 scenario 44.

45. What is one important AI-for-management consideration in Week 3 scenario 45?

A) It should replace managerial judgment in all cases.
B) It should rely only on hardware changes without process updates.
C) It should ignore governance, validation, and monitoring.
D) It should be aligned with business outcomes and decision quality.

Answer: D

Explanation: It should be aligned with business outcomes and decision quality. is correct because it best fits what is one important ai-for-management consideration in week 3 scenario 45.

46. What is one important AI-for-management consideration in Week 3 scenario 46?

A) It should ignore governance, validation, and monitoring.
B) It should replace managerial judgment in all cases.
C) It should rely only on hardware changes without process updates.
D) It should be aligned with business outcomes and decision quality.

Answer: D

Explanation: It should be aligned with business outcomes and decision quality. is correct because it best fits what is one important ai-for-management consideration in week 3 scenario 46.

47. What is one important AI-for-management consideration in Week 3 scenario 47?

A) It should be aligned with business outcomes and decision quality.
B) It should rely only on hardware changes without process updates.
C) It should ignore governance, validation, and monitoring.
D) It should replace managerial judgment in all cases.

Answer: A

Explanation: It should be aligned with business outcomes and decision quality. is correct because it best fits what is one important ai-for-management consideration in week 3 scenario 47.

48. What is one important AI-for-management consideration in Week 3 scenario 48?

A) It should rely only on hardware changes without process updates.
B) It should replace managerial judgment in all cases.
C) It should be aligned with business outcomes and decision quality.
D) It should ignore governance, validation, and monitoring.

Answer: C

Explanation: It should be aligned with business outcomes and decision quality. is correct because it best fits what is one important ai-for-management consideration in week 3 scenario 48.

49. What is one important AI-for-management consideration in Week 3 scenario 49?

A) It should be aligned with business outcomes and decision quality.
B) It should rely only on hardware changes without process updates.
C) It should replace managerial judgment in all cases.
D) It should ignore governance, validation, and monitoring.

Answer: A

Explanation: It should be aligned with business outcomes and decision quality. is correct because it best fits what is one important ai-for-management consideration in week 3 scenario 49.

50. What is one important AI-for-management consideration in Week 3 scenario 50?

A) It should replace managerial judgment in all cases.
B) It should be aligned with business outcomes and decision quality.
C) It should rely only on hardware changes without process updates.
D) It should ignore governance, validation, and monitoring.

Answer: B

Explanation: It should be aligned with business outcomes and decision quality. is correct because it best fits what is one important ai-for-management consideration in week 3 scenario 50.

Online Assessment Questions

1. Which strategy focuses on achieving competitive advantage by reducing cost?

A) Differentiation
B) Turnaround Strategy
C) Focus Strategy
D) Cost Leadership

Answer: D

Explanation: Cost Leadership is correct because it best fits which strategy focuses on achieving competitive advantage by reducing cost.

2. In the IT Strategic Grid, low strategic impact but high operational impact falls under:

A) Support
B) Factory
C) Turnaround
D) Strategic

Answer: B

Explanation: Factory is correct because it best fits the concept of in the it strategic grid, low strategic impact but high operational impact falls under.

3. In Porter’s Value Chain, Information Technology is classified as:

A) Primary activity
B) Support activity
C) Strategic activity
D) Outbound activity

Answer: B

Explanation: Support activity is correct because it best fits the concept of in porter's value chain, information technology is classified as.

4. The evolution of business models after the dot-com bubble emphasized:

A) Government regulation
B) Free services
C) Monetization clarity
D) User growth only

Answer: C

Explanation: Monetization clarity is correct because it best fits the concept of the evolution of business models after the dot-com bubble emphasized.

5. Which activities are primary in Porter's Value Chain? (Multiple correct)

A) Inbound Logistics
B) Technology Development
C) Human Resource Management
D) Service

Answer: A, D

Explanation: Inbound Logistics, Service are correct because each option captures a required part of which activities are primary in porter's value chain? (multiple correct).

6. The value of a platform increasing with more users is called:

A) Digital orchestration
B) Viral monetization
C) Market scaling
D) Network effect

Answer: D

Explanation: Network effect is correct because it best fits the concept of the value of a platform increasing with more users is called.

7. Who sets governance rules in a digital platform?

A) Users
B) Producers
C) Platform owner
D) Government

Answer: C

Explanation: Platform owner is correct because it best fits who sets governance rules in a digital platform.

8. Which AI business model relies on large volumes of curated data?

A) AI Platforms
B) AI Product as a Service
C) AI Data Monetization
D) Encryption Ecosystem

Answer: C

Explanation: AI Data Monetization is correct because it best fits which ai business model relies on large volumes of curated data.

9. LinkedIn adding social features mainly aims to:

A) Remove competitors
B) Change business model
C) Reduce costs
D) Increase engagement

Answer: D

Explanation: Increase engagement is correct because it best fits the concept of linkedin adding social features mainly aims to.

10. Frictionless entry in platforms leads to:

A) High costs
B) Rapid participation growth
C) Reduced users
D) Slower growth

Answer: B

Explanation: Rapid participation growth is correct because it best fits the concept of frictionless entry in platforms leads to.

Week 4

Concepts

1. ImageNet

Large labeled dataset -> boosted computer vision progress.

2. Unicorn Startup

Private startup valued > $1B.

3. Porter Competitive Advantage

Two types:
Cost Advantage
Differentiation

4. Algorithmic Decision Making

Simon model:
Intelligence -> Design -> Choice -> Implementation
Decisions:
Structured (routine)
Semi-structured
Unstructured (complex)

5. AI vs Human

AI: fast, scalable
Humans: intuition, handle ambiguity

6. Overfitting & Bias

Overfitting: model learns noise
Correlation ? causation

7. Cognitive Bias

Confirmation bias
Anchoring
Position bias

8. AI Risks

Hallucination
Jailbreaking
Data poisoning
Alignment problem

9. Bias in AI

Historical
Representation
Measurement
Aggregation
Deployment

10. Explainable AI (XAI)

Interpretability: understand directly

Explainability: approximate explanation

Tools: SHAP, LIME

Black box = not transparent

> *Example: In Week 4, a company can apply imagenet together with unicorn startup to improve decision quality and business outcomes.*

MCQs

1. IT in value chain is?

A) Outbound
B) Primary
C) Strategic
D) Support

Answer: D

Explanation: Support is correct because it best fits the concept of it in value chain is.

2. Platform value increases due to?

A) CPU
B) Network effect
C) Cost
D) Ads

Answer: B

Explanation: Network effect is correct because it best fits the concept of platform value increases due to.

3. What best describes ImageNet in AI for management?

A) ImageNet eliminates the need for managerial judgment and governance.
B) Large labeled dataset -> boosted computer vision progress.
C) ImageNet removes the need for data quality checks and monitoring.
D) ImageNet guarantees perfect outcomes in every scenario.

Answer: B

Explanation: Large labeled dataset -> boosted computer vision progress. is correct because it best fits what best describes imagenet in ai for management.

4. Which statement about Unicorn Startup is most accurate?

A) Private startup valued > $1B.
B) Unicorn Startup removes the need for data quality checks and monitoring.
C) Unicorn Startup guarantees perfect outcomes in every scenario.
D) Unicorn Startup eliminates the need for managerial judgment and governance.

Answer: A

Explanation: Private startup valued > $1B. is correct because it best fits which statement about unicorn startup is most accurate.

5. How does Porter Competitive Advantage support managerial decision-making?

A) Porter Competitive Advantage removes the need for data quality checks and monitoring.
B) Porter Competitive Advantage eliminates the need for managerial judgment and governance.
C) Cost Advantage
D) Porter Competitive Advantage guarantees perfect outcomes in every scenario.

Answer: C

Explanation: Cost Advantage is correct because it best fits how does porter competitive advantage support managerial decision-making.

6. What is a key business implication of Algorithmic Decision Making?

A) Algorithmic Decision Making eliminates the need for managerial judgment and governance.
B) Intelligence -> Design -> Choice -> Implementation
C) Algorithmic Decision Making removes the need for data quality checks and monitoring.
D) Algorithmic Decision Making guarantees perfect outcomes in every scenario.

Answer: B

Explanation: Intelligence -> Design -> Choice -> Implementation is correct because it best fits what is a key business implication of algorithmic decision making.

7. What best describes AI vs Human in AI for management?

A) AI vs Human guarantees perfect outcomes in every scenario.
B) AI: fast, scalable
C) AI vs Human removes the need for data quality checks and monitoring.
D) AI vs Human eliminates the need for managerial judgment and governance.

Answer: B

Explanation: AI: fast, scalable is correct because it best fits what best describes ai vs human in ai for management.

8. Which statement about Overfitting & Bias is most accurate?

A) Overfitting: model learns noise
B) Overfitting & Bias guarantees perfect outcomes in every scenario.
C) Overfitting & Bias eliminates the need for managerial judgment and governance.
D) Overfitting & Bias removes the need for data quality checks and monitoring.

Answer: A

Explanation: Overfitting: model learns noise is correct because it best fits which statement about overfitting & bias is most accurate.

9. How does Cognitive Bias support managerial decision-making?

A) Cognitive Bias removes the need for data quality checks and monitoring.
B) Cognitive Bias guarantees perfect outcomes in every scenario.
C) Confirmation bias
D) Cognitive Bias eliminates the need for managerial judgment and governance.

Answer: C

Explanation: Confirmation bias is correct because it best fits how does cognitive bias support managerial decision-making.

10. What is a key business implication of AI Risks?

A) AI Risks guarantees perfect outcomes in every scenario.
B) AI Risks removes the need for data quality checks and monitoring.
C) Data poisoning
D) AI Risks eliminates the need for managerial judgment and governance.

Answer: C

Explanation: Data poisoning is correct because it best fits what is a key business implication of ai risks.

11. What best describes Bias in AI in AI for management?

A) Bias in AI eliminates the need for managerial judgment and governance.
B) Bias in AI removes the need for data quality checks and monitoring.
C) Bias in AI guarantees perfect outcomes in every scenario.
D) Bias in AI

Answer: D

Explanation: Bias in AI is correct because it best fits what best describes bias in ai in ai for management.

12. Which statement about Explainable AI (XAI) is most accurate?

A) Explainable AI (XAI) eliminates the need for managerial judgment and governance.
B) Explainable AI (XAI) guarantees perfect outcomes in every scenario.
C) Explainable AI (XAI) removes the need for data quality checks and monitoring.
D) Interpretability: understand directly

Answer: D

Explanation: Interpretability: understand directly is correct because it best fits which statement about explainable ai (xai) is most accurate.

13. How does ImageNet support managerial decision-making?

A) ImageNet eliminates the need for managerial judgment and governance.
B) Large labeled dataset -> boosted computer vision progress.
C) ImageNet removes the need for data quality checks and monitoring.
D) ImageNet guarantees perfect outcomes in every scenario.

Answer: B

Explanation: Large labeled dataset -> boosted computer vision progress. is correct because it best fits how does imagenet support managerial decision-making.

14. What is a key business implication of Unicorn Startup?

A) Private startup valued > $1B.
B) Unicorn Startup eliminates the need for managerial judgment and governance.
C) Unicorn Startup guarantees perfect outcomes in every scenario.
D) Unicorn Startup removes the need for data quality checks and monitoring.

Answer: A

Explanation: Private startup valued > $1B. is correct because it best fits what is a key business implication of unicorn startup.

15. What best describes Porter Competitive Advantage in AI for management?

A) Porter Competitive Advantage eliminates the need for managerial judgment and governance.
B) Porter Competitive Advantage removes the need for data quality checks and monitoring.
C) Cost Advantage
D) Porter Competitive Advantage guarantees perfect outcomes in every scenario.

Answer: C

Explanation: Cost Advantage is correct because it best fits what best describes porter competitive advantage in ai for management.

16. Which statement about Algorithmic Decision Making is most accurate?

A) Algorithmic Decision Making eliminates the need for managerial judgment and governance.
B) Intelligence -> Design -> Choice -> Implementation
C) Algorithmic Decision Making guarantees perfect outcomes in every scenario.
D) Algorithmic Decision Making removes the need for data quality checks and monitoring.

Answer: B

Explanation: Intelligence -> Design -> Choice -> Implementation is correct because it best fits which statement about algorithmic decision making is most accurate.

17. How does AI vs Human support managerial decision-making?

A) AI vs Human guarantees perfect outcomes in every scenario.
B) AI: fast, scalable
C) AI vs Human eliminates the need for managerial judgment and governance.
D) AI vs Human removes the need for data quality checks and monitoring.

Answer: B

Explanation: AI: fast, scalable is correct because it best fits how does ai vs human support managerial decision-making.

18. What is a key business implication of Overfitting & Bias?

A) Overfitting & Bias removes the need for data quality checks and monitoring.
B) Overfitting: model learns noise
C) Overfitting & Bias eliminates the need for managerial judgment and governance.
D) Overfitting & Bias guarantees perfect outcomes in every scenario.

Answer: B

Explanation: Overfitting: model learns noise is correct because it best fits what is a key business implication of overfitting & bias.

19. What best describes Cognitive Bias in AI for management?

A) Cognitive Bias guarantees perfect outcomes in every scenario.
B) Cognitive Bias removes the need for data quality checks and monitoring.
C) Cognitive Bias eliminates the need for managerial judgment and governance.
D) Confirmation bias

Answer: D

Explanation: Confirmation bias is correct because it best fits what best describes cognitive bias in ai for management.

20. Which statement about AI Risks is most accurate?

A) AI Risks guarantees perfect outcomes in every scenario.
B) AI Risks eliminates the need for managerial judgment and governance.
C) Data poisoning
D) AI Risks removes the need for data quality checks and monitoring.

Answer: C

Explanation: Data poisoning is correct because it best fits which statement about ai risks is most accurate.

21. How does Bias in AI support managerial decision-making?

A) Bias in AI
B) Bias in AI eliminates the need for managerial judgment and governance.
C) Bias in AI guarantees perfect outcomes in every scenario.
D) Bias in AI removes the need for data quality checks and monitoring.

Answer: A

Explanation: Bias in AI is correct because it best fits how does bias in ai support managerial decision-making.

22. What is a key business implication of Explainable AI (XAI)?

A) Explainable AI (XAI) guarantees perfect outcomes in every scenario.
B) Explainable AI (XAI) eliminates the need for managerial judgment and governance.
C) Interpretability: understand directly
D) Explainable AI (XAI) removes the need for data quality checks and monitoring.

Answer: C

Explanation: Interpretability: understand directly is correct because it best fits what is a key business implication of explainable ai (xai).

23. What is one important AI-for-management consideration in Week 4 scenario 23?

A) It should replace managerial judgment in all cases.
B) It should ignore governance, validation, and monitoring.
C) It should rely only on hardware changes without process updates.
D) It should be aligned with business outcomes and decision quality.

Answer: D

Explanation: It should be aligned with business outcomes and decision quality. is correct because it best fits what is one important ai-for-management consideration in week 4 scenario 23.

24. What is one important AI-for-management consideration in Week 4 scenario 24?

A) It should replace managerial judgment in all cases.
B) It should rely only on hardware changes without process updates.
C) It should ignore governance, validation, and monitoring.
D) It should be aligned with business outcomes and decision quality.

Answer: D

Explanation: It should be aligned with business outcomes and decision quality. is correct because it best fits what is one important ai-for-management consideration in week 4 scenario 24.

25. What is one important AI-for-management consideration in Week 4 scenario 25?

A) It should ignore governance, validation, and monitoring.
B) It should be aligned with business outcomes and decision quality.
C) It should rely only on hardware changes without process updates.
D) It should replace managerial judgment in all cases.

Answer: B

Explanation: It should be aligned with business outcomes and decision quality. is correct because it best fits what is one important ai-for-management consideration in week 4 scenario 25.

26. What is one important AI-for-management consideration in Week 4 scenario 26?

A) It should replace managerial judgment in all cases.
B) It should be aligned with business outcomes and decision quality.
C) It should rely only on hardware changes without process updates.
D) It should ignore governance, validation, and monitoring.

Answer: B

Explanation: It should be aligned with business outcomes and decision quality. is correct because it best fits what is one important ai-for-management consideration in week 4 scenario 26.

27. What is one important AI-for-management consideration in Week 4 scenario 27?

A) It should be aligned with business outcomes and decision quality.
B) It should replace managerial judgment in all cases.
C) It should rely only on hardware changes without process updates.
D) It should ignore governance, validation, and monitoring.

Answer: A

Explanation: It should be aligned with business outcomes and decision quality. is correct because it best fits what is one important ai-for-management consideration in week 4 scenario 27.

28. What is one important AI-for-management consideration in Week 4 scenario 28?

A) It should replace managerial judgment in all cases.
B) It should ignore governance, validation, and monitoring.
C) It should be aligned with business outcomes and decision quality.
D) It should rely only on hardware changes without process updates.

Answer: C

Explanation: It should be aligned with business outcomes and decision quality. is correct because it best fits what is one important ai-for-management consideration in week 4 scenario 28.

29. What is one important AI-for-management consideration in Week 4 scenario 29?

A) It should replace managerial judgment in all cases.
B) It should ignore governance, validation, and monitoring.
C) It should be aligned with business outcomes and decision quality.
D) It should rely only on hardware changes without process updates.

Answer: C

Explanation: It should be aligned with business outcomes and decision quality. is correct because it best fits what is one important ai-for-management consideration in week 4 scenario 29.

30. What is one important AI-for-management consideration in Week 4 scenario 30?

A) It should replace managerial judgment in all cases.
B) It should rely only on hardware changes without process updates.
C) It should be aligned with business outcomes and decision quality.
D) It should ignore governance, validation, and monitoring.

Answer: C

Explanation: It should be aligned with business outcomes and decision quality. is correct because it best fits what is one important ai-for-management consideration in week 4 scenario 30.

31. What is one important AI-for-management consideration in Week 4 scenario 31?

A) It should ignore governance, validation, and monitoring.
B) It should rely only on hardware changes without process updates.
C) It should be aligned with business outcomes and decision quality.
D) It should replace managerial judgment in all cases.

Answer: C

Explanation: It should be aligned with business outcomes and decision quality. is correct because it best fits what is one important ai-for-management consideration in week 4 scenario 31.

32. What is one important AI-for-management consideration in Week 4 scenario 32?

A) It should replace managerial judgment in all cases.
B) It should be aligned with business outcomes and decision quality.
C) It should ignore governance, validation, and monitoring.
D) It should rely only on hardware changes without process updates.

Answer: B

Explanation: It should be aligned with business outcomes and decision quality. is correct because it best fits what is one important ai-for-management consideration in week 4 scenario 32.

33. What is one important AI-for-management consideration in Week 4 scenario 33?

A) It should ignore governance, validation, and monitoring.
B) It should replace managerial judgment in all cases.
C) It should be aligned with business outcomes and decision quality.
D) It should rely only on hardware changes without process updates.

Answer: C

Explanation: It should be aligned with business outcomes and decision quality. is correct because it best fits what is one important ai-for-management consideration in week 4 scenario 33.

34. What is one important AI-for-management consideration in Week 4 scenario 34?

A) It should be aligned with business outcomes and decision quality.
B) It should ignore governance, validation, and monitoring.
C) It should replace managerial judgment in all cases.
D) It should rely only on hardware changes without process updates.

Answer: A

Explanation: It should be aligned with business outcomes and decision quality. is correct because it best fits what is one important ai-for-management consideration in week 4 scenario 34.

35. What is one important AI-for-management consideration in Week 4 scenario 35?

A) It should replace managerial judgment in all cases.
B) It should rely only on hardware changes without process updates.
C) It should be aligned with business outcomes and decision quality.
D) It should ignore governance, validation, and monitoring.

Answer: C

Explanation: It should be aligned with business outcomes and decision quality. is correct because it best fits what is one important ai-for-management consideration in week 4 scenario 35.

36. What is one important AI-for-management consideration in Week 4 scenario 36?

A) It should replace managerial judgment in all cases.
B) It should ignore governance, validation, and monitoring.
C) It should be aligned with business outcomes and decision quality.
D) It should rely only on hardware changes without process updates.

Answer: C

Explanation: It should be aligned with business outcomes and decision quality. is correct because it best fits what is one important ai-for-management consideration in week 4 scenario 36.

37. What is one important AI-for-management consideration in Week 4 scenario 37?

A) It should replace managerial judgment in all cases.
B) It should rely only on hardware changes without process updates.
C) It should ignore governance, validation, and monitoring.
D) It should be aligned with business outcomes and decision quality.

Answer: D

Explanation: It should be aligned with business outcomes and decision quality. is correct because it best fits what is one important ai-for-management consideration in week 4 scenario 37.

38. What is one important AI-for-management consideration in Week 4 scenario 38?

A) It should ignore governance, validation, and monitoring.
B) It should rely only on hardware changes without process updates.
C) It should replace managerial judgment in all cases.
D) It should be aligned with business outcomes and decision quality.

Answer: D

Explanation: It should be aligned with business outcomes and decision quality. is correct because it best fits what is one important ai-for-management consideration in week 4 scenario 38.

39. What is one important AI-for-management consideration in Week 4 scenario 39?

A) It should ignore governance, validation, and monitoring.
B) It should be aligned with business outcomes and decision quality.
C) It should replace managerial judgment in all cases.
D) It should rely only on hardware changes without process updates.

Answer: B

Explanation: It should be aligned with business outcomes and decision quality. is correct because it best fits what is one important ai-for-management consideration in week 4 scenario 39.

40. What is one important AI-for-management consideration in Week 4 scenario 40?

A) It should ignore governance, validation, and monitoring.
B) It should replace managerial judgment in all cases.
C) It should rely only on hardware changes without process updates.
D) It should be aligned with business outcomes and decision quality.

Answer: D

Explanation: It should be aligned with business outcomes and decision quality. is correct because it best fits what is one important ai-for-management consideration in week 4 scenario 40.

41. What is one important AI-for-management consideration in Week 4 scenario 41?

A) It should ignore governance, validation, and monitoring.
B) It should be aligned with business outcomes and decision quality.
C) It should rely only on hardware changes without process updates.
D) It should replace managerial judgment in all cases.

Answer: B

Explanation: It should be aligned with business outcomes and decision quality. is correct because it best fits what is one important ai-for-management consideration in week 4 scenario 41.

42. What is one important AI-for-management consideration in Week 4 scenario 42?

A) It should be aligned with business outcomes and decision quality.
B) It should replace managerial judgment in all cases.
C) It should ignore governance, validation, and monitoring.
D) It should rely only on hardware changes without process updates.

Answer: A

Explanation: It should be aligned with business outcomes and decision quality. is correct because it best fits what is one important ai-for-management consideration in week 4 scenario 42.

43. What is one important AI-for-management consideration in Week 4 scenario 43?

A) It should be aligned with business outcomes and decision quality.
B) It should replace managerial judgment in all cases.
C) It should ignore governance, validation, and monitoring.
D) It should rely only on hardware changes without process updates.

Answer: A

Explanation: It should be aligned with business outcomes and decision quality. is correct because it best fits what is one important ai-for-management consideration in week 4 scenario 43.

44. What is one important AI-for-management consideration in Week 4 scenario 44?

A) It should replace managerial judgment in all cases.
B) It should be aligned with business outcomes and decision quality.
C) It should rely only on hardware changes without process updates.
D) It should ignore governance, validation, and monitoring.

Answer: B

Explanation: It should be aligned with business outcomes and decision quality. is correct because it best fits what is one important ai-for-management consideration in week 4 scenario 44.

45. What is one important AI-for-management consideration in Week 4 scenario 45?

A) It should ignore governance, validation, and monitoring.
B) It should be aligned with business outcomes and decision quality.
C) It should replace managerial judgment in all cases.
D) It should rely only on hardware changes without process updates.

Answer: B

Explanation: It should be aligned with business outcomes and decision quality. is correct because it best fits what is one important ai-for-management consideration in week 4 scenario 45.

46. What is one important AI-for-management consideration in Week 4 scenario 46?

A) It should be aligned with business outcomes and decision quality.
B) It should rely only on hardware changes without process updates.
C) It should ignore governance, validation, and monitoring.
D) It should replace managerial judgment in all cases.

Answer: A

Explanation: It should be aligned with business outcomes and decision quality. is correct because it best fits what is one important ai-for-management consideration in week 4 scenario 46.

47. What is one important AI-for-management consideration in Week 4 scenario 47?

A) It should ignore governance, validation, and monitoring.
B) It should replace managerial judgment in all cases.
C) It should be aligned with business outcomes and decision quality.
D) It should rely only on hardware changes without process updates.

Answer: C

Explanation: It should be aligned with business outcomes and decision quality. is correct because it best fits what is one important ai-for-management consideration in week 4 scenario 47.

48. What is one important AI-for-management consideration in Week 4 scenario 48?

A) It should rely only on hardware changes without process updates.
B) It should ignore governance, validation, and monitoring.
C) It should be aligned with business outcomes and decision quality.
D) It should replace managerial judgment in all cases.

Answer: C

Explanation: It should be aligned with business outcomes and decision quality. is correct because it best fits what is one important ai-for-management consideration in week 4 scenario 48.

49. What is one important AI-for-management consideration in Week 4 scenario 49?

A) It should rely only on hardware changes without process updates.
B) It should ignore governance, validation, and monitoring.
C) It should replace managerial judgment in all cases.
D) It should be aligned with business outcomes and decision quality.

Answer: D

Explanation: It should be aligned with business outcomes and decision quality. is correct because it best fits what is one important ai-for-management consideration in week 4 scenario 49.

50. What is one important AI-for-management consideration in Week 4 scenario 50?

A) It should rely only on hardware changes without process updates.
B) It should be aligned with business outcomes and decision quality.
C) It should ignore governance, validation, and monitoring.
D) It should replace managerial judgment in all cases.

Answer: B

Explanation: It should be aligned with business outcomes and decision quality. is correct because it best fits what is one important ai-for-management consideration in week 4 scenario 50.

Online Assessment Questions

1. An ecosystem business model is best defined as:

A) Single firm controlling value chain
B) Multiple stakeholders creating value together
C) Direct buyer-seller transactions
D) Data monetization model

Answer: B

Explanation: Multiple stakeholders creating value together is correct because it best fits the concept of an ecosystem business model is best defined as.

2. Short-form video platforms mainly use which filtering?

A) Collaborative filtering
B) Hybrid
C) Demographic filtering
D) Content-based filtering

Answer: D

Explanation: Content-based filtering is correct because it best fits the concept of short-form video platforms mainly use which filtering.

3. Which component powers AI training using tensor computations?

A) CPU
B) FPGA
C) ASIC
D) GPU

Answer: D

Explanation: GPU is correct because it best fits which component powers ai training using tensor computations.

4. Content-based filtering primarily tracks:

A) User behavior like watch time
B) Spending
C) Downloads
D) Similar users

Answer: A

Explanation: User behavior like watch time is correct because it best fits the concept of content-based filtering primarily tracks.

5. Growth of computer vision models was enabled by:

A) ImageNet
B) Common Crawl
C) WordNet
D) DBPedia

Answer: A

Explanation: ImageNet is correct because it best fits the concept of growth of computer vision models was enabled by.

6. Platform cost advantage comes from:

A) Ads
B) Government support
C) Infrastructure
D) User-generated content

Answer: D

Explanation: User-generated content is correct because it best fits the concept of platform cost advantage comes from.

7. Platform monetization methods include (Multiple correct):

A) Login fees
B) Embedded e-commerce
C) Branded challenges
D) Virtual gifts

Answer: B, C, D

Explanation: Embedded e-commerce, Branded challenges, Virtual gifts are correct because each option captures a required part of the concept of platform monetization methods include (multiple correct).

8. A Unicorn startup is:

A) AI company only
B) Global company
C) Private company > $1B
D) Public company > $1B

Answer: C

Explanation: Private company > $1B is correct because it best fits the concept of a unicorn startup is.

9. Porter's generic strategies focus on:

A) Technology and Data
B) Ads and Revenue
C) Market and Users
D) Cost and Differentiation

Answer: D

Explanation: Cost and Differentiation is correct because it best fits the concept of porter's generic strategies focus on.

10. Freemium model means:

A) Free basic + paid premium
B) Fully free
C) Fully paid
D) Ads only

Answer: A

Explanation: Free basic + paid premium is correct because it best fits the concept of freemium model means.

Week 5

Concepts

1. Business Value of AI

AI is useful only when it creates measurable value like cost savings or revenue increase.
Just building a model is not enough-business impact matters.

2. AI Value Drivers

AI creates value in 4 ways:

Automation -> replaces human work (reduces cost)

Anticipation -> predicts future (better decisions)

Coordination -> improves processes between teams

Personalization -> custom experience for users

3. Resource-Based View (RBV)

Just owning AI tech is useless. You need to convert it into capabilities using people, processes, and data. Valuable resources must be valuable, rare, and hard to copy.

4. Business Model (Supply Side)

Business model explains how a company:

creates value

delivers value

earns money

It focuses on the supply side, not customer demand.

5. Transaction Cost Economics (TCE)

Firms choose between:

doing work internally

outsourcing

Decision depends on total cost = production cost + transaction cost.

6. Transaction Costs

These include:

searching vendors

negotiating contracts

monitoring performance

Also includes risk (bias, failure, etc.)

7. AI-as-a-Product

Using external AI adds:

coordination cost

contract enforcement cost

8. Economies of Scale

When production increases:

cost per unit decreases

9. AI Capability Components

AI capability =
Technology
People
Data
Culture
Training & incentives

10. Performance Measurement

To check AI success:
Operational metrics -> speed, efficiency
Business metrics -> revenue, profit
Technical metrics -> accuracy

11. Key KPIs in AI Systems

Examples:
Escalation rate
Resolution time
Sentiment score

> *Example: In Week 5, a company can apply business value of ai together with ai value drivers to improve decision quality and business outcomes.*

MCQs

1. Which AI value driver reduces cost by replacing human work?

A) Anticipation
B) Personalization
C) Automation
D) Coordination

Answer: C

Explanation: Automation is correct because it best fits which ai value driver reduces cost by replacing human work.

2. Anticipation in AI is mainly about?

A) Predicting future outcomes
B) Managing teams
C) Reducing costs
D) Improving UI

Answer: A

Explanation: Predicting future outcomes is correct because it best fits the concept of anticipation in ai is mainly about.

3. What is Business model focuses on?

A) Customers only
B) External market only
C) Supply side
D) Demand side

Answer: C

Explanation: Supply side is correct because it best fits what is business model focuses on.

4. RBV says resources must be?

A) Valuable, rare, inimitable
B) Fast and efficient
C) Cheap and scalable
D) Technical only

Answer: A

Explanation: Valuable, rare, inimitable is correct because it best fits the concept of rbv says resources must be.

5. What is Total cost in TCE?

A) Revenue + cost
B) Fixed + variable
C) Production + transaction
D) Profit + loss

Answer: C

Explanation: Production + transaction is correct because it best fits what is total cost in tce.

6. What is Transaction cost includes?

A) Contracting and monitoring
B) Manufacturing
C) Coding
D) Marketing

Answer: A

Explanation: Contracting and monitoring is correct because it best fits what is transaction cost includes.

7. AI bias is an example of?

A) Fixed cost
B) Production cost
C) Transaction risk
D) Revenue

Answer: C

Explanation: Transaction risk is correct because it best fits the concept of ai bias is an example of.

8. What is Economies of scale means?

A) Lower cost per unit
B) Same cost always
C) No cost
D) Higher cost per unit

Answer: A

Explanation: Lower cost per unit is correct because it best fits what is economies of scale means.

9. What is AI capability requires?

A) Tech + people + process
B) Only money
C) Only data
D) Only algorithms

Answer: A

Explanation: Tech + people + process is correct because it best fits what is ai capability requires.

10. Which is NOT part of AI capability?

A) Training
B) Culture
C) Hashtags
D) People

Answer: C

Explanation: Hashtags is correct because it best fits which is not part of ai capability.

11. What is Coordination improves?

A) Hardware
B) Individual learning
C) Storage
D) Team processes

Answer: D

Explanation: Team processes is correct because it best fits what is coordination improves.

12. What is Personalization means?

A) Reducing cost
B) Training AI
C) Custom user experience
D) Mass production

Answer: C

Explanation: Custom user experience is correct because it best fits what is personalization means.

13. What is Anticipation uses?

A) Databases
B) Hardware
C) Predictive models
D) UI design

Answer: C

Explanation: Predictive models is correct because it best fits what is anticipation uses.

14. What is Operational metrics include?

A) Revenue
B) Market share
C) Profit
D) Speed

Answer: D

Explanation: Speed is correct because it best fits what is operational metrics include.

15. What is Business metrics include?

A) Speed
B) Revenue
C) Latency
D) Accuracy

Answer: B

Explanation: Revenue is correct because it best fits what is business metrics include.

16. What is Escalation rate measures?

A) Tickets going to humans
B) Revenue
C) Accuracy
D) Cost

Answer: A

Explanation: Tickets going to humans is correct because it best fits what is escalation rate measures.

17. What is Resolution time measures?

A) Data
B) Profit
C) Model size
D) Speed of solving issues

Answer: D

Explanation: Speed of solving issues is correct because it best fits what is resolution time measures.

18. What is Sentiment score measures?

A) Model accuracy
B) Cost
C) Speed
D) Customer emotion

Answer: D

Explanation: Customer emotion is correct because it best fits what is sentiment score measures.

19. What is AI replacing manual tasks?

A) Anticipation
B) Coordination
C) Automation
D) Personalization

Answer: C

Explanation: Automation is correct because it best fits what is ai replacing manual tasks.

20. What is Forecasting demand is?

A) Anticipation
B) Automation
C) Coordination
D) Scaling

Answer: A

Explanation: Anticipation is correct because it best fits what is forecasting demand is.

21. What is Business value requires?

A) measurable impact
B) What is Business value requires guarantees perfect outcomes in every scenario.
C) What is Business value requires removes the need for data quality checks and monitoring.
D) What is Business value requires eliminates the need for managerial judgment and governance.

Answer: A

Explanation: measurable impact is correct because it best fits what is business value requires.

22. AI alone guarantees value -> No (C)?

A) AI alone guarantees value -> No (C) guarantees perfect outcomes in every scenario.
B) AI alone guarantees value -> No (C) removes the need for data quality checks and monitoring.
C) AI alone guarantees value -> No (C) is correctly applied in business decisions
D) AI alone guarantees value -> No (C) eliminates the need for managerial judgment and governance.

Answer: C

Explanation: AI alone guarantees value -> No (C) is correctly applied in business decisions is correct because it best fits the concept of ai alone guarantees value -> no (c).

23. Vendor outsourcing adds -> transaction cost -> B?

A) Vendor outsourcing adds -> transaction cost -> B eliminates the need for managerial judgment and governance.
B) Vendor outsourcing adds -> transaction cost -> B is correctly applied in business decisions
C) Vendor outsourcing adds -> transaction cost -> B guarantees perfect outcomes in every scenario.
D) Vendor outsourcing adds -> transaction cost -> B removes the need for data quality checks and monitoring.

Answer: B

Explanation: Vendor outsourcing adds -> transaction cost -> B is correctly applied in business decisions is correct because it best fits the concept of vendor outsourcing adds -> transaction cost -> b.

24. Coordination cost includes -> monitoring -> C?

A) Coordination cost includes -> monitoring -> C guarantees perfect outcomes in every scenario.
B) Coordination cost includes -> monitoring -> C eliminates the need for managerial judgment and governance.
C) Coordination cost includes -> monitoring -> C removes the need for data quality checks and monitoring.
D) Coordination cost includes -> monitoring -> C is correctly applied in business decisions

Answer: D

Explanation: Coordination cost includes -> monitoring -> C is correctly applied in business decisions is correct because it best fits the concept of coordination cost includes -> monitoring -> c.

25. RBV focuses on -> internal strengths -> A?

A) RBV focuses on -> internal strengths -> A guarantees perfect outcomes in every scenario.
B) RBV focuses on -> internal strengths -> A removes the need for data quality checks and monitoring.
C) RBV focuses on -> internal strengths -> A is correctly applied in business decisions
D) RBV focuses on -> internal strengths -> A eliminates the need for managerial judgment and governance.

Answer: C

Explanation: RBV focuses on -> internal strengths -> A is correctly applied in business decisions is correct because it best fits the concept of rbv focuses on -> internal strengths -> a.

26. Supply side = firm activities -> B?

A) Supply side = firm activities -> B eliminates the need for managerial judgment and governance.
B) Supply side = firm activities -> B is correctly applied in business decisions
C) Supply side = firm activities -> B removes the need for data quality checks and monitoring.
D) Supply side = firm activities -> B guarantees perfect outcomes in every scenario.

Answer: B

Explanation: Supply side = firm activities -> B is correctly applied in business decisions is correct because it best fits the concept of supply side = firm activities -> b.

27. Economies of scale -> high volume -> A?

A) Economies of scale -> high volume -> A removes the need for data quality checks and monitoring.
B) Economies of scale -> high volume -> A is correctly applied in business decisions
C) Economies of scale -> high volume -> A eliminates the need for managerial judgment and governance.
D) Economies of scale -> high volume -> A guarantees perfect outcomes in every scenario.

Answer: B

Explanation: Economies of scale -> high volume -> A is correctly applied in business decisions is correct because it best fits the concept of economies of scale -> high volume -> a.

28. AI capability includes -> culture -> C?

A) AI capability includes -> culture -> C guarantees perfect outcomes in every scenario.
B) AI capability includes -> culture -> C eliminates the need for managerial judgment and governance.
C) AI capability includes -> culture -> C removes the need for data quality checks and monitoring.
D) AI capability includes -> culture -> C is correctly applied in business decisions

Answer: D

Explanation: AI capability includes -> culture -> C is correctly applied in business decisions is correct because it best fits the concept of ai capability includes -> culture -> c.

29. AI improves -> decision quality -> B?

A) AI improves -> decision quality -> B removes the need for data quality checks and monitoring.
B) AI improves -> decision quality -> B eliminates the need for managerial judgment and governance.
C) AI improves -> decision quality -> B is correctly applied in business decisions
D) AI improves -> decision quality -> B guarantees perfect outcomes in every scenario.

Answer: C

Explanation: AI improves -> decision quality -> B is correctly applied in business decisions is correct because it best fits the concept of ai improves -> decision quality -> b.

30. Transaction risk example -> bias -> A?

A) Transaction risk example -> bias -> A guarantees perfect outcomes in every scenario.
B) Transaction risk example -> bias -> A is correctly applied in business decisions
C) Transaction risk example -> bias -> A eliminates the need for managerial judgment and governance.
D) Transaction risk example -> bias -> A removes the need for data quality checks and monitoring.

Answer: B

Explanation: Transaction risk example -> bias -> A is correctly applied in business decisions is correct because it best fits the concept of transaction risk example -> bias -> a.

31. Predictive AI -> anticipation -> C?

A) Predictive AI -> anticipation -> C guarantees perfect outcomes in every scenario.
B) Predictive AI -> anticipation -> C is correctly applied in business decisions
C) Predictive AI -> anticipation -> C eliminates the need for managerial judgment and governance.
D) Predictive AI -> anticipation -> C removes the need for data quality checks and monitoring.

Answer: B

Explanation: Predictive AI -> anticipation -> C is correctly applied in business decisions is correct because it best fits the concept of predictive ai -> anticipation -> c.

32. Custom ads -> personalization -> B?

A) Custom ads -> personalization -> B eliminates the need for managerial judgment and governance.
B) Custom ads -> personalization -> B is correctly applied in business decisions
C) Custom ads -> personalization -> B removes the need for data quality checks and monitoring.
D) Custom ads -> personalization -> B guarantees perfect outcomes in every scenario.

Answer: B

Explanation: Custom ads -> personalization -> B is correctly applied in business decisions is correct because it best fits the concept of custom ads -> personalization -> b.

33. Workflow optimization -> coordination -> D?

A) Workflow optimization -> coordination -> D eliminates the need for managerial judgment and governance.
B) Workflow optimization -> coordination -> D removes the need for data quality checks and monitoring.
C) Workflow optimization -> coordination -> D guarantees perfect outcomes in every scenario.
D) Workflow optimization -> coordination -> D is correctly applied in business decisions

Answer: D

Explanation: Workflow optimization -> coordination -> D is correctly applied in business decisions is correct because it best fits the concept of workflow optimization -> coordination -> d.

34. Chatbot replacing agent -> automation -> A?

A) Chatbot replacing agent -> automation -> A is correctly applied in business decisions
B) Chatbot replacing agent -> automation -> A removes the need for data quality checks and monitoring.
C) Chatbot replacing agent -> automation -> A eliminates the need for managerial judgment and governance.
D) Chatbot replacing agent -> automation -> A guarantees perfect outcomes in every scenario.

Answer: A

Explanation: Chatbot replacing agent -> automation -> A is correctly applied in business decisions is correct because it best fits the concept of chatbot replacing agent -> automation -> a.

35. KPI example -> resolution time -> C?

A) KPI example -> resolution time -> C guarantees perfect outcomes in every scenario.
B) KPI example -> resolution time -> C removes the need for data quality checks and monitoring.
C) KPI example -> resolution time -> C eliminates the need for managerial judgment and governance.
D) KPI example -> resolution time -> C is correctly applied in business decisions

Answer: D

Explanation: KPI example -> resolution time -> C is correctly applied in business decisions is correct because it best fits the concept of kpi example -> resolution time -> c.

36. Business KPI -> profit -> D?

A) Business KPI -> profit -> D is correctly applied in business decisions
B) Business KPI -> profit -> D removes the need for data quality checks and monitoring.
C) Business KPI -> profit -> D eliminates the need for managerial judgment and governance.
D) Business KPI -> profit -> D guarantees perfect outcomes in every scenario.

Answer: A

Explanation: Business KPI -> profit -> D is correctly applied in business decisions is correct because it best fits the concept of business kpi -> profit -> d.

37. Operational KPI -> efficiency -> B?

A) Operational KPI -> efficiency -> B is correctly applied in business decisions
B) Operational KPI -> efficiency -> B removes the need for data quality checks and monitoring.
C) Operational KPI -> efficiency -> B eliminates the need for managerial judgment and governance.
D) Operational KPI -> efficiency -> B guarantees perfect outcomes in every scenario.

Answer: A

Explanation: Operational KPI -> efficiency -> B is correctly applied in business decisions is correct because it best fits the concept of operational kpi -> efficiency -> b.

38. Technical KPI -> accuracy -> A?

A) Technical KPI -> accuracy -> A is correctly applied in business decisions
B) Technical KPI -> accuracy -> A guarantees perfect outcomes in every scenario.
C) Technical KPI -> accuracy -> A eliminates the need for managerial judgment and governance.
D) Technical KPI -> accuracy -> A removes the need for data quality checks and monitoring.

Answer: A

Explanation: Technical KPI -> accuracy -> A is correctly applied in business decisions is correct because it best fits the concept of technical kpi -> accuracy -> a.

39. AI value driver count -> 4 -> C?

A) AI value driver count -> 4 -> C is correctly applied in business decisions
B) AI value driver count -> 4 -> C removes the need for data quality checks and monitoring.
C) AI value driver count -> 4 -> C eliminates the need for managerial judgment and governance.
D) AI value driver count -> 4 -> C guarantees perfect outcomes in every scenario.

Answer: A

Explanation: AI value driver count -> 4 -> C is correctly applied in business decisions is correct because it best fits the concept of ai value driver count -> 4 -> c.

40. Coordination reduces -> internal cost -> B?

A) Coordination reduces -> internal cost -> B is correctly applied in business decisions
B) Coordination reduces -> internal cost -> B removes the need for data quality checks and monitoring.
C) Coordination reduces -> internal cost -> B guarantees perfect outcomes in every scenario.
D) Coordination reduces -> internal cost -> B eliminates the need for managerial judgment and governance.

Answer: A

Explanation: Coordination reduces -> internal cost -> B is correctly applied in business decisions is correct because it best fits the concept of coordination reduces -> internal cost -> b.

41. AI training improves -> capability -> A?

A) AI training improves -> capability -> A removes the need for data quality checks and monitoring.
B) AI training improves -> capability -> A is correctly applied in business decisions
C) AI training improves -> capability -> A guarantees perfect outcomes in every scenario.
D) AI training improves -> capability -> A eliminates the need for managerial judgment and governance.

Answer: B

Explanation: AI training improves -> capability -> A is correctly applied in business decisions is correct because it best fits the concept of ai training improves -> capability -> a.

42. Incentives help -> adoption -> D?

A) Incentives help -> adoption -> D removes the need for data quality checks and monitoring.
B) Incentives help -> adoption -> D eliminates the need for managerial judgment and governance.
C) Incentives help -> adoption -> D guarantees perfect outcomes in every scenario.
D) Incentives help -> adoption -> D is correctly applied in business decisions

Answer: D

Explanation: Incentives help -> adoption -> D is correctly applied in business decisions is correct because it best fits the concept of incentives help -> adoption -> d.

43. Data needed for -> AI -> B?

A) Data needed for -> AI -> B is correctly applied in business decisions
B) Data needed for -> AI -> B guarantees perfect outcomes in every scenario.
C) Data needed for -> AI -> B removes the need for data quality checks and monitoring.
D) Data needed for -> AI -> B eliminates the need for managerial judgment and governance.

Answer: A

Explanation: Data needed for -> AI -> B is correctly applied in business decisions is correct because it best fits the concept of data needed for -> ai -> b.

44. AI model alone -> insufficient -> C?

A) AI model alone -> insufficient -> C guarantees perfect outcomes in every scenario.
B) AI model alone -> insufficient -> C eliminates the need for managerial judgment and governance.
C) AI model alone -> insufficient -> C is correctly applied in business decisions
D) AI model alone -> insufficient -> C removes the need for data quality checks and monitoring.

Answer: C

Explanation: AI model alone -> insufficient -> C is correctly applied in business decisions is correct because it best fits the concept of ai model alone -> insufficient -> c.

45. Scaling reduces -> cost -> A?

A) Scaling reduces -> cost -> A removes the need for data quality checks and monitoring.
B) Scaling reduces -> cost -> A is correctly applied in business decisions
C) Scaling reduces -> cost -> A guarantees perfect outcomes in every scenario.
D) Scaling reduces -> cost -> A eliminates the need for managerial judgment and governance.

Answer: B

Explanation: Scaling reduces -> cost -> A is correctly applied in business decisions is correct because it best fits the concept of scaling reduces -> cost -> a.

46. Monitoring vendors -> transaction -> D?

A) Monitoring vendors -> transaction -> D is correctly applied in business decisions
B) Monitoring vendors -> transaction -> D guarantees perfect outcomes in every scenario.
C) Monitoring vendors -> transaction -> D eliminates the need for managerial judgment and governance.
D) Monitoring vendors -> transaction -> D removes the need for data quality checks and monitoring.

Answer: A

Explanation: Monitoring vendors -> transaction -> D is correctly applied in business decisions is correct because it best fits the concept of monitoring vendors -> transaction -> d.

47. Business model defines -> monetization -> B?

A) Business model defines -> monetization -> B eliminates the need for managerial judgment and governance.
B) Business model defines -> monetization -> B guarantees perfect outcomes in every scenario.
C) Business model defines -> monetization -> B removes the need for data quality checks and monitoring.
D) Business model defines -> monetization -> B is correctly applied in business decisions

Answer: D

Explanation: Business model defines -> monetization -> B is correctly applied in business decisions is correct because it best fits the concept of business model defines -> monetization -> b.

48. AI improves -> efficiency -> A?

A) AI improves -> efficiency -> A is correctly applied in business decisions
B) AI improves -> efficiency -> A removes the need for data quality checks and monitoring.
C) AI improves -> efficiency -> A eliminates the need for managerial judgment and governance.
D) AI improves -> efficiency -> A guarantees perfect outcomes in every scenario.

Answer: A

Explanation: AI improves -> efficiency -> A is correctly applied in business decisions is correct because it best fits the concept of ai improves -> efficiency -> a.

49. AI reduces -> manual work -> C?

A) AI reduces -> manual work -> C eliminates the need for managerial judgment and governance.
B) AI reduces -> manual work -> C guarantees perfect outcomes in every scenario.
C) AI reduces -> manual work -> C removes the need for data quality checks and monitoring.
D) AI reduces -> manual work -> C is correctly applied in business decisions

Answer: D

Explanation: AI reduces -> manual work -> C is correctly applied in business decisions is correct because it best fits the concept of ai reduces -> manual work -> c.

50. AI value depends on -> usage -> D?

A) AI value depends on -> usage -> D eliminates the need for managerial judgment and governance.
B) AI value depends on -> usage -> D guarantees perfect outcomes in every scenario.
C) AI value depends on -> usage -> D is correctly applied in business decisions
D) AI value depends on -> usage -> D removes the need for data quality checks and monitoring.

Answer: C

Explanation: AI value depends on -> usage -> D is correctly applied in business decisions is correct because it best fits the concept of ai value depends on -> usage -> d.

Online Assessment Questions

1. Which AI value driver reduces manual work?

A) Coordination
B) Anticipation
C) Personalization
D) Automation

Answer: D

Explanation: Automation is correct because it best fits which ai value driver reduces manual work.

2. Anticipation in AI refers to:

A) Scaling
B) Automation
C) Coordination
D) Prediction

Answer: D

Explanation: Prediction is correct because it best fits the concept of anticipation in ai refers to.

3. Business model mainly focuses on:

A) Supply
B) Data
C) Customers
D) Demand

Answer: A

Explanation: Supply is correct because it best fits the concept of business model mainly focuses on.

4. Total cost includes:

A) Profit
B) Both
C) Production
D) Transaction

Answer: B

Explanation: Both is correct because it best fits the concept of total cost includes.

5. Transaction cost includes:

A) Training
B) Storage
C) Coding
D) Contracting

Answer: D

Explanation: Contracting is correct because it best fits the concept of transaction cost includes.

6. Economies of scale result in:

A) No change
B) Higher cost
C) Lower cost
D) Loss

Answer: C

Explanation: Lower cost is correct because it best fits the concept of economies of scale result in.

7. AI capability includes:

A) Tech + people + process
B) Only model
C) Only hardware
D) Only data

Answer: A

Explanation: Tech + people + process is correct because it best fits the concept of ai capability includes.

8. Operational metrics measure:

A) Accuracy
B) Profit
C) Efficiency
D) Data

Answer: C

Explanation: Efficiency is correct because it best fits the concept of operational metrics measure.

9. Business metrics include:

A) Recall
B) Precision
C) Revenue
D) Speed

Answer: C

Explanation: Revenue is correct because it best fits the concept of business metrics include.

10. Escalation rate measures:

A) Accuracy
B) Human intervention
C) Speed
D) Cost

Answer: B

Explanation: Human intervention is correct because it best fits the concept of escalation rate measures.

Week 6

Concepts

1. Financial vs Technical vs Operational Metrics

Technical metrics (Precision, Recall, AUC) tell how well the model works mathematically.

Operational metrics measure process efficiency (speed, automation).

Financial metrics (revenue, cost savings, CLV) show real business value -> most important for executives.

2. A/B Testing (Randomized Controlled Trial)

Users are randomly split into two groups: control vs treatment.

Only one change is applied -> ensures fair comparison.

Considered gold standard because it removes bias and proves causality.

3. Synthetic Control Method (SCM)

Used when A/B testing is not possible.

Creates a fake control group using weighted combination of other groups.

Helps estimate "what would have happened" without intervention.

4. Measuring AI Business Value

Biggest challenge: proving ROI to non-technical stakeholders.

Need clear, measurable impact (money, efficiency, outcomes).

Without this -> AI projects fail to scale.

5. Class Imbalance Problem

Happens when one class is very rare (e.g., fraud detection).

Accuracy becomes misleading (model predicts majority always).

Use better metrics: Precision, Recall, AUC, KS.

6. AUC (Area Under Curve)

Measures model’s ability to distinguish classes.

Higher AUC -> better model across all thresholds.

Useful when classes are imbalanced.

7. Precision & Recall

Precision: how many predicted positives are correct.

Recall: how many actual positives are captured.

Trade-off between both depending on problem.

8. KS Statistic

Measures separation between positive & negative distributions.

Higher KS -> better discrimination.

Common in credit scoring.

9. Hierarchy of Metrics

Technical -> Operational -> Financial

Business cares most about financial outcomes

Managers focus on operational metrics

10. Operational Metrics Examples

Process efficiency

Speed of delivery

Automation of tasks

These show practical system impact

11. Ranking Metrics (Search/Feed systems)

Used where order matters (top results important)

Metrics:

Mean Reciprocal Rank (MRR)

NDCG (Normalized Discounted Cumulative Gain)

12. Mean Reciprocal Rank (MRR)

Focuses on position of first correct result

Higher if correct answer appears early

13. NDCG

Rewards correct results higher in ranking

Penalizes relevant results appearing lower

Considers position importance

14. Swap Set Analysis (Credit Models)

Compares old vs new model decisions

Uses:

LTV (good customers)

LGD (bad customers)

Goal: maximize net profit

15. Financial Evaluation in AI

Best metric: net profit under constraints

Not just accuracy -> business impact matters

16. KS vs Accuracy

Accuracy depends on threshold

KS evaluates across all thresholds

Hence better for imbalanced datasets

Example: In Week 6, a company can apply financial vs technical vs operational metrics together with a/b testing (randomized controlled trial) to improve decision quality and business outcomes.

MCQs

1. Which metric is most important for executives?

A) Financial metrics
B) Recall
C) Precision
D) Accuracy

Answer: A

Explanation: Financial metrics is correct because it best fits which metric is most important for executives.

2. A/B testing is mainly used for?

A) Data cleaning
B) Feature selection
C) Measuring causality
D) Training models

Answer: C

Explanation: Measuring causality is correct because it best fits the concept of a/b testing is mainly used for.

3. What makes A/B testing reliable?

A) Visualization
B) Large data
C) Speed
D) Randomization

Answer: D

Explanation: Randomization is correct because it best fits what makes a/b testing reliable.

4. Synthetic Control Method is used when?

A) A/B testing not possible
B) Model is slow
C) Labels missing
D) Data is large

Answer: A

Explanation: A/B testing not possible is correct because it best fits the concept of synthetic control method is used when.

5. What is SCM creates?

A) Labels
B) Fake dataset
C) New model
D) Counterfactual control group

Answer: D

Explanation: Counterfactual control group is correct because it best fits what is scm creates.

6. Main challenge in AI adoption?

A) Hardware
B) Data storage
C) Proving ROI
D) Coding

Answer: C

Explanation: Proving ROI is correct because it best fits the concept of main challenge in ai adoption.

7. What is Class imbalance affects?

A) Accuracy reliability
B) Data size
C) Storage
D) Training time

Answer: A

Explanation: Accuracy reliability is correct because it best fits what is class imbalance affects.

8. What is Accuracy fails when?

A) Data small
B) Data balanced
C) Model linear
D) Classes imbalanced

Answer: D

Explanation: Classes imbalanced is correct because it best fits what is accuracy fails when.

9. What is AUC measures?

A) Separation ability
B) Error
C) Speed
D) Memory

Answer: A

Explanation: Separation ability is correct because it best fits what is auc measures.

10. What is Higher AUC means?

A) Slower training
B) Worse model
C) Less data
D) Better classification

Answer: D

Explanation: Better classification is correct because it best fits what is higher auc means.

11. What is Precision measures?

A) Total predictions
B) Errors
C) Speed
D) Correct positives

Answer: D

Explanation: Correct positives is correct because it best fits what is precision measures.

12. What is Recall measures?

A) Captured positives
B) Correct negatives
C) Loss
D) Speed

Answer: A

Explanation: Captured positives is correct because it best fits what is recall measures.

13. What is KS statistic measures?

A) Accuracy
B) Speed
C) Loss
D) Separation

Answer: D

Explanation: Separation is correct because it best fits what is ks statistic measures.

14. What is KS is common in?

A) NLP
B) Vision
C) Robotics
D) Credit scoring

Answer: D

Explanation: Credit scoring is correct because it best fits what is ks is common in.

15. What is Metric hierarchy highest level?

A) Operational
B) Model
C) Technical
D) Financial

Answer: D

Explanation: Financial is correct because it best fits what is metric hierarchy highest level.

16. What is Operational metrics focus on?

A) Accuracy
B) Efficiency
C) Profit
D) Data

Answer: B

Explanation: Efficiency is correct because it best fits what is operational metrics focus on.

17. What is Example of operational metric?

A) AUC
B) Speed
C) Revenue
D) Precision

Answer: B

Explanation: Speed is correct because it best fits what is example of operational metric.

18. Ranking metrics are used in?

A) Regression
B) Classification
C) Ranking systems
D) Clustering

Answer: C

Explanation: Ranking systems is correct because it best fits the concept of ranking metrics are used in.

19. What is MRR focuses on?

A) Errors
B) Last result
C) First correct result
D) All results

Answer: C

Explanation: First correct result is correct because it best fits what is mrr focuses on.

20. What is NDCG rewards?

A) Random order
B) Errors
C) Correct top ranking
D) Speed

Answer: C

Explanation: Correct top ranking is correct because it best fits what is ndcg rewards.

21. What is Swap set analysis compares?

A) Models
B) Inputs
C) Data
D) Features

Answer: A

Explanation: Models is correct because it best fits what is swap set analysis compares.

22. What is LTV is for?

A) Bad customers
B) Data
C) Good customers
D) Errors

Answer: C

Explanation: Good customers is correct because it best fits what is ltv is for.

23. What is LGD is for?

A) Errors
B) Good customers
C) Bad customers
D) Data

Answer: C

Explanation: Bad customers is correct because it best fits what is lgd is for.

24. What is Goal of swap analysis?

A) Accuracy
B) Speed
C) Profit
D) Memory

Answer: C

Explanation: Profit is correct because it best fits what is goal of swap analysis.

25. What is Financial metrics include?

A) Revenue
B) Speed
C) AUC
D) Recall

Answer: A

Explanation: Revenue is correct because it best fits what is financial metrics include.

26. What is KS works across?

A) No thresholds
B) All thresholds
C) Fixed value
D) Single threshold

Answer: B

Explanation: All thresholds is correct because it best fits what is ks works across.

27. What is Accuracy depends on?

A) Model type
B) Threshold
C) Speed
D) Data size

Answer: B

Explanation: Threshold is correct because it best fits what is accuracy depends on.

28. What is Best metric for imbalance?

A) Speed
B) Accuracy
C) KS
D) Size

Answer: C

Explanation: KS is correct because it best fits what is best metric for imbalance.

29. What is Operational level cares about?

A) Profit
B) Theory
C) Efficiency
D) Math

Answer: C

Explanation: Efficiency is correct because it best fits what is operational level cares about.

30. What is Technical metrics include?

A) Profit
B) Precision
C) Cost
D) Revenue

Answer: B

Explanation: Precision is correct because it best fits what is technical metrics include.

31. 31 A/B test removes -> Bias?

A) 31 A/B test removes -> Bias eliminates the need for managerial judgment and governance.
B) 31 A/B test removes -> Bias removes the need for data quality checks and monitoring.
C) 31 A/B test removes -> Bias is correctly applied in business decisions
D) 31 A/B test removes -> Bias guarantees perfect outcomes in every scenario.

Answer: C

Explanation: 31 A/B test removes -> Bias is correctly applied in business decisions is correct because it best fits the concept of 31 a/b test removes -> bias.

32. 32 SCM estimates -> Counterfactual?

A) 32 SCM estimates -> Counterfactual removes the need for data quality checks and monitoring.
B) 32 SCM estimates -> Counterfactual eliminates the need for managerial judgment and governance.
C) 32 SCM estimates -> Counterfactual is correctly applied in business decisions
D) 32 SCM estimates -> Counterfactual guarantees perfect outcomes in every scenario.

Answer: C

Explanation: 32 SCM estimates -> Counterfactual is correctly applied in business decisions is correct because it best fits the concept of 32 scm estimates -> counterfactual.

33. 33 Recall important for -> Rare events?

A) 33 Recall important for -> Rare events guarantees perfect outcomes in every scenario.
B) 33 Recall important for -> Rare events is correctly applied in business decisions
C) 33 Recall important for -> Rare events eliminates the need for managerial judgment and governance.
D) 33 Recall important for -> Rare events removes the need for data quality checks and monitoring.

Answer: B

Explanation: 33 Recall important for -> Rare events is correctly applied in business decisions is correct because it best fits the concept of 33 recall important for -> rare events.

34. 34 Precision important for -> False positives control?

A) 34 Precision important for -> False positives control is correctly applied in business decisions
B) 34 Precision important for -> False positives control removes the need for data quality checks and monitoring.
C) 34 Precision important for -> False positives control eliminates the need for managerial judgment and governance.
D) 34 Precision important for -> False positives control guarantees perfect outcomes in every scenario.

Answer: A

Explanation: 34 Precision important for -> False positives control is correctly applied in business decisions is correct because it best fits the concept of 34 precision important for -> false positives control.

35. 35 AUC range -> 0 to 1?

A) 35 AUC range -> 0 to 1 guarantees perfect outcomes in every scenario.
B) 35 AUC range -> 0 to 1 removes the need for data quality checks and monitoring.
C) 35 AUC range -> 0 to 1 eliminates the need for managerial judgment and governance.
D) 35 AUC range -> 0 to 1 is correctly applied in business decisions

Answer: D

Explanation: 35 AUC range -> 0 to 1 is correctly applied in business decisions is correct because it best fits the concept of 35 auc range -> 0 to 1.

36. 36 KS higher -> Better model?

A) 36 KS higher -> Better model guarantees perfect outcomes in every scenario.
B) 36 KS higher -> Better model removes the need for data quality checks and monitoring.
C) 36 KS higher -> Better model eliminates the need for managerial judgment and governance.
D) 36 KS higher -> Better model is correctly applied in business decisions

Answer: D

Explanation: 36 KS higher -> Better model is correctly applied in business decisions is correct because it best fits the concept of 36 ks higher -> better model.

37. 37 NDCG considers -> Position?

A) 37 NDCG considers -> Position is correctly applied in business decisions
B) 37 NDCG considers -> Position removes the need for data quality checks and monitoring.
C) 37 NDCG considers -> Position eliminates the need for managerial judgment and governance.
D) 37 NDCG considers -> Position guarantees perfect outcomes in every scenario.

Answer: A

Explanation: 37 NDCG considers -> Position is correctly applied in business decisions is correct because it best fits the concept of 37 ndcg considers -> position.

38. 38 MRR ignores -> All except first correct?

A) 38 MRR ignores -> All except first correct eliminates the need for managerial judgment and governance.
B) 38 MRR ignores -> All except first correct is correctly applied in business decisions
C) 38 MRR ignores -> All except first correct guarantees perfect outcomes in every scenario.
D) 38 MRR ignores -> All except first correct removes the need for data quality checks and monitoring.

Answer: B

Explanation: 38 MRR ignores -> All except first correct is correctly applied in business decisions is correct because it best fits the concept of 38 mrr ignores -> all except first correct.

39. 39 Financial metrics -> Business value?

A) 39 Financial metrics -> Business value eliminates the need for managerial judgment and governance.
B) 39 Financial metrics -> Business value is correctly applied in business decisions
C) 39 Financial metrics -> Business value guarantees perfect outcomes in every scenario.
D) 39 Financial metrics -> Business value removes the need for data quality checks and monitoring.

Answer: B

Explanation: 39 Financial metrics -> Business value is correctly applied in business decisions is correct because it best fits the concept of 39 financial metrics -> business value.

40. 40 Operational metrics -> Process improvement?

A) 40 Operational metrics -> Process improvement eliminates the need for managerial judgment and governance.
B) 40 Operational metrics -> Process improvement guarantees perfect outcomes in every scenario.
C) 40 Operational metrics -> Process improvement is correctly applied in business decisions
D) 40 Operational metrics -> Process improvement removes the need for data quality checks and monitoring.

Answer: C

Explanation: 40 Operational metrics -> Process improvement is correctly applied in business decisions is correct because it best fits the concept of 40 operational metrics -> process improvement.

41. 41 Technical metrics -> Model quality?

A) 41 Technical metrics -> Model quality guarantees perfect outcomes in every scenario.
B) 41 Technical metrics -> Model quality removes the need for data quality checks and monitoring.
C) 41 Technical metrics -> Model quality eliminates the need for managerial judgment and governance.
D) 41 Technical metrics -> Model quality is correctly applied in business decisions

Answer: D

Explanation: 41 Technical metrics -> Model quality is correctly applied in business decisions is correct because it best fits the concept of 41 technical metrics -> model quality.

42. 42 Class imbalance -> Rare class problem?

A) 42 Class imbalance -> Rare class problem is correctly applied in business decisions
B) 42 Class imbalance -> Rare class problem removes the need for data quality checks and monitoring.
C) 42 Class imbalance -> Rare class problem guarantees perfect outcomes in every scenario.
D) 42 Class imbalance -> Rare class problem eliminates the need for managerial judgment and governance.

Answer: A

Explanation: 42 Class imbalance -> Rare class problem is correctly applied in business decisions is correct because it best fits the concept of 42 class imbalance -> rare class problem.

43. 43 A/B requires -> Random split?

A) 43 A/B requires -> Random split guarantees perfect outcomes in every scenario.
B) 43 A/B requires -> Random split is correctly applied in business decisions
C) 43 A/B requires -> Random split eliminates the need for managerial judgment and governance.
D) 43 A/B requires -> Random split removes the need for data quality checks and monitoring.

Answer: B

Explanation: 43 A/B requires -> Random split is correctly applied in business decisions is correct because it best fits the concept of 43 a/b requires -> random split.

44. 44 SCM used when -> No randomization?

A) 44 SCM used when -> No randomization eliminates the need for managerial judgment and governance.
B) 44 SCM used when -> No randomization is correctly applied in business decisions
C) 44 SCM used when -> No randomization guarantees perfect outcomes in every scenario.
D) 44 SCM used when -> No randomization removes the need for data quality checks and monitoring.

Answer: B

Explanation: 44 SCM used when -> No randomization is correctly applied in business decisions is correct because it best fits the concept of 44 scm used when -> no randomization.

45. 45 Swap set -> Model comparison?

A) 45 Swap set -> Model comparison is correctly applied in business decisions
B) 45 Swap set -> Model comparison eliminates the need for managerial judgment and governance.
C) 45 Swap set -> Model comparison removes the need for data quality checks and monitoring.
D) 45 Swap set -> Model comparison guarantees perfect outcomes in every scenario.

Answer: A

Explanation: 45 Swap set -> Model comparison is correctly applied in business decisions is correct because it best fits the concept of 45 swap set -> model comparison.

46. 46 LTV high -> Good customer?

A) 46 LTV high -> Good customer guarantees perfect outcomes in every scenario.
B) 46 LTV high -> Good customer removes the need for data quality checks and monitoring.
C) 46 LTV high -> Good customer is correctly applied in business decisions
D) 46 LTV high -> Good customer eliminates the need for managerial judgment and governance.

Answer: C

Explanation: 46 LTV high -> Good customer is correctly applied in business decisions is correct because it best fits the concept of 46 ltv high -> good customer.

47. 47 LGD high -> Loss risk?

A) 47 LGD high -> Loss risk is correctly applied in business decisions
B) 47 LGD high -> Loss risk guarantees perfect outcomes in every scenario.
C) 47 LGD high -> Loss risk eliminates the need for managerial judgment and governance.
D) 47 LGD high -> Loss risk removes the need for data quality checks and monitoring.

Answer: A

Explanation: 47 LGD high -> Loss risk is correctly applied in business decisions is correct because it best fits the concept of 47 lgd high -> loss risk.

48. 48 KS compares -> Distributions?

A) 48 KS compares -> Distributions removes the need for data quality checks and monitoring.
B) 48 KS compares -> Distributions is correctly applied in business decisions
C) 48 KS compares -> Distributions guarantees perfect outcomes in every scenario.
D) 48 KS compares -> Distributions eliminates the need for managerial judgment and governance.

Answer: B

Explanation: 48 KS compares -> Distributions is correctly applied in business decisions is correct because it best fits the concept of 48 ks compares -> distributions.

49. 49 Accuracy misleading when -> Imbalance?

A) 49 Accuracy misleading when -> Imbalance guarantees perfect outcomes in every scenario.
B) 49 Accuracy misleading when -> Imbalance is correctly applied in business decisions
C) 49 Accuracy misleading when -> Imbalance eliminates the need for managerial judgment and governance.
D) 49 Accuracy misleading when -> Imbalance removes the need for data quality checks and monitoring.

Answer: B

Explanation: 49 Accuracy misleading when -> Imbalance is correctly applied in business decisions is correct because it best fits the concept of 49 accuracy misleading when -> imbalance.

50. 50 Best success metric -> Profit?

A) 50 Best success metric -> Profit is correctly applied in business decisions
B) 50 Best success metric -> Profit removes the need for data quality checks and monitoring.
C) 50 Best success metric -> Profit eliminates the need for managerial judgment and governance.
D) 50 Best success metric -> Profit guarantees perfect outcomes in every scenario.

Answer: A

Explanation: 50 Best success metric -> Profit is correctly applied in business decisions is correct because it best fits the concept of 50 best success metric -> profit.

Online Assessment Questions

1. Most important metric for executives:

A) Accuracy
B) Recall
C) Precision
D) Financial

Answer: D

Explanation: Financial is correct because it best fits the concept of most important metric for executives.

2. A/B testing is used to measure:

A) Speed
B) Storage
C) Training
D) Causality

Answer: D

Explanation: Causality is correct because it best fits the concept of a/b testing is used to measure.

3. A/B testing requires:

A) Randomization
B) Clustering
C) Scaling
D) Storage

Answer: A

Explanation: Randomization is correct because it best fits the concept of a/b testing requires.

4. Synthetic control method is used when:

A) A/B not possible
B) Model slow
C) Data missing
D) Data large

Answer: A

Explanation: A/B not possible is correct because it best fits the concept of synthetic control method is used when.

5. Accuracy fails when:

A) Imbalanced data
B) Small data
C) Balanced data
D) Large data

Answer: A

Explanation: Imbalanced data is correct because it best fits the concept of accuracy fails when.

6. AUC measures:

A) Cost
B) Storage
C) Classification ability
D) Speed

Answer: C

Explanation: Classification ability is correct because it best fits the concept of auc measures.

7. Precision measures:

A) Speed
B) Total predictions
C) Correct positives
D) Errors

Answer: C

Explanation: Correct positives is correct because it best fits the concept of precision measures.

8. Recall measures:

A) Loss
B) Speed
C) Captured positives
D) Correct negatives

Answer: C

Explanation: Captured positives is correct because it best fits the concept of recall measures.

9. KS statistic measures:

A) Separation
B) Storage
C) Speed
D) Cost

Answer: A

Explanation: Separation is correct because it best fits the concept of ks statistic measures.

10. Best metric for business success:

A) Profit
B) Precision
C) Recall
D) Accuracy

Answer: A

Explanation: Profit is correct because it best fits the concept of best metric for business success.

Week 7

Concepts

1. Levels of Management

Strategic: Top-level decisions (long-term goals, vision).
Tactical: Mid-level decisions (plans, strategies).
Operational: Day-to-day routine decisions (execution).

2. Types of Decisions

Structured: Routine, rules-based (easy to automate).
Semi-structured: Mix of logic + human judgment.
Unstructured: Complex, no fixed rules (requires thinking).

3. Examples

Payroll -> Structured
Marketing strategy -> Semi-structured
Investment decisions -> Unstructured

4. Business Intelligence (BI) Infrastructure

Systems that store + analyze data.
Includes:
Data warehouse
Dashboards
Reports
Helps in decision-making.

5. Data Warehouse

Central storage of structured data.
Used for analysis and reporting.
Core part of BI systems.

6. Explainable AI (XAI)

Helps understand how AI makes decisions.
Important for trust + transparency.
Tools: SHAP, LIME.

7. SHAP & LIME

Explain predictions by showing feature importance.
Used when model is complex (black box).
Helps stakeholders trust AI.

8. AI Assistant vs AI Agent

Assistant: Needs instructions, reactive.
Agent: Autonomous, takes actions, plans workflows.

9. AI Agents

Can design workflows.
Execute tasks independently.

Use tools and memory.

10. Correlation vs Causation

Correlation ? causation.

Just because two things relate doesn't mean one causes the other.

11. Bias in Decision Making

Humans often biased.

AI can also learn bias from data.

Must validate before decisions.

12. Anchoring Bias

First information influences decisions heavily.

Even if wrong, it affects thinking.

13. Confirmation Bias

People look for info that supports their beliefs.

Ignore conflicting evidence.

14. Governance in AI

Focuses on fairness, transparency, accountability.

Important in high-risk domains.

15. Model Simplicity vs Complexity

Complex models -> accurate but hard to explain.

Simple models -> easier to trust.

16. Interpretability Importance

Required in domains like finance, healthcare.

Helps justify decisions.

> *Example: In Week 7, a company can apply levels of management together with types of decisions to improve decision quality and business outcomes.*

MCQs

1. Which level handles routine decisions?

A) Strategic
B) Operational
C) Tactical
D) Executive

Answer: B

Explanation: Operational is correct because it best fits which level handles routine decisions.

2. What is Long-term decisions belong to?

A) Strategic
B) Operational
C) Tactical
D) Routine

Answer: A

Explanation: Strategic is correct because it best fits what is long-term decisions belong to.

3. What is Semi-structured decisions involve?

A) Only intuition
B) Both rules and judgment
C) No logic
D) Only rules

Answer: B

Explanation: Both rules and judgment is correct because it best fits what is semi-structured decisions involve.

4. What is Payroll processing is?

A) Semi-structured
B) Strategic
C) Unstructured
D) Structured

Answer: D

Explanation: Structured is correct because it best fits what is payroll processing is.

5. What is Marketing strategy is?

A) Semi-structured
B) Routine
C) Operational
D) Structured

Answer: A

Explanation: Semi-structured is correct because it best fits what is marketing strategy is.

6. What is Investment decision is?

A) Semi-structured
B) Unstructured
C) Structured
D) Routine

Answer: B

Explanation: Unstructured is correct because it best fits what is investment decision is.

7. BI systems are used for?

A) Networking
B) Gaming
C) Data analysis
D) Coding

Answer: C

Explanation: Data analysis is correct because it best fits the concept of bi systems are used for.

8. Core component of BI infrastructure?

A) Cache
B) RAM
C) CPU
D) Data warehouse

Answer: D

Explanation: Data warehouse is correct because it best fits the concept of core component of bi infrastructure.

9. What is Data warehouse stores?

A) Code
B) Images
C) Structured data
D) Models

Answer: C

Explanation: Structured data is correct because it best fits what is data warehouse stores.

10. What is XAI is used for?

A) Training
B) Explainability
C) Storage
D) Speed

Answer: B

Explanation: Explainability is correct because it best fits what is xai is used for.

11. What is SHAP is used for?

A) Explanation
B) Compression
C) Training
D) Storage

Answer: A

Explanation: Explanation is correct because it best fits what is shap is used for.

12. What is LIME explains?

A) Predictions
B) Network
C) Dataset
D) Hardware

Answer: A

Explanation: Predictions is correct because it best fits what is lime explains.

13. What is AI assistant is?

A) Independent
B) Reactive
C) Autonomous
D) Self-learning

Answer: B

Explanation: Reactive is correct because it best fits what is ai assistant is.

14. What is AI agent is?

A) Autonomous
B) Static
C) Passive
D) Manual

Answer: A

Explanation: Autonomous is correct because it best fits what is ai agent is.

15. What is AI agents can?

A) Only respond
B) Store files
C) Train models
D) Execute workflows

Answer: D

Explanation: Execute workflows is correct because it best fits what is ai agents can.

16. What is Correlation means?

A) Relation
B) Cause
C) Prediction
D) Bias

Answer: A

Explanation: Relation is correct because it best fits what is correlation means.

17. What is Correlation implies causation?

A) Never
B) Only AI
C) Always
D) Sometimes

Answer: A

Explanation: Never is correct because it best fits what is correlation implies causation.

18. What is Anchoring bias means?

A) Ignore info
B) First info influences
C) Data loss
D) Random thinking

Answer: B

Explanation: First info influences is correct because it best fits what is anchoring bias means.

19. What is Confirmation bias means?

A) Ignore data
B) Accept all
C) Reject all
D) Favor supporting info

Answer: D

Explanation: Favor supporting info is correct because it best fits what is confirmation bias means.

20. What is AI bias comes from?

A) Code
B) RAM
C) CPU
D) Data

Answer: D

Explanation: Data is correct because it best fits what is ai bias comes from.

21. What is Governance ensures?

A) Coding
B) Fairness
C) Storage
D) Speed

Answer: B

Explanation: Fairness is correct because it best fits what is governance ensures.

22. What is XAI important for?

A) Trust
B) Speed
C) Storage
D) Games

Answer: A

Explanation: Trust is correct because it best fits what is xai important for.

23. What is Simple models are?

A) Large
B) Complex
C) Interpretable
D) Slow

Answer: C

Explanation: Interpretable is correct because it best fits what is simple models are.

24. What is Complex models are?

A) Easy to explain
B) Simple
C) Small
D) Hard to explain

Answer: D

Explanation: Hard to explain is correct because it best fits what is complex models are.

25. What is Operational decisions are?

A) Daily
B) Strategic
C) Complex
D) Long-term

Answer: A

Explanation: Daily is correct because it best fits what is operational decisions are.

26. 26 Tactical decisions -> Middle level?

A) 26 Tactical decisions -> Middle level removes the need for data quality checks and monitoring.
B) 26 Tactical decisions -> Middle level guarantees perfect outcomes in every scenario.
C) 26 Tactical decisions -> Middle level is correctly applied in business decisions
D) 26 Tactical decisions -> Middle level eliminates the need for managerial judgment and governance.

Answer: C

Explanation: 26 Tactical decisions -> Middle level is correctly applied in business decisions is correct because it best fits the concept of 26 tactical decisions -> middle level.

27. 27 Strategic decisions -> Top level?

A) 27 Strategic decisions -> Top level removes the need for data quality checks and monitoring.
B) 27 Strategic decisions -> Top level is correctly applied in business decisions
C) 27 Strategic decisions -> Top level eliminates the need for managerial judgment and governance.
D) 27 Strategic decisions -> Top level guarantees perfect outcomes in every scenario.

Answer: B

Explanation: 27 Strategic decisions -> Top level is correctly applied in business decisions is correct because it best fits the concept of 27 strategic decisions -> top level.

28. 28 Structured decisions -> Routine?

A) 28 Structured decisions -> Routine is correctly applied in business decisions
B) 28 Structured decisions -> Routine removes the need for data quality checks and monitoring.
C) 28 Structured decisions -> Routine guarantees perfect outcomes in every scenario.
D) 28 Structured decisions -> Routine eliminates the need for managerial judgment and governance.

Answer: A

Explanation: 28 Structured decisions -> Routine is correctly applied in business decisions is correct because it best fits the concept of 28 structured decisions -> routine.

29. 29 Unstructured -> No rules?

A) 29 Unstructured -> No rules is correctly applied in business decisions
B) 29 Unstructured -> No rules eliminates the need for managerial judgment and governance.
C) 29 Unstructured -> No rules guarantees perfect outcomes in every scenario.
D) 29 Unstructured -> No rules removes the need for data quality checks and monitoring.

Answer: A

Explanation: 29 Unstructured -> No rules is correctly applied in business decisions is correct because it best fits the concept of 29 unstructured -> no rules.

30. 30 BI dashboards -> Visualization?

A) 30 BI dashboards -> Visualization is correctly applied in business decisions
B) 30 BI dashboards -> Visualization guarantees perfect outcomes in every scenario.
C) 30 BI dashboards -> Visualization eliminates the need for managerial judgment and governance.
D) 30 BI dashboards -> Visualization removes the need for data quality checks and monitoring.

Answer: A

Explanation: 30 BI dashboards -> Visualization is correctly applied in business decisions is correct because it best fits the concept of 30 bi dashboards -> visualization.

31. 31 Reports -> Data summary?

A) 31 Reports -> Data summary removes the need for data quality checks and monitoring.
B) 31 Reports -> Data summary is correctly applied in business decisions
C) 31 Reports -> Data summary eliminates the need for managerial judgment and governance.
D) 31 Reports -> Data summary guarantees perfect outcomes in every scenario.

Answer: B

Explanation: 31 Reports -> Data summary is correctly applied in business decisions is correct because it best fits the concept of 31 reports -> data summary.

32. 32 Data marts -> Subset data?

A) 32 Data marts -> Subset data is correctly applied in business decisions
B) 32 Data marts -> Subset data removes the need for data quality checks and monitoring.
C) 32 Data marts -> Subset data eliminates the need for managerial judgment and governance.
D) 32 Data marts -> Subset data guarantees perfect outcomes in every scenario.

Answer: A

Explanation: 32 Data marts -> Subset data is correctly applied in business decisions is correct because it best fits the concept of 32 data marts -> subset data.

33. 33 SHAP shows -> Feature impact?

A) 33 SHAP shows -> Feature impact is correctly applied in business decisions
B) 33 SHAP shows -> Feature impact eliminates the need for managerial judgment and governance.
C) 33 SHAP shows -> Feature impact guarantees perfect outcomes in every scenario.
D) 33 SHAP shows -> Feature impact removes the need for data quality checks and monitoring.

Answer: A

Explanation: 33 SHAP shows -> Feature impact is correctly applied in business decisions is correct because it best fits the concept of 33 shap shows -> feature impact.

34. 34 LIME -> Local explanation?

A) 34 LIME -> Local explanation eliminates the need for managerial judgment and governance.
B) 34 LIME -> Local explanation guarantees perfect outcomes in every scenario.
C) 34 LIME -> Local explanation is correctly applied in business decisions
D) 34 LIME -> Local explanation removes the need for data quality checks and monitoring.

Answer: C

Explanation: 34 LIME -> Local explanation is correctly applied in business decisions is correct because it best fits the concept of 34 lime -> local explanation.

35. 35 AI assistant -> Needs prompt?

A) 35 AI assistant -> Needs prompt is correctly applied in business decisions
B) 35 AI assistant -> Needs prompt guarantees perfect outcomes in every scenario.
C) 35 AI assistant -> Needs prompt eliminates the need for managerial judgment and governance.
D) 35 AI assistant -> Needs prompt removes the need for data quality checks and monitoring.

Answer: A

Explanation: 35 AI assistant -> Needs prompt is correctly applied in business decisions is correct because it best fits the concept of 35 ai assistant -> needs prompt.

36. 36 AI agent -> Self-driven?

A) 36 AI agent -> Self-driven guarantees perfect outcomes in every scenario.
B) 36 AI agent -> Self-driven removes the need for data quality checks and monitoring.
C) 36 AI agent -> Self-driven is correctly applied in business decisions
D) 36 AI agent -> Self-driven eliminates the need for managerial judgment and governance.

Answer: C

Explanation: 36 AI agent -> Self-driven is correctly applied in business decisions is correct because it best fits the concept of 36 ai agent -> self-driven.

37. 37 Bias affects -> Decisions?

A) 37 Bias affects -> Decisions guarantees perfect outcomes in every scenario.
B) 37 Bias affects -> Decisions is correctly applied in business decisions
C) 37 Bias affects -> Decisions eliminates the need for managerial judgment and governance.
D) 37 Bias affects -> Decisions removes the need for data quality checks and monitoring.

Answer: B

Explanation: 37 Bias affects -> Decisions is correctly applied in business decisions is correct because it best fits the concept of 37 bias affects -> decisions.

38. 38 Correlation mistake -> Wrong conclusions?

A) 38 Correlation mistake -> Wrong conclusions is correctly applied in business decisions
B) 38 Correlation mistake -> Wrong conclusions guarantees perfect outcomes in every scenario.
C) 38 Correlation mistake -> Wrong conclusions eliminates the need for managerial judgment and governance.
D) 38 Correlation mistake -> Wrong conclusions removes the need for data quality checks and monitoring.

Answer: A

Explanation: 38 Correlation mistake -> Wrong conclusions is correctly applied in business decisions is correct because it best fits the concept of 38 correlation mistake -> wrong conclusions.

39. What is 39 Governance -> Accountability?

A) What is 39 Governance -> Accountability is correctly applied in business decisions
B) What is 39 Governance -> Accountability removes the need for data quality checks and monitoring.
C) What is 39 Governance -> Accountability guarantees perfect outcomes in every scenario.
D) What is 39 Governance -> Accountability eliminates the need for managerial judgment and governance.

Answer: A

Explanation: What is 39 Governance -> Accountability is correctly applied in business decisions is correct because it best fits what is 39 governance -> accountability.

40. 40 Interpretability -> Understanding model?

A) 40 Interpretability -> Understanding model guarantees perfect outcomes in every scenario.
B) 40 Interpretability -> Understanding model eliminates the need for managerial judgment and governance.
C) 40 Interpretability -> Understanding model is correctly applied in business decisions
D) 40 Interpretability -> Understanding model removes the need for data quality checks and monitoring.

Answer: C

Explanation: 40 Interpretability -> Understanding model is correctly applied in business decisions is correct because it best fits the concept of 40 interpretability -> understanding model.

41. What is 41 Transparency -> Clarity?

A) What is 41 Transparency -> Clarity removes the need for data quality checks and monitoring.
B) What is 41 Transparency -> Clarity eliminates the need for managerial judgment and governance.
C) What is 41 Transparency -> Clarity guarantees perfect outcomes in every scenario.
D) What is 41 Transparency -> Clarity is correctly applied in business decisions

Answer: D

Explanation: What is 41 Transparency -> Clarity is correctly applied in business decisions is correct because it best fits what is 41 transparency -> clarity.

42. 42 AI agent vs assistant -> Autonomy difference?

A) 42 AI agent vs assistant -> Autonomy difference eliminates the need for managerial judgment and governance.
B) 42 AI agent vs assistant -> Autonomy difference removes the need for data quality checks and monitoring.
C) 42 AI agent vs assistant -> Autonomy difference is correctly applied in business decisions
D) 42 AI agent vs assistant -> Autonomy difference guarantees perfect outcomes in every scenario.

Answer: C

Explanation: 42 AI agent vs assistant -> Autonomy difference is correctly applied in business decisions is correct because it best fits the concept of 42 ai agent vs assistant -> autonomy difference.

43. 43 BI helps -> Decision making?

A) 43 BI helps -> Decision making is correctly applied in business decisions
B) 43 BI helps -> Decision making removes the need for data quality checks and monitoring.
C) 43 BI helps -> Decision making eliminates the need for managerial judgment and governance.
D) 43 BI helps -> Decision making guarantees perfect outcomes in every scenario.

Answer: A

Explanation: 43 BI helps -> Decision making is correctly applied in business decisions is correct because it best fits the concept of 43 bi helps -> decision making.

44. 44 Warehouse -> Central storage?

A) 44 Warehouse -> Central storage removes the need for data quality checks and monitoring.
B) 44 Warehouse -> Central storage eliminates the need for managerial judgment and governance.
C) 44 Warehouse -> Central storage guarantees perfect outcomes in every scenario.
D) 44 Warehouse -> Central storage is correctly applied in business decisions

Answer: D

Explanation: 44 Warehouse -> Central storage is correctly applied in business decisions is correct because it best fits the concept of 44 warehouse -> central storage.

45. 45 Bias reduction -> Data cleaning?

A) 45 Bias reduction -> Data cleaning is correctly applied in business decisions
B) 45 Bias reduction -> Data cleaning guarantees perfect outcomes in every scenario.
C) 45 Bias reduction -> Data cleaning removes the need for data quality checks and monitoring.
D) 45 Bias reduction -> Data cleaning eliminates the need for managerial judgment and governance.

Answer: A

Explanation: 45 Bias reduction -> Data cleaning is correctly applied in business decisions is correct because it best fits the concept of 45 bias reduction -> data cleaning.

46. 46 Explainability -> Trust building?

A) 46 Explainability -> Trust building eliminates the need for managerial judgment and governance.
B) 46 Explainability -> Trust building removes the need for data quality checks and monitoring.
C) 46 Explainability -> Trust building is correctly applied in business decisions
D) 46 Explainability -> Trust building guarantees perfect outcomes in every scenario.

Answer: C

Explanation: 46 Explainability -> Trust building is correctly applied in business decisions is correct because it best fits the concept of 46 explainability -> trust building.

47. 47 Decision types -> 3 types?

A) 47 Decision types -> 3 types guarantees perfect outcomes in every scenario.
B) 47 Decision types -> 3 types eliminates the need for managerial judgment and governance.
C) 47 Decision types -> 3 types removes the need for data quality checks and monitoring.
D) 47 Decision types -> 3 types is correctly applied in business decisions

Answer: D

Explanation: 47 Decision types -> 3 types is correctly applied in business decisions is correct because it best fits the concept of 47 decision types -> 3 types.

48. What is 48 Semi-structured -> Hybrid?

A) What is 48 Semi-structured -> Hybrid is correctly applied in business decisions
B) What is 48 Semi-structured -> Hybrid removes the need for data quality checks and monitoring.
C) What is 48 Semi-structured -> Hybrid guarantees perfect outcomes in every scenario.
D) What is 48 Semi-structured -> Hybrid eliminates the need for managerial judgment and governance.

Answer: A

Explanation: What is 48 Semi-structured -> Hybrid is correctly applied in business decisions is correct because it best fits what is 48 semi-structured -> hybrid.

49. 49 AI decisions -> Data-driven?

A) 49 AI decisions -> Data-driven is correctly applied in business decisions
B) 49 AI decisions -> Data-driven guarantees perfect outcomes in every scenario.
C) 49 AI decisions -> Data-driven eliminates the need for managerial judgment and governance.
D) 49 AI decisions -> Data-driven removes the need for data quality checks and monitoring.

Answer: A

Explanation: 49 AI decisions -> Data-driven is correctly applied in business decisions is correct because it best fits the concept of 49 ai decisions -> data-driven.

50. 50 Best practice -> Validate models?

A) 50 Best practice -> Validate models removes the need for data quality checks and monitoring.
B) 50 Best practice -> Validate models is correctly applied in business decisions
C) 50 Best practice -> Validate models guarantees perfect outcomes in every scenario.
D) 50 Best practice -> Validate models eliminates the need for managerial judgment and governance.

Answer: B

Explanation: 50 Best practice -> Validate models is correctly applied in business decisions is correct because it best fits the concept of 50 best practice -> validate models.

Online Assessment Questions

1. Operational decisions are:

A) Complex
B) Strategic
C) Routine
D) Tactical

Answer: C

Explanation: Routine is correct because it best fits the concept of operational decisions are.

2. Strategic decisions are:

A) Long-term
B) Routine
C) Simple
D) Short-term

Answer: A

Explanation: Long-term is correct because it best fits the concept of strategic decisions are.

3. Semi-structured decisions involve:

A) Judgment only
B) None
C) Both
D) Rules only

Answer: C

Explanation: Both is correct because it best fits the concept of semi-structured decisions involve.

4. Payroll is an example of:

A) Structured
B) Strategic
C) Semi
D) Unstructured

Answer: A

Explanation: Structured is correct because it best fits the concept of payroll is an example of.

5. BI systems are used for:

A) Data analysis
B) Networking
C) Coding
D) Gaming

Answer: A

Explanation: Data analysis is correct because it best fits the concept of bi systems are used for.

6. Data warehouse stores:

A) Structured data
B) Models
C) Images
D) Code

Answer: A

Explanation: Structured data is correct because it best fits the concept of data warehouse stores.

7. AI agent is:

A) Static
B) Manual
C) Reactive
D) Autonomous

Answer: D

Explanation: Autonomous is correct because it best fits the concept of ai agent is.

8. AI assistant is:

A) Autonomous
B) Self-learning
C) Independent
D) Reactive

Answer: D

Explanation: Reactive is correct because it best fits the concept of ai assistant is.

9. Correlation implies causation?

A) Always
B) Sometimes
C) Yes
D) No

Answer: D

Explanation: No is correct because it best fits the concept of correlation implies causation.

10. Anchoring bias means:

A) Random
B) Data error
C) First info influence
D) Ignore info

Answer: C

Explanation: First info influence is correct because it best fits the concept of anchoring bias means.

Week 8

Concepts

1. Feedforward Neural Networks (FNN)

Data moves only in one direction (input -> output).
No memory of past inputs.
Used for basic classification/regression.

2. Vanishing Gradient Problem

In deep networks, gradients become very small during backpropagation.
Earlier layers learn very slowly or stop learning.
Limits deep neural network training.

3. Convolutional Neural Networks (CNNs)

Designed for image data.
Use filters to detect patterns like edges, shapes.
Widely used in computer vision.

4. Transformers

Modern architecture for sequence data (text).
Uses attention mechanism instead of recurrence.
Handles long-range dependencies better.

5. Self-Attention Mechanism

Helps model understand which words are important.
Each word looks at other words in the sentence.
Captures context effectively.

6. Multi-Head Attention

Multiple attention layers run in parallel.
Each learns different relationships.
Improves model understanding.

7. Positional Encoding

Adds order information to input tokens.
Needed because transformers process data in parallel.
Without it -> model doesn't know sequence order.

8. Word Embeddings

Converts words into vectors (numbers).
Similar words have similar vectors.
Captures meaning.

9. Encoder-Decoder Architecture

Encoder processes input.
Decoder generates output.
Used in translation, summarization.

10. Fine-tuning

Adapting a pretrained model to a specific task.
Uses smaller dataset.
Saves time and improves performance.

11. Recurrent Neural Networks (RNNs)

Have memory (past inputs influence current output).
Good for sequence data (text, time series).
Uses hidden state.

12. Hidden State in RNN

Stores past information.
Passed from one time step to next.
Acts like memory.

13. Weight Sharing in RNN

Same weights used across all time steps.
Reduces complexity.
Improves generalization.

14. Encoder Block (Transformer)

Contains:
Multi-head attention
Feedforward layer
Processes input representation.

15. Decoder Block

Generates output sequence.
Uses encoder output + attention.

16. LSTM (Long Short-Term Memory)

Improved RNN to fix vanishing gradient.
Uses gates:
Input gate
Forget gate
Output gate

17. LSTM Gates

Forget gate -> what to remove
Input gate -> what to add
Output gate -> what to show

18. Output Gate (Important)

Controls what part of memory is used as output.
Important when predictions are noisy.

> *Example: In Week 8, a company can apply feedforward neural networks (fnn) together with vanishing gradient problem to improve decision quality and business outcomes.*

MCQs

1. What is Feedforward networks have?

A) Cycles
B) Feedback loops
C) Memory
D) One-way flow

Answer: D

Explanation: One-way flow is correct because it best fits what is feedforward networks have.

2. Main issue in deep networks?

A) Overfitting
B) Noise
C) Data loss
D) Vanishing gradient

Answer: D

Explanation: Vanishing gradient is correct because it best fits the concept of main issue in deep networks.

3. What is Vanishing gradient affects?

A) Input layer
B) Data
C) Early layers
D) Output layer

Answer: C

Explanation: Early layers is correct because it best fits what is vanishing gradient affects.

4. What is CNNs are best for?

A) Audio
B) Tables
C) Images
D) Text

Answer: C

Explanation: Images is correct because it best fits what is cnns are best for.

5. What is CNN detects?

A) Patterns
B) Noise
C) Numbers
D) Words

Answer: A

Explanation: Patterns is correct because it best fits what is cnn detects.

6. What is Transformers use?

A) Loops
B) Memory
C) Attention
D) Recursion

Answer: C

Explanation: Attention is correct because it best fits what is transformers use.

7. What is Attention helps?

A) Focus on important parts
B) Speed
C) Training
D) Storage

Answer: A

Explanation: Focus on important parts is correct because it best fits what is attention helps.

8. What is Multi-head attention?

A) No learning
B) One focus
C) Fixed output
D) Multiple perspectives

Answer: D

Explanation: Multiple perspectives is correct because it best fits what is multi-head attention.

9. What is Positional encoding adds?

A) Noise
B) Speed
C) Data
D) Order

Answer: D

Explanation: Order is correct because it best fits what is positional encoding adds.

10. What is Without positional encoding?

A) Better output
B) More data
C) No order info
D) Model faster

Answer: C

Explanation: No order info is correct because it best fits what is without positional encoding.

11. What is Word embeddings represent?

A) Images
B) Words as text
C) Words as vectors
D) Audio

Answer: C

Explanation: Words as vectors is correct because it best fits what is word embeddings represent.

12. What is Similar words have?

A) Same text
B) Same size
C) Same frequency
D) Similar vectors

Answer: D

Explanation: Similar vectors is correct because it best fits what is similar words have.

13. What is Encoder does?

A) Input processing
B) Data cleaning
C) Storage
D) Output generation

Answer: A

Explanation: Input processing is correct because it best fits what is encoder does.

14. What is Decoder does?

A) Training
B) Input
C) Storage
D) Output generation

Answer: D

Explanation: Output generation is correct because it best fits what is decoder does.

15. What is Fine-tuning means?

A) Data cleaning
B) Removing layers
C) Training from scratch
D) Adjust pretrained model

Answer: D

Explanation: Adjust pretrained model is correct because it best fits what is fine-tuning means.

16. What is RNN is used for?

A) Tables
B) Images only
C) Sequential data
D) Static data

Answer: C

Explanation: Sequential data is correct because it best fits what is rnn is used for.

17. What is RNN memory stored in?

A) Hidden state
B) Loss
C) Input
D) Output

Answer: A

Explanation: Hidden state is correct because it best fits what is rnn memory stored in.

18. What is Hidden state represents?

A) Errors
B) Future
C) Speed
D) Past info

Answer: D

Explanation: Past info is correct because it best fits what is hidden state represents.

19. What is Weight sharing means?

A) No weights
B) Different weights
C) Random weights
D) Same weights

Answer: D

Explanation: Same weights is correct because it best fits what is weight sharing means.

20. What is Encoder block contains?

A) CNN
B) RNN
C) Attention + FFN
D) Loss

Answer: C

Explanation: Attention + FFN is correct because it best fits what is encoder block contains.

21. What is Decoder block uses?

A) Only loss
B) Encoder output
C) No data
D) Only input

Answer: B

Explanation: Encoder output is correct because it best fits what is decoder block uses.

22. What is LSTM solves?

A) Overfitting
B) Noise
C) Vanishing gradient
D) Data loss

Answer: C

Explanation: Vanishing gradient is correct because it best fits what is lstm solves.

23. What is LSTM has?

A) 4 gates
B) 1 gate
C) 3 gates
D) 2 gates

Answer: C

Explanation: 3 gates is correct because it best fits what is lstm has.

24. What is Forget gate does?

A) Add info
B) Store
C) Remove info
D) Output

Answer: C

Explanation: Remove info is correct because it best fits what is forget gate does.

25. What is Input gate does?

A) Remove
B) Add info
C) Ignore
D) Output

Answer: B

Explanation: Add info is correct because it best fits what is input gate does.

26. What is Output gate does?

A) Control output
B) Store
C) Train
D) Remove

Answer: A

Explanation: Control output is correct because it best fits what is output gate does.

27. What is Transformer advantage?

A) Small
B) Limited context
C) Parallel processing
D) Slow

Answer: C

Explanation: Parallel processing is correct because it best fits what is transformer advantage.

28. What is Attention score uses?

A) Dot product
B) Sum
C) Subtraction
D) Division

Answer: A

Explanation: Dot product is correct because it best fits what is attention score uses.

29. What is Softmax is used to?

A) Remove
B) Divide
C) Multiply
D) Normalize

Answer: D

Explanation: Normalize is correct because it best fits what is softmax is used to.

30. What is Scaling factor in attention?

A) 1
B) dk†
C) sqrtdk
D) n

Answer: C

Explanation: sqrtdk is correct because it best fits what is scaling factor in attention.

31. What is 31-50 (continuation properly)?

A) What is 31-50 (continuation properly) eliminates the need for managerial judgment and governance.
B) What is 31-50 (continuation properly) removes the need for data quality checks and monitoring.
C) What is 31-50 (continuation properly) is correctly applied in business decisions
D) What is 31-50 (continuation properly) guarantees perfect outcomes in every scenario.

Answer: C

Explanation: What is 31-50 (continuation properly) is correctly applied in business decisions is correct because it best fits what is 31-50 (continuation properly).

32. What is CNN layers detect ->?

A) Noise
B) Words
C) Edges
D) Speed

Answer: C

Explanation: Edges is correct because it best fits what is cnn layers detect ->.

33. RNN passes info through ->?

A) Nodes
B) Layers
C) Data
D) Time steps

Answer: D

Explanation: Time steps is correct because it best fits the concept of rnn passes info through ->.

34. Transformer removes need for ->?

A) Output
B) Attention
C) Recurrence
D) Data

Answer: C

Explanation: Recurrence is correct because it best fits the concept of transformer removes need for ->.

35. What is Embeddings capture ->?

A) Loss
B) Speed
C) Syntax only
D) Meaning

Answer: D

Explanation: Meaning is correct because it best fits what is embeddings capture ->.

36. Encoder output goes to ->?

A) Decoder
B) Storage
C) Input
D) Loss

Answer: A

Explanation: Decoder is correct because it best fits the concept of encoder output goes to ->.

37. What is Fine-tuning uses ->?

A) No data
B) Big data
C) Random
D) Small dataset

Answer: D

Explanation: Small dataset is correct because it best fits what is fine-tuning uses ->.

38. What is RNN limitation ->?

A) Memory
B) Speed
C) Storage
D) Vanishing gradient

Answer: D

Explanation: Vanishing gradient is correct because it best fits what is rnn limitation ->.

39. What is LSTM improves ->?

A) Input
B) Memory retention
C) Storage
D) Speed

Answer: B

Explanation: Memory retention is correct because it best fits what is lstm improves ->.

40. What is Output gate controls ->?

A) Output
B) Memory delete
C) Input
D) Loss

Answer: A

Explanation: Output is correct because it best fits what is output gate controls ->.

41. What is Multi-head attention gives ->?

A) Single view
B) No view
C) Multiple views
D) Random

Answer: C

Explanation: Multiple views is correct because it best fits what is multi-head attention gives ->.

42. What is Transformer best for ->?

A) Tables
B) Images
C) Text
D) Audio

Answer: C

Explanation: Text is correct because it best fits what is transformer best for ->.

43. What is CNN pooling ->?

A) Add noise
B) Increase data
C) Train
D) Reduce size

Answer: D

Explanation: Reduce size is correct because it best fits what is cnn pooling ->.

44. Hidden state updated using ->?

A) None
B) Future state
C) Previous state
D) Random

Answer: C

Explanation: Previous state is correct because it best fits the concept of hidden state updated using ->.

45. What is Weight sharing helps ->?

A) Increase cost
B) Noise
C) Error
D) Reduce parameters

Answer: D

Explanation: Reduce parameters is correct because it best fits what is weight sharing helps ->.

46. What is Attention focuses on ->?

A) None
B) All tokens equally
C) Important tokens
D) Random

Answer: C

Explanation: Important tokens is correct because it best fits what is attention focuses on ->.

47. What is Decoder predicts ->?

A) Error
B) Input
C) Loss
D) Output sequence

Answer: D

Explanation: Output sequence is correct because it best fits what is decoder predicts ->.

48. What is Softmax output ->?

A) Probabilities
B) Noise
C) Raw values
D) Errors

Answer: A

Explanation: Probabilities is correct because it best fits what is softmax output ->.

49. LSTM memory cell stores ->?

A) Noise
B) Errors
C) Speed
D) Long-term info

Answer: D

Explanation: Long-term info is correct because it best fits the concept of lstm memory cell stores ->.

50. What is Transformer handles ->?

A) None
B) Short only
C) Long dependencies
D) Noise

Answer: C

Explanation: Long dependencies is correct because it best fits what is transformer handles ->.

Online Assessment Questions

1. CNN is used for:

A) Text
B) Tables
C) Images
D) Audio

Answer: C

Explanation: Images is correct because it best fits the concept of cnn is used for.

2. Transformer uses:

A) Memory
B) Recurrence
C) Attention
D) Pooling

Answer: C

Explanation: Attention is correct because it best fits the concept of transformer uses.

3. Positional encoding provides:

A) Speed
B) Noise
C) Order
D) Output

Answer: C

Explanation: Order is correct because it best fits the concept of positional encoding provides.

4. RNN has:

A) No memory
B) Static
C) No training
D) Memory

Answer: D

Explanation: Memory is correct because it best fits the concept of rnn has.

5. LSTM solves:

A) Noise
B) Vanishing gradient
C) Speed
D) Overfitting

Answer: B

Explanation: Vanishing gradient is correct because it best fits the concept of lstm solves.

6. Word embeddings represent:

A) Images
B) Vectors
C) Audio
D) Text

Answer: B

Explanation: Vectors is correct because it best fits the concept of word embeddings represent.

7. Encoder does:

A) Output
B) Training
C) Input processing
D) Storage

Answer: C

Explanation: Input processing is correct because it best fits the concept of encoder does.

8. Decoder does:

A) Storage
B) Cleaning
C) Input
D) Output generation

Answer: D

Explanation: Output generation is correct because it best fits the concept of decoder does.

9. Attention helps model:

A) Focus
B) Ignore
C) Remove
D) Add noise

Answer: A

Explanation: Focus is correct because it best fits the concept of attention helps model.

10. Multi-head attention gives:

A) Random
B) Multiple views
C) Single view
D) No learning

Answer: B

Explanation: Multiple views is correct because it best fits the concept of multi-head attention gives.

Week 9

Concepts

1. Risk Management Process

Standard flow: Risk Identification -> Risk Assessment -> Risk Control
First find risks -> then evaluate -> then take action
Used in all systems including AI

2. Risk Identification

Finding possible threats/problems
Example: system failure, bias, attack
First and most important step

3. Risk Assessment

Evaluating how serious the risk is
Based on probability + impact
Helps prioritize risks

4. Risk Control

Taking actions to reduce risk
Includes prevention, mitigation, transfer
Final step in process

5. Risk Formula (Quantitative)

Risk = Threat Probability ×Success Probability ×Exposure ×Asset Value
Shows both likelihood and impact
Used in decision-making

6. Measurement Error

Difference between estimated risk and actual risk
Comes from uncertainty or wrong assumptions
Important in risk models

7. Levels of Risk Assessment

Individual (user-level risk)
Organizational (company-level)
National (society-level)

8. Types of Threats

Malicious -> intentional attacks
Unintentional -> errors, bugs
Both are important in AI systems

9. Tangible vs Intangible Impact

Tangible -> measurable (money loss)
Intangible -> reputation, trust
Exams usually focus on tangible

10. AI System Failure Impact

Most measurable impact -> financial cost
Example: wrong loan approvals, fraud loss

11. Threat Categories

Malicious (attacks)
Unintentional (mistakes)
Core classification in security

12. Malicious Attacks in ML

Designed to harm models
Includes:
Data poisoning
Adversarial attacks

13. Data Poisoning

Injecting bad data during training
Model learns wrong patterns
Happens before deployment

14. Adversarial Attacks

Modify input slightly to fool model
Happens during runtime
Causes wrong predictions

15. Adversarial Inputs

Small changes invisible to humans
Big impact on AI output
Example: image misclassification

16. Environmental Risks of AI

Training models consumes huge energy
Produces carbon emissions
Important ethical concern

17. Training Data Issues

AI trained on internet data
May not represent all populations
Causes bias

18. Lack of Diversity in Data

Some groups underrepresented
Leads to unfair outcomes
Key limitation of AI systems

19. AI Bias

Comes from:
Data
Labels
Design
Can cause discrimination

20. Risk in AI Systems

Comes from:

Technical issues

Data issues

Human misuse

Example: In Week 9, a company can apply risk management process together with risk identification to improve decision quality and business outcomes.

MCQs

1. First step in risk management?

A) Risk mitigation
B) Risk identification
C) Risk assessment
D) Risk control

Answer: B

Explanation: Risk identification is correct because it best fits the concept of first step in risk management.

2. Correct sequence of risk process?

A) Identification -> Assessment -> Control
B) Assessment -> Identification -> Control
C) Control -> Assessment -> Identification
D) Assessment -> Control -> Identification

Answer: A

Explanation: Identification -> Assessment -> Control is correct because it best fits the concept of correct sequence of risk process.

3. What is Risk assessment focuses on?

A) Coding
B) Detection
C) Removal
D) Evaluation

Answer: D

Explanation: Evaluation is correct because it best fits what is risk assessment focuses on.

4. What is Risk control means?

A) Measuring risks
B) Ignoring risks
C) Reducing risks
D) Finding risks

Answer: C

Explanation: Reducing risks is correct because it best fits what is risk control means.

5. What is Risk formula includes?

A) Only probability
B) Only cost
C) Both probability and impact
D) Only impact

Answer: C

Explanation: Both probability and impact is correct because it best fits what is risk formula includes.

6. What is Measurement error refers to?

A) Uncertainty in estimation
B) Speed
C) Model size
D) Data size

Answer: A

Explanation: Uncertainty in estimation is correct because it best fits what is measurement error refers to.

7. What is Risk levels include?

A) Individual, organizational, national
B) Only company
C) Local only
D) Only global

Answer: A

Explanation: Individual, organizational, national is correct because it best fits what is risk levels include.

8. What is Malicious threats are?

A) Random
B) Accidental
C) Natural
D) Intentional

Answer: D

Explanation: Intentional is correct because it best fits what is malicious threats are.

9. What is Unintentional threats are?

A) Designed
B) Accidental
C) Planned
D) Malicious

Answer: B

Explanation: Accidental is correct because it best fits what is unintentional threats are.

10. What is Tangible impact means?

A) Emotional
B) Financial
C) Ethical
D) Social

Answer: B

Explanation: Financial is correct because it best fits what is tangible impact means.

11. What is Example of tangible impact?

A) Trust loss
B) Financial loss
C) Reputation
D) Stress

Answer: B

Explanation: Financial loss is correct because it best fits what is example of tangible impact.

12. What is Data poisoning occurs during?

A) Training
B) Testing
C) Deployment
D) Output

Answer: A

Explanation: Training is correct because it best fits what is data poisoning occurs during.

13. What is Adversarial attack occurs during?

A) Runtime
B) Cleaning
C) Training
D) Storage

Answer: A

Explanation: Runtime is correct because it best fits what is adversarial attack occurs during.

14. What is Adversarial input is?

A) Training data
B) Labels
C) Clean data
D) Modified input

Answer: D

Explanation: Modified input is correct because it best fits what is adversarial input is.

15. What is AI environmental issue?

A) Accuracy
B) Storage
C) Speed
D) Energy consumption

Answer: D

Explanation: Energy consumption is correct because it best fits what is ai environmental issue.

16. AI bias mainly comes from?

A) RAM
B) GPU
C) CPU
D) Data

Answer: D

Explanation: Data is correct because it best fits the concept of ai bias mainly comes from.

17. What is Training data limitation?

A) Too clean
B) Lack of diversity
C) Too large
D) Too fast

Answer: B

Explanation: Lack of diversity is correct because it best fits what is training data limitation.

18. What is Risk?

A) Speed
B) Data
C) Probability ×Impact
D) Cost

Answer: C

Explanation: Probability ×Impact is correct because it best fits what is risk.

19. What is Threat probability refers to?

A) Data
B) Chance of attack
C) Loss
D) Cost

Answer: B

Explanation: Chance of attack is correct because it best fits what is threat probability refers to.

20. What is Exposure means?

A) Output
B) Speed
C) Data
D) System vulnerability

Answer: D

Explanation: System vulnerability is correct because it best fits what is exposure means.

21. What is Asset value refers to?

A) Model
B) Data size
C) Importance
D) Speed

Answer: C

Explanation: Importance is correct because it best fits what is asset value refers to.

22. What is AI failure leads to?

A) Financial loss
B) No impact
C) Only errors
D) Speed loss

Answer: A

Explanation: Financial loss is correct because it best fits what is ai failure leads to.

23. What is Malicious attack example?

A) Noise
B) Bug
C) Data poisoning
D) Error

Answer: C

Explanation: Data poisoning is correct because it best fits what is malicious attack example.

24. What is Unintentional threat example?

A) Attack
B) Malware
C) Hack
D) Bug

Answer: D

Explanation: Bug is correct because it best fits what is unintentional threat example.

25. What is Risk control includes?

A) Data
B) Mitigation
C) Training
D) Ignore

Answer: B

Explanation: Mitigation is correct because it best fits what is risk control includes.

26. 26 Risk assessment -> Evaluate severity?

A) 26 Risk assessment -> Evaluate severity is correctly applied in business decisions
B) 26 Risk assessment -> Evaluate severity eliminates the need for managerial judgment and governance.
C) 26 Risk assessment -> Evaluate severity guarantees perfect outcomes in every scenario.
D) 26 Risk assessment -> Evaluate severity removes the need for data quality checks and monitoring.

Answer: A

Explanation: 26 Risk assessment -> Evaluate severity is correctly applied in business decisions is correct because it best fits the concept of 26 risk assessment -> evaluate severity.

27. 27 Identification -> Find risks?

A) 27 Identification -> Find risks removes the need for data quality checks and monitoring.
B) 27 Identification -> Find risks eliminates the need for managerial judgment and governance.
C) 27 Identification -> Find risks guarantees perfect outcomes in every scenario.
D) 27 Identification -> Find risks is correctly applied in business decisions

Answer: D

Explanation: 27 Identification -> Find risks is correctly applied in business decisions is correct because it best fits the concept of 27 identification -> find risks.

28. 28 Control -> Reduce risks?

A) 28 Control -> Reduce risks guarantees perfect outcomes in every scenario.
B) 28 Control -> Reduce risks removes the need for data quality checks and monitoring.
C) 28 Control -> Reduce risks is correctly applied in business decisions
D) 28 Control -> Reduce risks eliminates the need for managerial judgment and governance.

Answer: C

Explanation: 28 Control -> Reduce risks is correctly applied in business decisions is correct because it best fits the concept of 28 control -> reduce risks.

29. 29 Adversarial -> Runtime attack?

A) 29 Adversarial -> Runtime attack guarantees perfect outcomes in every scenario.
B) 29 Adversarial -> Runtime attack is correctly applied in business decisions
C) 29 Adversarial -> Runtime attack eliminates the need for managerial judgment and governance.
D) 29 Adversarial -> Runtime attack removes the need for data quality checks and monitoring.

Answer: B

Explanation: 29 Adversarial -> Runtime attack is correctly applied in business decisions is correct because it best fits the concept of 29 adversarial -> runtime attack.

30. 30 Poisoning -> Training attack?

A) 30 Poisoning -> Training attack is correctly applied in business decisions
B) 30 Poisoning -> Training attack eliminates the need for managerial judgment and governance.
C) 30 Poisoning -> Training attack removes the need for data quality checks and monitoring.
D) 30 Poisoning -> Training attack guarantees perfect outcomes in every scenario.

Answer: A

Explanation: 30 Poisoning -> Training attack is correctly applied in business decisions is correct because it best fits the concept of 30 poisoning -> training attack.

31. 31 Bias -> Unfair outcomes?

A) 31 Bias -> Unfair outcomes eliminates the need for managerial judgment and governance.
B) 31 Bias -> Unfair outcomes is correctly applied in business decisions
C) 31 Bias -> Unfair outcomes removes the need for data quality checks and monitoring.
D) 31 Bias -> Unfair outcomes guarantees perfect outcomes in every scenario.

Answer: B

Explanation: 31 Bias -> Unfair outcomes is correctly applied in business decisions is correct because it best fits the concept of 31 bias -> unfair outcomes.

32. 32 Environmental risk -> Carbon emissions?

A) 32 Environmental risk -> Carbon emissions eliminates the need for managerial judgment and governance.
B) 32 Environmental risk -> Carbon emissions is correctly applied in business decisions
C) 32 Environmental risk -> Carbon emissions guarantees perfect outcomes in every scenario.
D) 32 Environmental risk -> Carbon emissions removes the need for data quality checks and monitoring.

Answer: B

Explanation: 32 Environmental risk -> Carbon emissions is correctly applied in business decisions is correct because it best fits the concept of 32 environmental risk -> carbon emissions.

33. 33 AI risk source -> Data + system?

A) 33 AI risk source -> Data + system eliminates the need for managerial judgment and governance.
B) 33 AI risk source -> Data + system is correctly applied in business decisions
C) 33 AI risk source -> Data + system removes the need for data quality checks and monitoring.
D) 33 AI risk source -> Data + system guarantees perfect outcomes in every scenario.

Answer: B

Explanation: 33 AI risk source -> Data + system is correctly applied in business decisions is correct because it best fits the concept of 33 ai risk source -> data + system.

34. 34 Measurement error -> Uncertainty?

A) 34 Measurement error -> Uncertainty eliminates the need for managerial judgment and governance.
B) 34 Measurement error -> Uncertainty removes the need for data quality checks and monitoring.
C) 34 Measurement error -> Uncertainty is correctly applied in business decisions
D) 34 Measurement error -> Uncertainty guarantees perfect outcomes in every scenario.

Answer: C

Explanation: 34 Measurement error -> Uncertainty is correctly applied in business decisions is correct because it best fits the concept of 34 measurement error -> uncertainty.

35. 35 Threat types -> 2 types?

A) 35 Threat types -> 2 types is correctly applied in business decisions
B) 35 Threat types -> 2 types removes the need for data quality checks and monitoring.
C) 35 Threat types -> 2 types guarantees perfect outcomes in every scenario.
D) 35 Threat types -> 2 types eliminates the need for managerial judgment and governance.

Answer: A

Explanation: 35 Threat types -> 2 types is correctly applied in business decisions is correct because it best fits the concept of 35 threat types -> 2 types.

36. 36 Malicious -> Intentional harm?

A) 36 Malicious -> Intentional harm removes the need for data quality checks and monitoring.
B) 36 Malicious -> Intentional harm guarantees perfect outcomes in every scenario.
C) 36 Malicious -> Intentional harm is correctly applied in business decisions
D) 36 Malicious -> Intentional harm eliminates the need for managerial judgment and governance.

Answer: C

Explanation: 36 Malicious -> Intentional harm is correctly applied in business decisions is correct because it best fits the concept of 36 malicious -> intentional harm.

37. 37 Unintentional -> Accidental harm?

A) 37 Unintentional -> Accidental harm is correctly applied in business decisions
B) 37 Unintentional -> Accidental harm eliminates the need for managerial judgment and governance.
C) 37 Unintentional -> Accidental harm guarantees perfect outcomes in every scenario.
D) 37 Unintentional -> Accidental harm removes the need for data quality checks and monitoring.

Answer: A

Explanation: 37 Unintentional -> Accidental harm is correctly applied in business decisions is correct because it best fits the concept of 37 unintentional -> accidental harm.

38. What is 38 Tangible -> Measurable?

A) What is 38 Tangible -> Measurable eliminates the need for managerial judgment and governance.
B) What is 38 Tangible -> Measurable removes the need for data quality checks and monitoring.
C) What is 38 Tangible -> Measurable is correctly applied in business decisions
D) What is 38 Tangible -> Measurable guarantees perfect outcomes in every scenario.

Answer: C

Explanation: What is 38 Tangible -> Measurable is correctly applied in business decisions is correct because it best fits what is 38 tangible -> measurable.

39. What is 39 Intangible -> Non-measurable?

A) What is 39 Intangible -> Non-measurable guarantees perfect outcomes in every scenario.
B) What is 39 Intangible -> Non-measurable is correctly applied in business decisions
C) What is 39 Intangible -> Non-measurable eliminates the need for managerial judgment and governance.
D) What is 39 Intangible -> Non-measurable removes the need for data quality checks and monitoring.

Answer: B

Explanation: What is 39 Intangible -> Non-measurable is correctly applied in business decisions is correct because it best fits what is 39 intangible -> non-measurable.

40. 40 AI failure -> Cost impact?

A) 40 AI failure -> Cost impact guarantees perfect outcomes in every scenario.
B) 40 AI failure -> Cost impact is correctly applied in business decisions
C) 40 AI failure -> Cost impact eliminates the need for managerial judgment and governance.
D) 40 AI failure -> Cost impact removes the need for data quality checks and monitoring.

Answer: B

Explanation: 40 AI failure -> Cost impact is correctly applied in business decisions is correct because it best fits the concept of 40 ai failure -> cost impact.

41. 41 Risk formula -> Multiplicative?

A) 41 Risk formula -> Multiplicative guarantees perfect outcomes in every scenario.
B) 41 Risk formula -> Multiplicative removes the need for data quality checks and monitoring.
C) 41 Risk formula -> Multiplicative eliminates the need for managerial judgment and governance.
D) 41 Risk formula -> Multiplicative is correctly applied in business decisions

Answer: D

Explanation: 41 Risk formula -> Multiplicative is correctly applied in business decisions is correct because it best fits the concept of 41 risk formula -> multiplicative.

42. 42 Exposure -> System weakness?

A) 42 Exposure -> System weakness is correctly applied in business decisions
B) 42 Exposure -> System weakness guarantees perfect outcomes in every scenario.
C) 42 Exposure -> System weakness removes the need for data quality checks and monitoring.
D) 42 Exposure -> System weakness eliminates the need for managerial judgment and governance.

Answer: A

Explanation: 42 Exposure -> System weakness is correctly applied in business decisions is correct because it best fits the concept of 42 exposure -> system weakness.

43. 43 Asset value -> Importance?

A) 43 Asset value -> Importance guarantees perfect outcomes in every scenario.
B) 43 Asset value -> Importance removes the need for data quality checks and monitoring.
C) 43 Asset value -> Importance eliminates the need for managerial judgment and governance.
D) 43 Asset value -> Importance is correctly applied in business decisions

Answer: D

Explanation: 43 Asset value -> Importance is correctly applied in business decisions is correct because it best fits the concept of 43 asset value -> importance.

44. 44 Diversity issue -> Bias risk?

A) 44 Diversity issue -> Bias risk is correctly applied in business decisions
B) 44 Diversity issue -> Bias risk eliminates the need for managerial judgment and governance.
C) 44 Diversity issue -> Bias risk removes the need for data quality checks and monitoring.
D) 44 Diversity issue -> Bias risk guarantees perfect outcomes in every scenario.

Answer: A

Explanation: 44 Diversity issue -> Bias risk is correctly applied in business decisions is correct because it best fits the concept of 44 diversity issue -> bias risk.

45. 45 Data source -> Internet?

A) 45 Data source -> Internet is correctly applied in business decisions
B) 45 Data source -> Internet removes the need for data quality checks and monitoring.
C) 45 Data source -> Internet eliminates the need for managerial judgment and governance.
D) 45 Data source -> Internet guarantees perfect outcomes in every scenario.

Answer: A

Explanation: 45 Data source -> Internet is correctly applied in business decisions is correct because it best fits the concept of 45 data source -> internet.

46. 46 Attack goal -> Wrong output?

A) 46 Attack goal -> Wrong output guarantees perfect outcomes in every scenario.
B) 46 Attack goal -> Wrong output is correctly applied in business decisions
C) 46 Attack goal -> Wrong output eliminates the need for managerial judgment and governance.
D) 46 Attack goal -> Wrong output removes the need for data quality checks and monitoring.

Answer: B

Explanation: 46 Attack goal -> Wrong output is correctly applied in business decisions is correct because it best fits the concept of 46 attack goal -> wrong output.

47. 47 Risk model -> Estimate risk?

A) 47 Risk model -> Estimate risk removes the need for data quality checks and monitoring.
B) 47 Risk model -> Estimate risk guarantees perfect outcomes in every scenario.
C) 47 Risk model -> Estimate risk eliminates the need for managerial judgment and governance.
D) 47 Risk model -> Estimate risk is correctly applied in business decisions

Answer: D

Explanation: 47 Risk model -> Estimate risk is correctly applied in business decisions is correct because it best fits the concept of 47 risk model -> estimate risk.

48. 48 AI risk -> Multi-source?

A) 48 AI risk -> Multi-source guarantees perfect outcomes in every scenario.
B) 48 AI risk -> Multi-source is correctly applied in business decisions
C) 48 AI risk -> Multi-source eliminates the need for managerial judgment and governance.
D) 48 AI risk -> Multi-source removes the need for data quality checks and monitoring.

Answer: B

Explanation: 48 AI risk -> Multi-source is correctly applied in business decisions is correct because it best fits the concept of 48 ai risk -> multi-source.

49. 49 Security focus -> Threat handling?

A) 49 Security focus -> Threat handling is correctly applied in business decisions
B) 49 Security focus -> Threat handling eliminates the need for managerial judgment and governance.
C) 49 Security focus -> Threat handling guarantees perfect outcomes in every scenario.
D) 49 Security focus -> Threat handling removes the need for data quality checks and monitoring.

Answer: A

Explanation: 49 Security focus -> Threat handling is correctly applied in business decisions is correct because it best fits the concept of 49 security focus -> threat handling.

50. 50 Best practice -> Assess + control?

A) 50 Best practice -> Assess + control eliminates the need for managerial judgment and governance.
B) 50 Best practice -> Assess + control removes the need for data quality checks and monitoring.
C) 50 Best practice -> Assess + control is correctly applied in business decisions
D) 50 Best practice -> Assess + control guarantees perfect outcomes in every scenario.

Answer: C

Explanation: 50 Best practice -> Assess + control is correctly applied in business decisions is correct because it best fits the concept of 50 best practice -> assess + control.

Online Assessment Questions

1. First step in risk management:

A) Mitigation
B) Assessment
C) Control
D) Identification

Answer: D

Explanation: Identification is correct because it best fits the concept of first step in risk management.

2. Risk formula includes:

A) Both
B) Probability
C) None
D) Impact

Answer: A

Explanation: Both is correct because it best fits the concept of risk formula includes.

3. Data poisoning occurs during:

A) Output
B) Training
C) Testing
D) Deployment

Answer: B

Explanation: Training is correct because it best fits the concept of data poisoning occurs during.

4. Adversarial attack occurs during:

A) Runtime
B) Storage
C) Training
D) Cleaning

Answer: A

Explanation: Runtime is correct because it best fits the concept of adversarial attack occurs during.

5. Tangible impact refers to:

A) Ethical
B) Financial
C) Reputation
D) Social

Answer: B

Explanation: Financial is correct because it best fits the concept of tangible impact refers to.

6. Malicious threat is:

A) Natural
B) Accidental
C) Random
D) Intentional

Answer: D

Explanation: Intentional is correct because it best fits the concept of malicious threat is.

7. Unintentional threat is:

A) Planned
B) Malicious
C) Designed
D) Accidental

Answer: D

Explanation: Accidental is correct because it best fits the concept of unintentional threat is.

8. AI environmental issue:

A) Storage
B) Accuracy
C) Energy use
D) Speed

Answer: C

Explanation: Energy use is correct because it best fits the concept of ai environmental issue.

9. Bias mainly comes from:

A) Data
B) GPU
C) RAM
D) CPU

Answer: A

Explanation: Data is correct because it best fits the concept of bias mainly comes from.

10. Risk control means:

A) Evaluate
B) Reduce
C) Find
D) Ignore

Answer: B

Explanation: Reduce is correct because it best fits the concept of risk control means.

Week 10

Concepts

1. Sources of Bias in AI

Bias can enter at multiple stages of AI pipeline.
Not just model -> also data, labeling, and deployment.
Important to identify where bias originates.

2. Historical Bias

Exists in real-world data before AI.
AI learns existing social inequalities.
Example: hiring bias in past data.

3. Representation Bias

Occurs when dataset is not representative.
Some groups under/over-represented.
Leads to unfair predictions.

4. Measurement Bias

Happens during data collection/labeling.
Labels may be inaccurate or subjective.
Affects model learning.

5. Aggregation Bias

One model applied to diverse groups.
Assumes same behavior for all.
Reduces fairness.

6. Deployment Bias

Occurs when model used in wrong context.
Humans may misinterpret outputs.
Happens after model is built.

7. "More Data" Myth

More data doesn’t always fix bias.
If labels are biased -> problem remains.
Data quality matters more than quantity.

8. Annotator Bias

Human labelers introduce bias.
Their opinions affect dataset.
Impacts model behavior.

9. Robustness in AI

Ability to perform well in different conditions.
Includes:
Different users
Different environments

Important for real-world systems.

10. Overfitting

Model learns noise instead of pattern.
Performs well on training but ????? on real data.
Causes poor generalization.

11. Spurious Correlation

False relationships in data.
Model learns wrong patterns.
Leads to incorrect predictions.

12. AI Hallucination

Model generates false but confident output.
Happens in LLMs.
Caused by poor training or complexity.

13. Jailbreaking

User tricks AI to bypass restrictions.
Exploits model weaknesses.
Security issue.

14. Prompt Injection

Malicious instructions hidden in input.
Model follows wrong instructions.
Dangerous in AI systems.

15. Indirect Prompt Injection

Attack hidden in external data (emails, docs).
Model unknowingly executes it.
Hard to detect.

16. Model Stealing

Attacker extracts model behavior.
Uses repeated queries.
Steals intellectual property.

17. Adversarial Attacks (Revisited)

Small input change -> wrong output.
Exploits model sensitivity.

18. Data Poisoning (Revisited)

Training data manipulated.
Model learns incorrect patterns.

19. Bias vs Variance Tradeoff

Bias -> oversimplified model
Variance -> overfitting
Need balance for best performance

20. Explainability Importance

Required for:
Trust

Regulation
Debugging
Especially in critical systems

Example: In Week 10, a company can apply sources of bias in ai together with historical bias to improve decision quality and business outcomes.

MCQs

1. What is Bias can enter at?

A) Only output
B) Only model
C) Only data
D) Multiple stages

Answer: D

Explanation: Multiple stages is correct because it best fits what is bias can enter at.

2. What is Historical bias comes from?

A) Real world
B) Testing
C) Data cleaning
D) Model

Answer: A

Explanation: Real world is correct because it best fits what is historical bias comes from.

3. What is Representation bias means?

A) Perfect data
B) Fast data
C) Unbalanced dataset
D) Clean data

Answer: C

Explanation: Unbalanced dataset is correct because it best fits what is representation bias means.

4. What is Measurement bias occurs during?

A) Testing
B) Training
C) Labeling
D) Deployment

Answer: C

Explanation: Labeling is correct because it best fits what is measurement bias occurs during.

5. What is Aggregation bias means?

A) No model
B) Data split
C) One model for all
D) Many models

Answer: C

Explanation: One model for all is correct because it best fits what is aggregation bias means.

6. What is Deployment bias occurs during?

A) Usage
B) Data cleaning
C) Storage
D) Training

Answer: A

Explanation: Usage is correct because it best fits what is deployment bias occurs during.

7. More data always fixes bias?

A) Only AI
B) No
C) Yes
D) Sometimes

Answer: B

Explanation: No is correct because it best fits the concept of more data always fixes bias.

8. What is Annotator bias comes from?

A) Machine
B) Human labelers
C) Code
D) Data

Answer: B

Explanation: Human labelers is correct because it best fits what is annotator bias comes from.

9. What is Robustness means?

A) Cost
B) Speed
C) Performance across conditions
D) Size

Answer: C

Explanation: Performance across conditions is correct because it best fits what is robustness means.

10. What is Overfitting means?

A) Data cleaning
B) Speed
C) Memorizing noise
D) Under learning

Answer: C

Explanation: Memorizing noise is correct because it best fits what is overfitting means.

11. What is Spurious correlation is?

A) Data
B) True relation
C) False relation
D) Strong relation

Answer: C

Explanation: False relation is correct because it best fits what is spurious correlation is.

12. What is AI hallucination means?

A) Fake output
B) Correct output
C) Slow output
D) No output

Answer: A

Explanation: Fake output is correct because it best fits what is ai hallucination means.

13. What is Jailbreaking is?

A) Training
B) Data cleaning
C) Storage
D) Security bypass

Answer: D

Explanation: Security bypass is correct because it best fits what is jailbreaking is.

14. What is Prompt injection is?

A) Malicious input
B) Output
C) Clean input
D) Data

Answer: A

Explanation: Malicious input is correct because it best fits what is prompt injection is.

15. What is Indirect injection occurs via?

A) Output
B) Training
C) External data
D) Model

Answer: C

Explanation: External data is correct because it best fits what is indirect injection occurs via.

16. What is Model stealing aims to?

A) Train model
B) Copy model
C) Improve model
D) Delete model

Answer: B

Explanation: Copy model is correct because it best fits what is model stealing aims to.

17. What is Adversarial attack changes?

A) Output
B) Model
C) Input
D) Loss

Answer: C

Explanation: Input is correct because it best fits what is adversarial attack changes.

18. What is Data poisoning affects?

A) Training
B) Testing
C) Output
D) Storage

Answer: A

Explanation: Training is correct because it best fits what is data poisoning affects.

19. Bias vs variance tradeoff balances?

A) Accuracy
B) Speed
C) Data
D) Underfit & overfit

Answer: D

Explanation: Underfit & overfit is correct because it best fits the concept of bias vs variance tradeoff balances.

20. What is High bias means?

A) Underfitting
B) Fast
C) Overfitting
D) Perfect

Answer: A

Explanation: Underfitting is correct because it best fits what is high bias means.

21. What is High variance means?

A) Overfit
B) Stable
C) Perfect
D) Underfit

Answer: A

Explanation: Overfit is correct because it best fits what is high variance means.

22. What is Explainability helps?

A) Storage
B) Trust
C) Speed
D) Cost

Answer: B

Explanation: Trust is correct because it best fits what is explainability helps.

23. What is Bias from sampling stage?

A) Representation
B) Deployment
C) Measurement
D) Historical

Answer: A

Explanation: Representation is correct because it best fits what is bias from sampling stage.

24. What is Bias from labeling stage?

A) Measurement
B) Deployment
C) Historical
D) Representation

Answer: A

Explanation: Measurement is correct because it best fits what is bias from labeling stage.

25. What is Bias from model design?

A) Measurement
B) Deployment
C) Aggregation
D) Historical

Answer: C

Explanation: Aggregation is correct because it best fits what is bias from model design.

26. What is 26-50 (complete properly)?

A) What is 26-50 (complete properly) guarantees perfect outcomes in every scenario.
B) What is 26-50 (complete properly) eliminates the need for managerial judgment and governance.
C) What is 26-50 (complete properly) removes the need for data quality checks and monitoring.
D) What is 26-50 (complete properly) is correctly applied in business decisions

Answer: D

Explanation: What is 26-50 (complete properly) is correctly applied in business decisions is correct because it best fits what is 26-50 (complete properly).

27. What is Deployment bias source?

A) What is Deployment bias source guarantees perfect outcomes in every scenario.
B) What is Deployment bias source removes the need for data quality checks and monitoring.
C) What is Deployment bias source eliminates the need for managerial judgment and governance.
D) Usage

Answer: D

Explanation: Usage is correct because it best fits what is deployment bias source.

28. What is Spurious pattern?

A) What is Spurious pattern removes the need for data quality checks and monitoring.
B) False
C) What is Spurious pattern eliminates the need for managerial judgment and governance.
D) What is Spurious pattern guarantees perfect outcomes in every scenario.

Answer: B

Explanation: False is correct because it best fits what is spurious pattern.

29. What is Robust model?

A) Stable
B) What is Robust model eliminates the need for managerial judgment and governance.
C) What is Robust model guarantees perfect outcomes in every scenario.
D) What is Robust model removes the need for data quality checks and monitoring.

Answer: A

Explanation: Stable is correct because it best fits what is robust model.

30. What is Hallucination risk?

A) What is Hallucination risk eliminates the need for managerial judgment and governance.
B) What is Hallucination risk removes the need for data quality checks and monitoring.
C) What is Hallucination risk guarantees perfect outcomes in every scenario.
D) LLM

Answer: D

Explanation: LLM is correct because it best fits what is hallucination risk.

31. What is Prompt injection?

A) Security risk
B) What is Prompt injection guarantees perfect outcomes in every scenario.
C) What is Prompt injection removes the need for data quality checks and monitoring.
D) What is Prompt injection eliminates the need for managerial judgment and governance.

Answer: A

Explanation: Security risk is correct because it best fits what is prompt injection.

32. What is Model stealing?

A) IP theft
B) What is Model stealing removes the need for data quality checks and monitoring.
C) What is Model stealing eliminates the need for managerial judgment and governance.
D) What is Model stealing guarantees perfect outcomes in every scenario.

Answer: A

Explanation: IP theft is correct because it best fits what is model stealing.

33. What is Bias affects?

A) Fairness
B) What is Bias affects eliminates the need for managerial judgment and governance.
C) What is Bias affects removes the need for data quality checks and monitoring.
D) What is Bias affects guarantees perfect outcomes in every scenario.

Answer: A

Explanation: Fairness is correct because it best fits what is bias affects.

34. What is Data quality >?

A) What is Data quality > removes the need for data quality checks and monitoring.
B) What is Data quality > guarantees perfect outcomes in every scenario.
C) What is Data quality > eliminates the need for managerial judgment and governance.
D) Quantity

Answer: D

Explanation: Quantity is correct because it best fits what is data quality >.

35. What is Overfitting?

A) What is Overfitting removes the need for data quality checks and monitoring.
B) What is Overfitting guarantees perfect outcomes in every scenario.
C) What is Overfitting eliminates the need for managerial judgment and governance.
D) Poor generalization

Answer: D

Explanation: Poor generalization is correct because it best fits what is overfitting.

36. What is Underfitting?

A) High bias
B) What is Underfitting removes the need for data quality checks and monitoring.
C) What is Underfitting eliminates the need for managerial judgment and governance.
D) What is Underfitting guarantees perfect outcomes in every scenario.

Answer: A

Explanation: High bias is correct because it best fits what is underfitting.

37. What is Variance?

A) Sensitivity
B) What is Variance eliminates the need for managerial judgment and governance.
C) What is Variance guarantees perfect outcomes in every scenario.
D) What is Variance removes the need for data quality checks and monitoring.

Answer: A

Explanation: Sensitivity is correct because it best fits what is variance.

38. What is Annotation?

A) What is Annotation eliminates the need for managerial judgment and governance.
B) What is Annotation guarantees perfect outcomes in every scenario.
C) Labeling
D) What is Annotation removes the need for data quality checks and monitoring.

Answer: C

Explanation: Labeling is correct because it best fits what is annotation.

39. What is Bias pipeline?

A) What is Bias pipeline removes the need for data quality checks and monitoring.
B) What is Bias pipeline eliminates the need for managerial judgment and governance.
C) Multi-stage
D) What is Bias pipeline guarantees perfect outcomes in every scenario.

Answer: A

Explanation: What is Bias pipeline removes the need for data quality checks and monitoring.
is correct because it best fits what is bias pipeline.

40. What is Hallucination?

A) What is Hallucination guarantees perfect outcomes in every scenario.
B) What is Hallucination eliminates the need for managerial judgment and governance.
C) What is Hallucination removes the need for data quality checks and monitoring.
D) Wrong output

Answer: D

Explanation: Wrong output is correct because it best fits what is hallucination.

41. What is Injection?

A) What is Injection eliminates the need for managerial judgment and governance.
B) What is Injection guarantees perfect outcomes in every scenario.
C) Malicious
D) What is Injection removes the need for data quality checks and monitoring.

Answer: C

Explanation: Malicious is correct because it best fits what is injection.

42. What is External attack?

A) What is External attack guarantees perfect outcomes in every scenario.
B) What is External attack removes the need for data quality checks and monitoring.
C) What is External attack eliminates the need for managerial judgment and governance.
D) Indirect injection

Answer: D

Explanation: Indirect injection is correct because it best fits what is external attack.

43. What is Data poisoning?

A) What is Data poisoning eliminates the need for managerial judgment and governance.
B) What is Data poisoning removes the need for data quality checks and monitoring.
C) Training phase
D) What is Data poisoning guarantees perfect outcomes in every scenario.

Answer: C

Explanation: Training phase is correct because it best fits what is data poisoning.

44. What is Adversarial?

A) Runtime
B) What is Adversarial removes the need for data quality checks and monitoring.
C) What is Adversarial eliminates the need for managerial judgment and governance.
D) What is Adversarial guarantees perfect outcomes in every scenario.

Answer: A

Explanation: Runtime is correct because it best fits what is adversarial.

45. What is Explainability?

A) What is Explainability guarantees perfect outcomes in every scenario.
B) What is Explainability eliminates the need for managerial judgment and governance.
C) What is Explainability removes the need for data quality checks and monitoring.
D) Transparency

Answer: D

Explanation: Transparency is correct because it best fits what is explainability.

46. What is Bias fix?

A) Data improvement
B) What is Bias fix eliminates the need for managerial judgment and governance.
C) What is Bias fix removes the need for data quality checks and monitoring.
D) What is Bias fix guarantees perfect outcomes in every scenario.

Answer: A

Explanation: Data improvement is correct because it best fits what is bias fix.

47. What is Representation issue?

A) What is Representation issue eliminates the need for managerial judgment and governance.
B) What is Representation issue guarantees perfect outcomes in every scenario.
C) Sampling
D) What is Representation issue removes the need for data quality checks and monitoring.

Answer: C

Explanation: Sampling is correct because it best fits what is representation issue.

48. What is Aggregation?

A) What is Aggregation removes the need for data quality checks and monitoring.
B) What is Aggregation eliminates the need for managerial judgment and governance.
C) What is Aggregation guarantees perfect outcomes in every scenario.
D) One model

Answer: D

Explanation: One model is correct because it best fits what is aggregation.

49. What is Deployment?

A) What is Deployment removes the need for data quality checks and monitoring.
B) What is Deployment eliminates the need for managerial judgment and governance.
C) Usage stage
D) What is Deployment guarantees perfect outcomes in every scenario.

Answer: C

Explanation: Usage stage is correct because it best fits what is deployment.

50. What is Robustness?

A) Generalization
B) What is Robustness guarantees perfect outcomes in every scenario.
C) What is Robustness eliminates the need for managerial judgment and governance.
D) What is Robustness removes the need for data quality checks and monitoring.

Answer: A

Explanation: Generalization is correct because it best fits what is robustness.

Online Assessment Questions

1. Representation bias occurs due to:

A) Speed
B) Balanced data
C) Sampling issue
D) Storage

Answer: C

Explanation: Sampling issue is correct because it best fits the concept of representation bias occurs due to.

2. Measurement bias occurs during:

A) Labeling
B) Deployment
C) Testing
D) Training

Answer: A

Explanation: Labeling is correct because it best fits the concept of measurement bias occurs during.

3. Aggregation bias means:

A) One model for all
B) Multiple models
C) No model
D) Data removal

Answer: A

Explanation: One model for all is correct because it best fits the concept of aggregation bias means.

4. Overfitting means:

A) Memorizing noise
B) Speed
C) Under learning
D) Perfect learning

Answer: A

Explanation: Memorizing noise is correct because it best fits the concept of overfitting means.

5. AI hallucination means:

A) Correct output
B) No output
C) False output
D) Fast output

Answer: C

Explanation: False output is correct because it best fits the concept of ai hallucination means.

6. Prompt injection is:

A) Output
B) Malicious input
C) Clean data
D) Safe input

Answer: B

Explanation: Malicious input is correct because it best fits the concept of prompt injection is.

7. Model stealing means:

A) Copying model
B) Storage
C) Training
D) Cleaning

Answer: A

Explanation: Copying model is correct because it best fits the concept of model stealing means.

8. Bias vs variance tradeoff balances:

A) Accuracy
B) Underfit & overfit
C) Speed
D) Data

Answer: B

Explanation: Underfit & overfit is correct because it best fits the concept of bias vs variance tradeoff balances.

9. High bias means:

A) Fast
B) Overfit
C) Underfit
D) Perfect

Answer: C

Explanation: Underfit is correct because it best fits the concept of high bias means.

10. Robustness means:

A) Size
B) Cost
C) Stability
D) Speed

Answer: C

Explanation: Stability is correct because it best fits the concept of robustness means.

Week 11

Concepts

1. Interpretability vs Explainability

Interpretability -> model itself is understandable (like linear regression).
Explainability -> using extra methods to explain complex models.
Simple = interpretable, Complex = needs explanation.

2. Black Box Models

Models like deep neural networks.
Internal working not understandable by humans.
High accuracy but low transparency.

3. Need for Explainable AI (XAI)

Required for:
Trust
Accountability
Legal compliance
Important in healthcare, finance.

4. Transparency in AI

Ability to understand how decisions are made.
Helps debugging and trust.

5. Feature Attribution

Shows which input features influenced output.
Example: which pixel or word mattered most.

6. Saliency Maps

Highlight important parts of input.
Used in images (important pixels).
Limitation: shows "where" not "how".

7. Perturbation Methods

Change input slightly and observe output.
Helps understand model behavior.

8. LIME (Local Interpretable Model-Agnostic Explanations)

Explains single prediction locally.
Uses simpler model around data point.

9. SHAP (SHapley Additive Explanations)

Based on game theory.
Shows contribution of each feature.
More consistent than LIME.

10. Local vs Global Explanation

Local -> explains one prediction
Global -> explains entire model behavior

11. Fidelity in Explanation

How accurate explanation reflects real model

Low fidelity -> misleading explanation

12. Black Box Problem

Hard to know why model made decision

Risk in critical applications

13. Epistemic Problem

Humans don't understand model logic

Due to complexity or lack of knowledge

14. "Many Things" Problem

System failures due to multiple interacting factors

Hard to identify single cause

15. Tradeoff: Accuracy vs Interpretability

Complex models -> accurate but opaque

Simple models -> interpretable but less powerful

16. Rudin's Argument

No need to sacrifice interpretability for accuracy

Interpretable models can perform equally well

17. Model Debugging using XAI

Helps find errors in model

Identifies wrong patterns learned

18. Ethical Importance of XAI

Ensures fairness

Prevents discrimination

Enables accountability

Example: In Week 11, a company can apply interpretability vs explainability together with black box models to improve decision quality and business outcomes.

MCQs

1. What is Interpretability means?

A) External explanation
B) Speed
C) Training
D) Model understandable

Answer: D

Explanation: Model understandable is correct because it best fits what is interpretability means.

2. What is Explainability means?

A) Training
B) External explanation
C) No explanation
D) Simple model

Answer: B

Explanation: External explanation is correct because it best fits what is explainability means.

3. What is Black box model is?

A) Linear
B) Complex
C) Simple
D) Transparent

Answer: B

Explanation: Complex is correct because it best fits what is black box model is.

4. What is XAI is needed for?

A) Storage
B) Hardware
C) Trust
D) Speed

Answer: C

Explanation: Trust is correct because it best fits what is xai is needed for.

5. What is Transparency helps in?

A) Understanding
B) Storage
C) Speed
D) Gaming

Answer: A

Explanation: Understanding is correct because it best fits what is transparency helps in.

6. What is Feature attribution shows?

A) Feature importance
B) Memory
C) Data size
D) Speed

Answer: A

Explanation: Feature importance is correct because it best fits what is feature attribution shows.

7. What is Saliency maps highlight?

A) Data
B) Code
C) Output
D) Important inputs

Answer: D

Explanation: Important inputs is correct because it best fits what is saliency maps highlight.

8. What is Limitation of saliency maps?

A) Complex
B) No "how" explanation
C) Slow
D) No output

Answer: B

Explanation: No "how" explanation is correct because it best fits what is limitation of saliency maps.

9. What is Perturbation means?

A) Input change
B) Data storage
C) Output change
D) Training

Answer: A

Explanation: Input change is correct because it best fits what is perturbation means.

10. What is LIME explains?

A) Local prediction
B) Data
C) Hardware
D) Whole model

Answer: A

Explanation: Local prediction is correct because it best fits what is lime explains.

11. What is SHAP is based on?

A) Physics
B) Biology
C) Game theory
D) Math only

Answer: C

Explanation: Game theory is correct because it best fits what is shap is based on.

12. What is SHAP shows?

A) Data
B) Loss
C) Model speed
D) Feature contribution

Answer: D

Explanation: Feature contribution is correct because it best fits what is shap shows.

13. What is Local explanation means?

A) One prediction
B) Dataset
C) Whole model
D) Code

Answer: A

Explanation: One prediction is correct because it best fits what is local explanation means.

14. What is Global explanation means?

A) Entire model
B) Code
C) Data
D) One point

Answer: A

Explanation: Entire model is correct because it best fits what is global explanation means.

15. What is Fidelity means?

A) Accuracy of explanation
B) Speed
C) Data
D) Storage

Answer: A

Explanation: Accuracy of explanation is correct because it best fits what is fidelity means.

16. Black box problem refers to?

A) Storage
B) Speed
C) Lack of understanding
D) Cost

Answer: C

Explanation: Lack of understanding is correct because it best fits the concept of black box problem refers to.

17. What is Epistemic problem means?

A) Storage
B) Speed
C) Data issue
D) Knowledge gap

Answer: D

Explanation: Knowledge gap is correct because it best fits what is epistemic problem means.

18. What is Many things problem means?

A) Multiple causes
B) No cause
C) Data issue
D) Single cause

Answer: A

Explanation: Multiple causes is correct because it best fits what is many things problem means.

19. What is Accuracy vs interpretability?

A) No relation
B) Tradeoff
C) Same
D) Fixed

Answer: B

Explanation: Tradeoff is correct because it best fits what is accuracy vs interpretability.

20. What is Complex models are?

A) Simple
B) Opaque
C) Linear
D) Interpretable

Answer: B

Explanation: Opaque is correct because it best fits what is complex models are.

21. What is Simple models are?

A) Deep
B) Complex
C) Opaque
D) Interpretable

Answer: D

Explanation: Interpretable is correct because it best fits what is simple models are.

22. What is Rudin argues?

A) Accuracy > interpretability
B) No tradeoff needed
C) Tradeoff needed
D) Ignore models

Answer: B

Explanation: No tradeoff needed is correct because it best fits what is rudin argues.

23. What is XAI helps in?

A) Training
B) Speed
C) Debugging
D) Storage

Answer: C

Explanation: Debugging is correct because it best fits what is xai helps in.

24. What is Ethical importance of XAI?

A) Cost
B) Speed
C) Fairness
D) Storage

Answer: C

Explanation: Fairness is correct because it best fits what is ethical importance of xai.

25. What is Feature attribution method?

A) RNN
B) SHAP
C) LSTM
D) CNN

Answer: B

Explanation: SHAP is correct because it best fits what is feature attribution method.

26. What is LIME is?

A) Static
B) None
C) Local
D) Global

Answer: C

Explanation: Local is correct because it best fits what is lime is.

27. What is SHAP is?

A) Local + global
B) None
C) Only global
D) Only local

Answer: A

Explanation: Local + global is correct because it best fits what is shap is.

28. What is Perturbation methods work by?

A) Input variation
B) Output fixing
C) Storage
D) Training

Answer: A

Explanation: Input variation is correct because it best fits what is perturbation methods work by.

29. What is Saliency maps used in?

A) Data
B) Text only
C) Audio
D) Images

Answer: D

Explanation: Images is correct because it best fits what is saliency maps used in.

30. What is Explainability needed for?

A) Storage
B) Games
C) Speed
D) High-risk systems

Answer: D

Explanation: High-risk systems is correct because it best fits what is explainability needed for.

31. Black box models are risky because?

A) Hard to understand
B) Cheap
C) Fast
D) Small

Answer: A

Explanation: Hard to understand is correct because it best fits the concept of black box models are risky because.

32. What is Transparency ensures?

A) Trust
B) Cost
C) Speed
D) Storage

Answer: A

Explanation: Trust is correct because it best fits what is transparency ensures.

33. What is Explanation quality depends on?

A) Size
B) Speed
C) Fidelity
D) Storage

Answer: C

Explanation: Fidelity is correct because it best fits what is explanation quality depends on.

34. What is Local explanation example?

A) One output
B) Code
C) Entire model
D) Dataset

Answer: A

Explanation: One output is correct because it best fits what is local explanation example.

35. What is Global explanation example?

A) One case
B) Data
C) Code
D) Model behavior

Answer: D

Explanation: Model behavior is correct because it best fits what is global explanation example.

36. Epistemic issue occurs due to?

A) Data
B) Lack of knowledge
C) Storage
D) Speed

Answer: B

Explanation: Lack of knowledge is correct because it best fits the concept of epistemic issue occurs due to.

37. Many things problem leads to?

A) Hard debugging
B) Easy debugging
C) Small model
D) Fast model

Answer: A

Explanation: Hard debugging is correct because it best fits the concept of many things problem leads to.

38. What is SHAP advantage?

A) Consistency
B) Random
C) Size
D) Speed

Answer: A

Explanation: Consistency is correct because it best fits what is shap advantage.

39. What is LIME limitation?

A) Global only
B) No explanation
C) Fast
D) Local only

Answer: D

Explanation: Local only is correct because it best fits what is lime limitation.

40. What is XAI improves?

A) Storage
B) Cost
C) Trust
D) Speed

Answer: C

Explanation: Trust is correct because it best fits what is xai improves.

41. What is Feature importance shows?

A) Size
B) Influence
C) Data
D) Speed

Answer: B

Explanation: Influence is correct because it best fits what is feature importance shows.

42. What is Saliency maps show?

A) When
B) Where
C) How
D) Why

Answer: B

Explanation: Where is correct because it best fits what is saliency maps show.

43. What is Complex models need?

A) No explanation
B) Data
C) Speed
D) XAI

Answer: D

Explanation: XAI is correct because it best fits what is complex models need.

44. What is Debugging model uses?

A) Storage
B) Data
C) Speed
D) XAI

Answer: D

Explanation: XAI is correct because it best fits what is debugging model uses.

45. What is Accountability requires?

A) Storage
B) Cost
C) Speed
D) Explainability

Answer: D

Explanation: Explainability is correct because it best fits what is accountability requires.

46. What is Ethical AI requires?

A) Speed
B) Cost
C) Transparency
D) Storage

Answer: C

Explanation: Transparency is correct because it best fits what is ethical ai requires.

47. What is Best explanation is?

A) Small
B) Random
C) High fidelity
D) Fast

Answer: C

Explanation: High fidelity is correct because it best fits what is best explanation is.

48. What is Main goal of XAI?

A) Cost
B) Speed
C) Storage
D) Understanding

Answer: D

Explanation: Understanding is correct because it best fits what is main goal of xai.

49. What best describes Interpretability vs Explainability in AI for management?

A) Interpretability vs Explainability removes the need for data quality checks and monitoring.
B) Interpretability vs Explainability eliminates the need for managerial judgment and governance.
C) Interpretability vs Explainability guarantees perfect outcomes in every scenario.
D) Interpretability -> model itself is understandable (like linear regression).

Answer: D

Explanation: Interpretability -> model itself is understandable (like linear regression). is correct because it best fits what best describes interpretability vs explainability in ai for management.

50. Which statement about Black Box Models is most accurate?

A) Black Box Models removes the need for data quality checks and monitoring.
B) Black Box Models guarantees perfect outcomes in every scenario.
C) Models like deep neural networks.
D) Black Box Models eliminates the need for managerial judgment and governance.

Answer: C

Explanation: Models like deep neural networks. is correct because it best fits which statement about black box models is most accurate.

Online Assessment Questions

1. Interpretability means:

A) Data
B) Internal understanding
C) External explanation
D) Training

Answer: B

Explanation: Internal understanding is correct because it best fits the concept of interpretability means.

2. Explainability means:

A) External explanation
B) Training
C) Internal
D) Data

Answer: A

Explanation: External explanation is correct because it best fits the concept of explainability means.

3. SHAP is based on:

A) Biology
B) Game theory
C) Data
D) Physics

Answer: B

Explanation: Game theory is correct because it best fits the concept of shap is based on.

4. LIME explains:

A) Data
B) Code
C) Single prediction
D) Whole model

Answer: C

Explanation: Single prediction is correct because it best fits the concept of lime explains.

5. Saliency maps show:

A) Why
B) How
C) When
D) Where

Answer: D

Explanation: Where is correct because it best fits the concept of saliency maps show.

6. Fidelity means:

A) Storage
B) Speed
C) Explanation accuracy
D) Data

Answer: C

Explanation: Explanation accuracy is correct because it best fits the concept of fidelity means.

7. Black box problem is:

A) Lack of transparency
B) Speed
C) Cost
D) Storage

Answer: A

Explanation: Lack of transparency is correct because it best fits the concept of black box problem is.

8. Perturbation method works by:

A) Training
B) Storage
C) Output change
D) Input change

Answer: D

Explanation: Input change is correct because it best fits the concept of perturbation method works by.

9. XAI helps in:

A) Speed
B) Cost
C) Storage
D) Trust

Answer: D

Explanation: Trust is correct because it best fits the concept of xai helps in.

10. Rudin argues:

A) Tradeoff needed
B) Use deep models
C) No tradeoff
D) Ignore models

Answer: C

Explanation: No tradeoff is correct because it best fits the concept of rudin argues.

Week 12

Concepts

1. Fairness in AI

Fairness means AI decisions should not discriminate between groups.
Important in hiring, loans, healthcare.
Ensures equal treatment.

2. Types of Fairness

Individual fairness -> similar people treated similarly
Group fairness -> different groups treated equally
Both are important but hard to balance

3. Equality of Opportunity

People should have equal chance regardless of background.
Focus on fairness in outcomes.

4. Narrow View of Equality

Only current qualifications matter.
Ignores past disadvantages.

5. Middle View

Supports interventions like affirmative action.
Helps reduce inequality at decision stage.

6. Broad View

Focuses on root causes in society.
Long-term fairness approach.

7. Systematic Disadvantage

Some groups face disadvantages due to social factors.
Not related to ability or performance.

8. Discrimination in AI

Happens when model treats groups unfairly.
Can be intentional or unintentional.

9. Unintentional Bias

AI may produce unfair results without intention.
Comes from biased data or design.

10. Diversity vs Inclusion

Diversity -> different people included
Inclusion -> equal participation and access

11. Equity vs Equality

Equality -> same treatment
Equity -> fair treatment based on need

12. Justice as Fairness (Rawls)

Fair distribution of resources
Based on impartial decisions

13. Ethical Responsibility in AI

AI cannot be morally responsible
Humans must take responsibility

14. Responsibility Gap

Happens when no one is clearly responsible for AI decisions
Due to automation and complexity

15. Distributed Responsibility

Many people involved in AI system
Responsibility shared across them

16. Many Hands Problem

Hard to assign blame when many actors involved
Common in AI failures

17. Many Things Problem

Failure caused by multiple factors
Difficult to identify root cause

18. Transparency in Ethics

Important for accountability
Users should understand decisions

19. Importance of Fair AI

Prevents discrimination
Builds trust
Required for regulation

20. Ethical AI Goals

Fairness
Accountability
Transparency
Safety

> *Example: In Week 12, a company can apply fairness in ai together with types of fairness to improve decision quality and business outcomes.*

MCQs

1. What is Fairness in AI means?

A) Storage
B) No discrimination
C) Accuracy
D) Speed

Answer: B

Explanation: No discrimination is correct because it best fits what is fairness in ai means.

2. What is Individual fairness means?

A) Group equality
B) No fairness
C) Similar people treated similarly
D) Random decisions

Answer: C

Explanation: Similar people treated similarly is correct because it best fits what is individual fairness means.

3. What is Group fairness means?

A) Random
B) Same individuals
C) Equal treatment of groups
D) No groups

Answer: C

Explanation: Equal treatment of groups is correct because it best fits what is group fairness means.

4. What is Equality of opportunity ensures?

A) Equal chance
B) Same data
C) Same outcome
D) Same speed

Answer: A

Explanation: Equal chance is correct because it best fits what is equality of opportunity ensures.

5. What is Narrow view focuses on?

A) Random
B) Past inequality
C) Future
D) Current qualifications

Answer: D

Explanation: Current qualifications is correct because it best fits what is narrow view focuses on.

6. What is Middle view supports?

A) Ignore bias
B) Affirmative action
C) No change
D) Remove data

Answer: B

Explanation: Affirmative action is correct because it best fits what is middle view supports.

7. What is Broad view focuses on?

A) Immediate decision
B) Root causes
C) Speed
D) Data

Answer: B

Explanation: Root causes is correct because it best fits what is broad view focuses on.

8. What is Systematic disadvantage is?

A) Social disadvantage
B) Random
C) Technical
D) Fair

Answer: A

Explanation: Social disadvantage is correct because it best fits what is systematic disadvantage is.

9. What is AI discrimination comes from?

A) Storage
B) Bias
C) Speed
D) Hardware

Answer: B

Explanation: Bias is correct because it best fits what is ai discrimination comes from.

10. What is Unintentional bias means?

A) None
B) Malicious
C) Accidental
D) Planned

Answer: C

Explanation: Accidental is correct because it best fits what is unintentional bias means.

11. What is Diversity means?

A) No people
B) Same people
C) Random
D) Different people

Answer: D

Explanation: Different people is correct because it best fits what is diversity means.

12. What is Inclusion means?

A) Speed
B) Data
C) Equal participation
D) Exclusion

Answer: C

Explanation: Equal participation is correct because it best fits what is inclusion means.

13. What is Equality means?

A) Fair treatment
B) Random
C) Same treatment
D) Bias

Answer: C

Explanation: Same treatment is correct because it best fits what is equality means.

14. What is Equity means?

A) Random
B) Fair treatment
C) None
D) Same treatment

Answer: B

Explanation: Fair treatment is correct because it best fits what is equity means.

15. Justice as fairness proposed by?

A) Rawls
B) Newton
C) Turing
D) Aristotle

Answer: A

Explanation: Rawls is correct because it best fits the concept of justice as fairness proposed by.

16. AI moral responsibility lies with?

A) Data
B) Model
C) Humans
D) AI

Answer: C

Explanation: Humans is correct because it best fits the concept of ai moral responsibility lies with.

17. What is Responsibility gap means?

A) No clear blame
B) Clear blame
C) Fast model
D) Small model

Answer: A

Explanation: No clear blame is correct because it best fits what is responsibility gap means.

18. What is Distributed responsibility means?

A) Random
B) Multiple actors
C) No actors
D) Single person

Answer: B

Explanation: Multiple actors is correct because it best fits what is distributed responsibility means.

19. What is Many hands problem?

A) Many people involved
B) One cause
C) No people
D) Data

Answer: A

Explanation: Many people involved is correct because it best fits what is many hands problem.

20. What is Many things problem?

A) No factor
B) Single factor
C) Multiple factors
D) Random

Answer: C

Explanation: Multiple factors is correct because it best fits what is many things problem.

21. What is Transparency helps?

A) Storage
B) Speed
C) Cost
D) Accountability

Answer: D

Explanation: Accountability is correct because it best fits what is transparency helps.

22. What is Ethical AI ensures?

A) Fairness
B) Speed
C) Cost
D) Storage

Answer: A

Explanation: Fairness is correct because it best fits what is ethical ai ensures.

23. What is AI fairness needed in?

A) Hiring
B) Storage
C) Speed
D) Games

Answer: A

Explanation: Hiring is correct because it best fits what is ai fairness needed in.

24. What is Bias leads to?

A) Fairness
B) Speed
C) Discrimination
D) Storage

Answer: C

Explanation: Discrimination is correct because it best fits what is bias leads to.

25. What is Ethical AI includes?

A) Storage
B) Speed
C) Transparency
D) Cost

Answer: C

Explanation: Transparency is correct because it best fits what is ethical ai includes.

26. What is Equality focuses on?

A) Fair treatment
B) Random
C) Same treatment
D) None

Answer: C

Explanation: Same treatment is correct because it best fits what is equality focuses on.

27. What is Equity focuses on?

A) Random
B) None
C) Fair outcome
D) Same treatment

Answer: C

Explanation: Fair outcome is correct because it best fits what is equity focuses on.

28. What is Narrow view ignores?

A) Present
B) Past inequality
C) Speed
D) Data

Answer: B

Explanation: Past inequality is correct because it best fits what is narrow view ignores.

29. What is Middle view intervention at?

A) Speed
B) Data
C) Society
D) Decision stage

Answer: D

Explanation: Decision stage is correct because it best fits what is middle view intervention at.

30. What is Broad view intervention at?

A) Data
B) Model
C) Society level
D) Decision

Answer: C

Explanation: Society level is correct because it best fits what is broad view intervention at.

31. What is Inclusion ensures?

A) Random
B) Bias
C) Equal access
D) Exclusion

Answer: C

Explanation: Equal access is correct because it best fits what is inclusion ensures.

32. What is Diversity ensures?

A) Same group
B) Random
C) Different groups
D) No groups

Answer: C

Explanation: Different groups is correct because it best fits what is diversity ensures.

33. What is AI fairness improves?

A) Error
B) Bias
C) Trust
D) Noise

Answer: C

Explanation: Trust is correct because it best fits what is ai fairness improves.

34. Responsibility gap occurs due to?

A) Simplicity
B) Speed
C) Complexity
D) Storage

Answer: C

Explanation: Complexity is correct because it best fits the concept of responsibility gap occurs due to.

35. What is Ethical AI requires?

A) Storage
B) Accountability
C) Speed
D) Cost

Answer: B

Explanation: Accountability is correct because it best fits what is ethical ai requires.

36. AI cannot be responsible because?

A) No consciousness
B) Small
C) Fast
D) Cheap

Answer: A

Explanation: No consciousness is correct because it best fits the concept of ai cannot be responsible because.

37. What is Fair AI ensures?

A) Storage
B) Bias
C) No discrimination
D) Speed

Answer: C

Explanation: No discrimination is correct because it best fits what is fair ai ensures.

38. What is Ethical principles include?

A) Fairness
B) Speed
C) Storage
D) Cost

Answer: A

Explanation: Fairness is correct because it best fits what is ethical principles include.

39. What is Accountability means?

A) Responsibility
B) Storage
C) Speed
D) No blame

Answer: A

Explanation: Responsibility is correct because it best fits what is accountability means.

40. What is Transparency means?

A) Clear
B) Random
C) None
D) Hidden

Answer: A

Explanation: Clear is correct because it best fits what is transparency means.

41. What is AI bias should be?

A) Stored
B) Increased
C) Ignored
D) Reduced

Answer: D

Explanation: Reduced is correct because it best fits what is ai bias should be.

42. What is Ethical AI improves?

A) Error
B) Trust
C) Noise
D) Bias

Answer: B

Explanation: Trust is correct because it best fits what is ethical ai improves.

43. What is Discrimination is?

A) Unfair treatment
B) Random
C) None
D) Fair

Answer: A

Explanation: Unfair treatment is correct because it best fits what is discrimination is.

44. What is Fairness includes?

A) Equality
B) Bias
C) Error
D) Noise

Answer: A

Explanation: Equality is correct because it best fits what is fairness includes.

45. What is AI decisions must be?

A) Random
B) Biased
C) Fair
D) Fast

Answer: C

Explanation: Fair is correct because it best fits what is ai decisions must be.

46. What is Ethical AI avoids?

A) Fairness
B) Accountability
C) Transparency
D) Discrimination

Answer: D

Explanation: Discrimination is correct because it best fits what is ethical ai avoids.

47. What is Responsibility shared means?

A) One person
B) Many people
C) AI
D) No one

Answer: B

Explanation: Many people is correct because it best fits what is responsibility shared means.

48. What is Root cause focus ->?

A) Broad
B) None
C) Middle
D) Narrow

Answer: A

Explanation: Broad is correct because it best fits what is root cause focus ->.

49. Final goal of ethical AI?

A) Speed
B) Storage
C) Cost
D) Fair + safe systems

Answer: D

Explanation: Fair + safe systems is correct because it best fits the concept of final goal of ethical ai.

50. What best describes Fairness in AI in AI for management?

A) Fairness in AI eliminates the need for managerial judgment and governance.
B) Fairness in AI guarantees perfect outcomes in every scenario.
C) Fairness in AI removes the need for data quality checks and monitoring.
D) Fairness means AI decisions should not discriminate between groups.

Answer: D

Explanation: Fairness means AI decisions should not discriminate between groups. is correct because it best fits what best describes fairness in ai in ai for management.

Online Assessment Questions

1. Fairness means:

A) Cost
B) Storage
C) Speed
D) No discrimination

Answer: D

Explanation: No discrimination is correct because it best fits the concept of fairness means.

2. Individual fairness means:

A) Similar people treated similarly
B) Random
C) None
D) Group equality

Answer: A

Explanation: Similar people treated similarly is correct because it best fits the concept of individual fairness means.

3. Equity means:

A) Fair treatment
B) Random
C) None
D) Same treatment

Answer: A

Explanation: Fair treatment is correct because it best fits the concept of equity means.

4. Responsibility gap means:

A) No clear blame
B) Speed
C) Storage
D) Clear blame

Answer: A

Explanation: No clear blame is correct because it best fits the concept of responsibility gap means.

5. Diversity means:

A) Different
B) Random
C) Same
D) None

Answer: A

Explanation: Different is correct because it best fits the concept of diversity means.

6. Inclusion ensures:

A) Speed
B) Exclusion
C) Participation
D) Bias

Answer: C

Explanation: Participation is correct because it best fits the concept of inclusion ensures.

7. Narrow view focuses on:

A) Present qualification
B) Past
C) Future
D) None

Answer: A

Explanation: Present qualification is correct because it best fits the concept of narrow view focuses on.

8. Broad view focuses on:

A) Immediate
B) Data
C) Root causes
D) Speed

Answer: C

Explanation: Root causes is correct because it best fits the concept of broad view focuses on.

9. Many hands problem means:

A) Many actors
B) No actors
C) One person
D) Random

Answer: A

Explanation: Many actors is correct because it best fits the concept of many hands problem means.

10. Ethical AI ensures:

A) Fair systems
B) Cost
C) Speed
D) Storage

Answer: A

Explanation: Fair systems is correct because it best fits the concept of ethical ai ensures.